

Chapter 1

Historical Conditions of the Contemporary Economy

This chapter establishes the historical ground from which Generative Relational Economics (GRE) becomes intelligible as a contemporary economic research programme. Its role within Part I is preparatory and diagnostic. Before the book reconstructs the intellectual lineages of GRE in Chapter 2 or states its disciplinary constitution in Chapter 3, the present chapter asks what historical transformations make a relational and generative reconsideration of economic life increasingly salient. The objective is therefore to locate GRE within a changing material, institutional, epistemic, and technical environment. The chapter treats economic categories, economic subjects, and economic objects as historically stabilized formations whose analytical usefulness depends on the conditions under which they are produced, recognized, and governed.

The discussion follows a historical-material orientation without reducing economic thought to a mechanical reflection of productive conditions. Economic concepts emerge within relations among productive forces, institutions, law, administrative practice, scientific techniques, political norms, and historically specific problems of coordination and allocation. Categories such as labor, capital, property, firm, household, scarcity, and allocation acquired particular analytical force because modern economic orders made corresponding subjects, resources, and transactions increasingly legible and actionable. Entity-centered and allocation-centered approaches therefore appear in this chapter as historically successful abstractions with identifiable scope conditions. Their historical success forms part of the explanation for GRE's emergence, because a new framework has to account for the achievements of inherited abstractions as well as the situations in which those abstractions become strained.

The chapter develops its argument through three connected movements. Sections 1–7 reconstruct the historical formation of economic categories, economic problematics, modern institutional entities, allocation mechanisms, and differentiated sources of economic power. Sections 8–18 then follow the expansion of relational production through organizational interdependence, transnational networks, digital mediation, artificial intelligence, changing forms of scarcity, capability asymmetry, symbolic abundance, and the destabilization of inherited economic subject positions. This middle movement gradually shifts the analytical focus from

entities considered in relative isolation toward relations that constitute subjects, distribute capacities, and alter future possibilities. Sections 19–27 examine the scientific, computational, evidential, political, legal, and institutional conditions that make richer relational and generative analysis increasingly feasible, while also identifying the epistemic and governance limits that accompany this expansion of analytical capacity. Sections 28–29 synthesize these developments into an account of the historical opening, possibilities, and limits of GRE.

A central distinction throughout the chapter concerns the difference between relational importance and generative importance. Economic life has always depended on relations. The contemporary issue is that more of what matters economically is becoming difficult to understand through recipient-level states, isolated transactions, or stable entity boundaries alone. Relations can organize access, recognition, dependence, knowledge, authority, and productive coordination; they can also transform the capacities of the participants and the conditions under which later relations become possible. Generative analysis begins when the object of inquiry includes these downstream transformations: the production of new subjects, capabilities, constraints, institutional forms, option spaces, and transformation rules. GRE therefore enters the chapter gradually, after the historical and institutional conditions that motivate this extension have been reconstructed.

The same historical transformations that enlarge the relevance of relational and generative analysis also enlarge the risks of overclaiming. Digital infrastructures increase observability while creating new possibilities for surveillance and classification. Computational capacity enables larger and more heterogeneous models while increasing dependence on representation, parameterization, and model selection. Artificial intelligence expands productive and analytical capacity while intensifying questions concerning concentration, epistemic dependence, subjecthood, and the conversion of capability asymmetry into economic power. Political participation, human-rights recognition, capability-oriented development, commons institutions, and environmental or intergenerational concerns broaden the range of economically relevant relations and values, while preserving normative plurality. The chapter therefore treats technical possibility, evidential possibility, institutional possibility, and normative justification as distinct dimensions.

The resulting historical claim is modest in form and substantial in consequence. GRE is situated within a period in which relational and generative processes have become increasingly consequential for production, allocation, subject formation, value creation, knowledge, and power, while contemporary analytical infrastructures make those processes more accessible to systematic inquiry. This conjunction supplies neither a proof of GRE nor a prediction that economics must converge toward it. It establishes a problem-space in which GRE can be developed, compared with existing theories, tested against evidence, and revised through later inquiry. The next chapter turns from these historical conditions to the intellectual lineages and theoretical resources through which such a programme becomes conceptually possible.

1.1 Material Conditions of Economic Thought

This section establishes the historical-material premise of the chapter. Its role is to explain how economic categories, analytical objects, and preferred abstractions acquire their scope within historically specific social orders. The objective is narrower than a general philosophy of economic knowledge: the discussion identifies material, institutional, and epistemic conditions that make particular forms of economic reasoning possible, useful, and credible at particular moments. The argument proceeds in five stages. It first treats economic categories as historical formations; it then considers how productive capacities alter the practical and epistemic environment of inquiry, how relations of production become stabilized through institutions, how abstraction depends on historically available techniques of representation, and how the objects recognized by economics can change with these conditions. The method is historical and interpretive. Several established traditions are used as partial resources for specifying the problem, while the chapter keeps their stronger ontological and teleological commitments open for later examination.

The guiding position is that economic thought develops within the same world it seeks to describe. Changes in production, administration, law, measurement, communication, and institutional organization therefore have a double relevance: they alter economic life, and they alter the evidence, categories, and representations through which economic life becomes intelligible. Marx's materialist formulation is important for the first dimension because it connects productive forces, relations of production, and forms of social consciousness while locating legal and political forms within a wider material configuration (Marx 1859). Historical studies of quantification and statistics are important for the second dimension because they show that objectivity, classification, and measurement acquire authority through specific institutional settings and practical demands (Porter 1995; Desrosières 1998). GRE takes these contributions as reasons to historicize its own concepts. A theory that treats institutions, subjects, and values as generated within relations should also expect its categories to bear traces of the historical conditions under which they became thinkable.

1.1.1 Economic Categories as Historical Formations

The first analytical task is to distinguish the recurrence of an economic phenomenon from the historical stability of the category used to describe it. Exchange, provisioning, obligation, accumulation, scarcity, and coordination occur across very different social formations. Their recurrence leaves open whether the same analytical category has the same extension, institutional support, or causal role in each setting. A category such as labor, property, market, capital, household, or economy becomes analytically determinate through a historically specific configuration of practices, institutions, legal forms, measurement procedures, and relations among participants. Historical ontology provides a useful philosophical vocabulary for this point because concepts and objects can emerge together with new styles of reasoning and new practices of classification (Hacking 2002). The implication for GRE is methodological: historical inquiry concerns the formation and stabilization of an economic object together with the vocabulary and practices through which that object becomes describable.

Marx's critique of political economy offers an early and influential formulation of this historical specificity. In the 1859 Preface, relations of production are linked to a definite stage in the development of material productive forces, and legal and political forms are treated as historically connected with that economic structure (Marx 1859). The relevance here lies in the mode of inquiry. Stage claims and teleological implications associated with later readings of historical materialism remain separate questions. Economic categories acquire their content within determinate social relations. Capital, for example, is analytically different from a generic stock of useful objects because the category refers to a structured relation among property, production, labor, exchange, and accumulation. The same discipline applies to categories that appear more neutral. A household, a firm, a worker, or a market becomes an analytical unit through institutional and representational practices that determine which relations are treated as internal, external, relevant, or invisible.

Polanyi's historical account strengthens this point from a different direction. His analysis of nineteenth-century market society treats the expansion of market relations as part of a historically specific reorganization of economic and social relations (Polanyi 1944). This perspective matters for GRE because it shifts attention from the mere presence of exchange to the organization of exchange within a wider relational order. A market category may be analytically adequate in one domain and much less adequate when generalized across activities organized by reciprocity, household provision, public obligation, care, or other institutional forms. Historical specificity therefore constrains both the object under analysis and the permissible range of abstraction.

The historical character of categories also applies to the category of "the economy" itself. Mitchell reconstructs the modern understanding of the economy as a bounded totality of production, distribution, and consumption as a product of a mid-twentieth-century transformation in economic representation (Mitchell 1998). This example is important because changes in economic theory can accompany changes in the scale and boundary of the object itself. Once a national economy is represented as an integrated totality, questions about aggregate output, demand, employment, growth, and management become newly central, while many relations remain outside the principal frame. The formation of an economic category can therefore be an epistemic gain and a selective construction of visibility at the same time.

GRE adopts this historical stance while preserving standards of analytical adequacy. Categories can be more or less adequate to the relations they are used to analyze, and historically formed categories can still identify durable mechanisms or recurrent structures. The relevant discipline is to separate analytical usefulness from presumed timelessness. A category earns its place through the work it performs in a specified relational and historical domain, and its continued use remains open to revision when the domain, evidence, or relations change.

1.1.2 Productive Forces and Forms of Economic Knowledge

The second analytical task is to identify how transformations in productive capacity alter the environment in which economic knowledge is produced. Productive forces are used here in a broad historical sense that includes material techniques,

energy systems, transport, communication, computational capacity, organizational techniques, and the skills through which production can be coordinated. Their development changes the practical problems that demand explanation and the epistemic possibilities available for explaining them. GRE treats the relation between productive capacity and theory as historically mediated and contingent. Marx's formulation remains useful because it directs attention to the relation between changing productive capacities and changing relations of production while leaving room for more specific mechanisms through which knowledge responds (Marx 1859).

Changes in productive capacity can affect economic knowledge through several channels. New forms of production can generate new contractual relations, organizational units, and coordination problems. Changes in transport and communication can enlarge the spatial scale at which economic activity is coordinated. Changes in recording and computation can alter temporal resolution, making processes observable continuously or at shorter intervals. Productive development can also create new evidential infrastructures. Accounting systems, censuses, surveys, standardized classifications, and computational records participate in determining which economic differences can be aggregated, compared, and acted upon.

The history of quantification makes this point especially clear. Porter argues that the authority of quantitative objectivity developed partly in response to institutional demands for impersonal and standardized judgment across settings such as government, business, engineering, and social inquiry (Porter 1995). Desrosières reconstructs statistical reasoning through the history of classification, coding, aggregation, correlation, and modeling, emphasizing the conventions through which heterogeneous cases become comparable (Desrosières 1998). These histories place measurement inside the institutional history of economic knowledge. Measurement practices can become part of economic reality because decisions, eligibility, budgets, targets, and expectations are organized around the quantities they produce.

This observation is central to the historical possibility of GRE. Earlier economic thought analyzed relations with the conceptual and technical resources then available. Contemporary computation changes the tractability of relational questions by making large networks, longitudinal records, higher-order interactions, and multi-scale simulations technically accessible. A relation-centered or generative analysis therefore has a historical condition of possibility in the development of computational and evidential infrastructures, while its concepts retain analytical content beyond the infrastructures that support their investigation. Contemporary productive forces enlarge the possibility of GRE by making certain relational and generative questions more systematically researchable than under earlier technical conditions.

The same development creates new epistemic risks. Expanding measurement can encourage the identification of an object with its measurable manifestation. A productive system may become legible through variables that are administratively stable while remaining partial descriptions of its relational structure. Large computational representations can intensify this tendency by creating an appearance of completeness through scale. GRE therefore treats technical capacity as an expansion of possible inquiry accompanied by a stronger requirement to distinguish relational reality, manifestation, and representation. Productive forces transform both the reach of economic knowledge and the forms of error that economic knowledge can institu-

tionalize.

1.1.3 Relations of Production and Institutional Form

The third analytical task is to connect economic relations with the institutional forms that stabilize them. Relations of production concern organized positions in production, access to resources, authority over decisions, claims on output, and the conditions under which participants enter, remain within, or leave a productive arrangement. These relations become durable through legal, organizational, administrative, cultural, and infrastructural forms. Institutional form therefore participates in constituting the economic relation and the terms under which that relation can be recognized.

The materialist tradition provides one route into this problem by linking property relations and relations of production with stages of productive development (Marx 1859). Polanyi provides another by showing that market society depended on institutional transformations that reorganized the relation between economic exchange and wider social life (Polanyi 1944). Read together while preserving their differences, these approaches support a modest claim useful to GRE: productive activity is mediated by historically specific institutional arrangements, and changes in those arrangements can transform the economic meaning of otherwise similar acts. The transfer of an object, the performance of labor, or the provision of care can occupy different economic positions depending on property rules, contractual status, household organization, public provision, professional classification, and institutional recognition.

Institutional form also affects the scale at which economic entities appear coherent. A corporation can be treated as a unit in accounting, law, taxation, and contract because multiple internal relations are condensed into a recognized organizational person. A household can appear as a consumption unit, a care relation, a property-sharing arrangement, or an internally differentiated organization depending on the institutional and analytical frame. The same nominal entity can therefore support multiple relational descriptions. GRE later develops this point as a problem of economic architecture; its historical implication belongs here because the apparent stability of entities is partly an achievement of institutional form.

The relation between institution and economic knowledge is recursive. Institutions stabilize categories, and categories feed back into institutions. Once an administrative system recognizes a class of worker, firm, asset, household, industry, or transaction, the category can organize reporting, eligibility, taxation, regulation, contracting, and comparison. Desrosières' history of statistical classification is relevant because classification and encoding are practical operations that help create standardized populations for analysis (Desrosières 1998). Hacking's historical ontology supplies a broader philosophical formulation: classifications and styles of reasoning can participate in the historical emergence of the objects to which they refer (Hacking 2002). In economic life, institutional categories can therefore become causally effective while remaining partial representations of the relational realities they organize.

GRE approaches institutional form through two linked concerns. Stabilization concerns the rules, classifications, infrastructures, and expectations that make a relation durable enough to function as an economic object. Transformation concerns changes

in those supports that alter the relation itself. This dual focus treats institutions as historically generated relational arrangements that condition later economic action while remaining open to reproduction, contestation, and change.

1.1.4 Economic Abstraction under Historical Constraints

The fourth analytical task concerns abstraction. Abstraction is constitutive of economic inquiry because comparison, aggregation, and explanation require some differences to be suppressed. The historical question concerns which abstractions become credible, operational, and institutionally consequential under particular conditions, together with the relations that disappear when those abstractions are generalized beyond their domain of adequacy.

Quantification provides a clear example. A numerical category can coordinate judgment across distance by replacing local variation with standardized comparison. Porter emphasizes that quantitative objectivity gained authority partly because it could support impersonal forms of decision under institutional conditions in which reliance on individual discretion was difficult or contested (Porter 1995). Desrosières shows that statistical reasoning depends on conventions of equivalence through which heterogeneous cases are classified and made comparable (Desrosières 1998). These accounts reveal abstraction as an institutional achievement. A statistical aggregate becomes useful because differences judged irrelevant to a task have been suppressed in a repeatable way.

Economic abstraction is constrained by the historical availability of such operations. A national aggregate presupposes classifications, territorial boundaries, reporting systems, institutional capacities, and conventions through which dispersed activities can be treated as commensurable. Mitchell's account of the emergence of "the economy" demonstrates how representation helped establish a national economic totality as an object of expertise and intervention (Mitchell 1998). The abstraction made aggregate economic management thinkable, and the gain depended on a selective boundary around the relations represented. Every successful abstraction therefore establishes an analytical field together with a remainder.

For GRE, the remainder has methodological importance. Entity-centered accounting can be highly effective for questions concerning ownership, liability, taxation, or the direct recipient of a transfer. The same abstraction can be insufficient when the research question concerns how a transfer changes trust, dependence, coordination, knowledge, or future relational capacity. A graph representation may reveal connectivity while suppressing legal meaning; an ethnographic reconstruction may recover meaning while offering weak estimates of prevalence; a macroeconomic aggregate may represent total activity while hiding the distribution of relational conditions that generated it. The appropriate response is explicit specification of the abstraction's object, excluded differences, and historical conditions of usability.

This perspective also changes the interpretation of theoretical disagreement. Two economic theories can appear to conflict because they abstract at different scales, stabilize different objects, or preserve different relations. A theory centered on individual choice and a theory centered on institutions may produce incompatible descriptions when each is treated as complete, while still identifying real structures at

different levels. GRE later develops translation among theories as a distinct research programme. The material point established here is prior: abstraction is historically conditioned, and theoretical categories should be evaluated partly by the relations they preserve and the relations they suppress.

1.1.5 Transformation of Economic Objects across Historical Periods

The final analytical task draws the preceding arguments together into a historical view of changing economic objects. Economic history includes changes in quantities and institutions, and it also includes changes in what can appear as an economic object in the first place. New forms of production, administration, measurement, and political organization can stabilize entities and aggregates absent from earlier analytical languages or differently organized within them. Hacking's account of historically emergent concepts and objects provides a general philosophical frame for this possibility (Hacking 2002). Mitchell's reconstruction of the twentieth-century emergence of "the economy" provides a specifically economic example in which a new bounded object of macroeconomic representation became available to expertise and intervention (Mitchell 1998).

This transformation concerns relational visibility and institutional coordination. Activities later assembled into a national economy existed before the aggregate became a dominant object of analysis. The historical change lay in the representational and institutional configuration through which those activities could be organized as one object, measured through common categories, and treated as a target of coordinated intervention. A similar caution applies to newer objects. Data, attention, platform intermediation, algorithmic labor, knowledge infrastructures, and artificial productive capacity can become increasingly salient to economics because productive relations and measurement systems make their economic effects more visible and more consequential. Their appearance as economic objects therefore reflects both underlying changes in relations and changes in the apparatus through which those relations are represented.

The historical transformation of objects has two consequences for GRE. First, the field should expect its own object list to remain open. Relations, capacities, constraints, value forms, and generative possibilities that are analytically peripheral today may become central under different productive and institutional conditions. Second, economic theory should preserve the historical provenance of its categories. A concept developed for industrial wage labor, territorially organized markets, or nationally bounded accounting may continue to be useful under digital and transnational conditions, yet its range of adequacy requires renewed examination when the relations denoted by the term change. Historical continuity of terminology can conceal transformation in the underlying relational structure.

The historical-material argument developed in this section therefore establishes a limited premise: economic categories and objects are historically conditioned, and transformations in productive forces, institutional relations, and representational capacities can change both the economy and the forms of knowledge through which it becomes intelligible. The subsequent sections examine how modern entity-centered economic structures were historically stabilized, how digitalization and artificial in-

telligence strain some inherited abstractions, and why contemporary scientific and computational conditions create a specific opening for relational and generative economic analysis.

1.2 Genealogy of Economic Problematics

This section reconstructs a selective genealogy of the problems that have organized modern economic inquiry. Its role is to show how the center of economic analysis has shifted as productive arrangements, institutions, and techniques of representation changed. The objective is analytical rather than encyclopedic: the discussion follows five recurring problem-fields—wealth and exchange, production and division of labor, labor and capital, allocation and welfare, and information and institutions—and examines how each made particular economic relations visible while placing others in the background. The method is genealogical and comparative. Canonical works are used as indicators of changing problem-spaces, without treating the history of economics as a linear succession in which one problematic simply replaces another. Earlier questions persist, are reformulated, and often return inside later frameworks with altered meanings.

A genealogy of problematics differs from a chronology of doctrines. Chronology asks which theory came first and which school followed. Genealogy asks what had to become practically consequential and analytically representable for a particular question to acquire disciplinary centrality. The relevant movement is therefore one of changing emphasis. Wealth remains relevant after production becomes central; production remains relevant after allocation is formalized; allocation remains relevant when information and institutions enter the analysis. What changes is the configuration of problems through which economic reality is rendered intelligible. This section prepares the later historical argument by showing that modern economics was repeatedly reorganized around newly salient relations rather than by a single permanent definition of its subject matter.

1.2.1 Wealth and Exchange

The first problematic considered here concerns the production, circulation, and composition of wealth. In early modern political economy, wealth was inseparable from questions of trade, money, revenue, taxation, and the conditions under which national production could expand. Adam Smith's *An Inquiry into the Nature and Causes of the Wealth of Nations* makes the scale of this problematic explicit in its title and organization. The opening books move from the productive powers of labor through division of labor, money, prices, wages, profits, and rent before extending the analysis to accumulation, trade, public revenue, and state expenditure (Smith 1776). The object called wealth was therefore already relationally distributed across production, exchange, class incomes, public institutions, and international commerce.

Exchange acquired analytical importance because it offered a visible interface among otherwise heterogeneous activities. Money prices, market transactions, and trade flows provided forms in which dispersed production could be compared and coordinated. Smith's analysis of money and market exchange did not reduce social

wealth to circulation alone, yet exchange supplied a tractable surface on which value, price, and distribution could be investigated. The practical importance of expanding markets reinforced this analytical role. Exchange connected local productive activity to wider systems of demand, specialization, taxation, and accumulation, making the relation between individual transactions and national prosperity a durable economic question.

Ricardo's *On the Principles of Political Economy and Taxation* sharpened this problematic by organizing analysis around value, rent, wages, profits, foreign trade, taxation, machinery, and the distribution of the social product (Ricardo 1817). The book's structure reveals a significant shift in emphasis. Wealth and exchange remain central, yet their analysis increasingly depends on the distributional relations among classes and on the conditions of production that determine rents, wages, and profits. A transaction can no longer be understood solely as an exchange between two owners; it participates in a wider distributional structure whose effects become visible only when production and class positions are included.

For GRE, the significance of this early problematic lies in the relation between visibility and abstraction. Exchange produces highly legible manifestations: prices, quantities, payments, debts, and transfers. These manifestations can be recorded, compared, and aggregated. They therefore provide a powerful analytical interface with economic reality. The same visibility can privilege relations that culminate in exchange over relations whose effects are indirect, delayed, or institutionally unpriced. Care, trust, knowledge formation, environmental reproduction, and the maintenance of social capacities may condition exchange while appearing only weakly within the transaction itself. The genealogy of wealth and exchange thus supplies an early example of a recurring theme in this book: analytically successful manifestations can organize a field of inquiry while leaving part of the generative structure outside the dominant representation.

1.2.2 Production and Division of Labor

The production problematic shifts attention from the circulation of finished goods toward the organization of productive activity. Smith's treatment of the division of labor is a canonical expression of this move. He attributes major improvements in the productive powers of labor to specialization and links the extent of specialization to the extent of the market (Smith 1776). Productivity, on this account, is already relational in a straightforward sense: the output of an individual worker depends on an organized division of tasks, the circulation of intermediate outputs, accumulated skills, tools, and a market large enough to sustain specialization.

The analytical importance of division of labor extends beyond the familiar claim that specialization raises productivity. It changes the unit through which productive capacity is understood. A worker's output cannot be read only from individual effort or skill because productivity is partly constituted by the position occupied within a differentiated productive arrangement. The pin factory is analytically important precisely because the productive effect appears at the level of coordination among tasks. Individual capacities matter, yet their economic effect depends on sequencing, complementarity, repetition, tools, supervision, and the relation between the workshop

and its market.

This transformation also introduces dependence as an economic fact. Greater specialization increases productive power while making participants more reliant on activities performed elsewhere in the system. The same relation that raises output can narrow self-sufficiency. A productive order therefore generates capacities and dependencies together. Classical political economy did not need the later language of networks or relational fields to identify this structure; division of labor made interdependence visible through the organization of production itself.

The problem of coordination follows immediately. Once production is divided, economic analysis must explain how differentiated activities become mutually compatible. Market exchange supplies one answer, managerial organization another, custom and institution others. The division of labor therefore sits at an intersection between production and coordination. It makes clear that an economy cannot be represented only as a collection of independent producers because the productivity of each participant depends on relations that organize the activity of others.

GRE later develops this point in stronger form by asking which relations generate capacities that cannot be assigned to any isolated participant. The present historical section makes a more limited claim. The rise of the production problematic already shifted economic inquiry toward arrangements in which outcomes are properties of organized relations. This prepared the ground for later theories of the firm, industrial organization, networks, institutions, and collective productive capacity.

1.2.3 Labor, Capital, and Industrial Production

Industrialization deepened the production problematic by stabilizing labor and capital as central analytical categories. Ricardo's analysis of wages, profits, and rent placed distribution among classes close to the center of political economy (Ricardo 1817). Marx's *Capital* subsequently reorganized the field around the production process of capital, including the purchase of labor-power, the labor process, surplus value, cooperation, manufacture, machinery, and modern industry (Marx 1867). The economic object under investigation became increasingly processual: capital was analyzed through accumulation and reproduction, while labor appeared both as human activity and as a socially organized relation within production.

The shift matters because industrial production makes the economic effect of institutional position difficult to ignore. The worker and the owner enter production with different claims over productive assets, different forms of dependence, and different capacities to appropriate the resulting output. These positions are sustained by property, contract, labor markets, organizational authority, and the material concentration of productive means. Industrial economic analysis therefore cannot remain at the level of exchange among formally equivalent owners. The exchange of a wage for labor-power is embedded within a productive structure that determines what each party can contribute, control, and claim.

Marx's analysis is especially important for the genealogy pursued here because it treats production relations as constitutive of the economic process rather than as external conditions surrounding exchange. Cooperation and machinery do more than raise a given worker's productivity. They reorganize the labor process, concentrate

control, alter skill requirements, and transform the relation between workers and the means of production (Marx 1867). Industrial production consequently generates new economic capacities together with new asymmetries. The productive effect of machinery belongs to a configuration involving capital investment, technical knowledge, labor organization, ownership, and markets; its distributional effects depend on the relations through which that configuration is governed.

This industrial problematic also strengthens the importance of reproduction through time. Capital must be maintained and accumulated; labor capacity must be reproduced; firms and institutions must persist; technical systems require replacement, learning, and coordination. Economic analysis therefore acquires an increasingly temporal object. Current output depends on inherited productive structures, and current investment changes the conditions under which later production occurs. The distinction between a present allocation and a future productive capacity begins to matter even before generativity is made explicit as a theoretical category.

GRE does not inherit any single classical or Marxian theory of value, class, or historical development as a complete account. The genealogical lesson is narrower and more durable. Industrial political economy made visible that economic outcomes arise within structured relations among labor, capital, technology, ownership, and organization. Those relations generate capacities and asymmetries that persist beyond the individual transaction. Later sections of this chapter examine how modern institutions stabilized these relations into recognizable economic subjects such as worker, owner, household, firm, and corporation.

1.2.4 Allocation, Distribution, and Welfare

The allocation problematic reorganizes economic inquiry around choice among alternative uses of scarce means, the distribution of resulting goods and incomes, and the evaluation of economic arrangements. Marshall's *Principles of Economics* retains the older language of production, distribution, exchange, and consumption while developing a more unified analysis around demand, supply, value, and marginal adjustment (Marshall 1890). His treatment illustrates a transition in which economic reasoning increasingly abstracts from the full historical structure of production in order to analyze choice and adjustment within a specified field of alternatives.

Robbins later gave the scarcity formulation a particularly influential methodological expression by defining the economic aspect of conduct in terms of ends and scarce means with alternative uses (Robbins 1932). The significance of this formulation for the present genealogy lies in its portability. Once the economic problem is specified through scarcity and alternative use, the same analytical grammar can travel across households, firms, public budgets, time allocation, and other domains. The gain is considerable: heterogeneous substantive problems can be represented through a common structure of constrained choice. The corresponding methodological question concerns what must already be specified before the allocation problem can be solved—the relevant ends, available means, feasible alternatives, boundaries of the decision, and the units whose welfare or claims enter the analysis.

Welfare economics made these evaluative conditions more explicit. Pigou's *The Economics of Welfare* links economic welfare to the national dividend and examines

situations in which private and social net products diverge, thereby making the relation between decentralized economic activity and social evaluation a central problem (Pigou 1920). Allocation consequently becomes more than the placement of scarce resources. It becomes a question about the criteria under which one arrangement is judged preferable to another and about the institutional conditions under which private decisions produce socially consequential effects.

This movement supplied economics with powerful tools for policy analysis because it made trade-offs expressible within comparatively standardized frameworks. Budgets could be allocated, marginal changes compared, and welfare effects discussed with greater analytical discipline. The abstraction also encouraged a recipient-centered representation of policy: resources are assigned to persons, households, firms, sectors, or jurisdictions, and evaluation asks how outcomes change for those units. Under many administrative conditions this is both necessary and effective. Legal responsibility, taxation, eligibility, accounting, and payment systems all require identifiable recipients.

The GRE problem begins where the consequences of allocation exceed the recipient representation. A transfer to a person can change household dependence, trust, mobility, bargaining power, or participation. Funding to an organization can alter internal authority, external alliances, knowledge flows, and future access to institutions. Infrastructure expenditure can generate option spaces for actors who were not direct recipients. These effects do not invalidate allocation analysis; they reveal an additional layer in which allocation acts through relational structure. The historical importance of the allocation problematic is therefore twofold. It created a highly general language for disciplined choice under scarcity, and it made increasingly visible the need to specify which consequences and relations enter the evaluative frame.

1.2.5 Information, Institutions, and Coordination

Twentieth-century economic inquiry increasingly treated information, organization, and institutions as central to coordination. Hayek's analysis of dispersed knowledge reformulated the coordination problem by emphasizing that the information required for economic adjustment is distributed across participants and is never available in fully concentrated form to a single decision-maker (Hayek 1945). The economic problem thereby acquires an epistemic dimension: coordination depends on how locally held and frequently changing knowledge enters practical adjustment.

Coase's theory of the firm introduced a complementary organizational question. If market exchange coordinates activity through prices, the existence of firms requires an account of why some transactions are organized through authority and internal direction. Coase's answer directs attention to the costs of using the price mechanism and to the boundary between market and organization (Coase 1937). The firm becomes more than a production function or legal owner of assets; it becomes an institutional arrangement that coordinates relations differently from a sequence of market contracts.

Institutional economics extends this shift by treating rules, norms, enforcement, and organizational arrangements as constitutive of economic performance. North characterizes institutions as humanly devised constraints that structure interaction

and argues that institutional change shapes economic performance through time (North 1990). This moves the coordination problematic beyond the instantaneous matching of decentralized choices. Economic action occurs within rule systems that shape incentives, transaction costs, expectations, learning, and the feasible forms of organization. History enters because institutional arrangements persist, change incrementally, and condition later choices.

The convergence of information and institutional problems expands the range of relations that economics must represent. Coordination depends on prices, yet also on authority, rules, conventions, trust, expertise, legal recognition, and infrastructures through which information can move. Different arrangements solve different coordination problems and distribute knowledge, discretion, and power differently. A price system can aggregate some dispersed information while leaving organizational knowledge local; a firm can coordinate complex tasks while concentrating authority; an institution can reduce uncertainty while stabilizing rules that later become costly to change.

This expanded problematic provides an important bridge toward GRE. The genealogy traced across this section shows successive enlargements of the economic field: from wealth and exchange, to organized production, to industrial relations among labor and capital, to allocation and welfare, and then to information, institutions, and coordination. The sequence is cumulative rather than substitutive. Contemporary economic life contains all of these problematics simultaneously, and each preserves analytical strengths developed under different historical conditions.

The implication for GRE is methodological and historical. A generative relational approach should enter this accumulated field as another language of inquiry rather than as a declaration that earlier economics studied the wrong objects. Its distinctive question concerns what becomes visible when relations are examined for the capacities, constraints, subjects, values, and future possibilities they generate. That question arises from the tensions internal to the inherited problematics themselves: exchange depends on production; production depends on coordination; allocation changes relations; institutions shape feasible action; and information is distributed through structures that themselves evolve. The remaining sections of Chapter 1 reconstruct the historical order in which these relations became increasingly consequential and increasingly difficult to treat as fixed background conditions.

1.3 Industrial Modernity and the Modern Economic Order

This section reconstructs the institutional architecture that made the modern economic order simultaneously productive, administratively legible, and analytically tractable. Its role in the chapter is to connect the genealogy of economic problematics developed in Section 1.2 with the later construction of recognizable economic subjects and entities. The objective is to identify the historical configuration in which industrial production, wage labor, property, firms, markets, and states became mutually reinforcing components of economic organization. The discussion proceeds through five dimensions: material production and scarcity, wage labor and social reproduction, property and productive assets, firms and administrative organization,

and the coordination of market and state. The method is historical and institutional. The section does not treat industrial modernity as a single globally uniform stage; it uses the term for a family of arrangements whose concrete forms varied across jurisdictions and periods while sharing several recurrent organizational features.

The central analytical point is that the modern economic order stabilized relations that could increasingly be represented through identifiable entities, contracts, accounts, incomes, assets, and administrative categories. Industrial production concentrated productive means and coordinated differentiated labor. Wage relations connected participation in production to monetary income. Property regimes stabilized claims over productive assets. Firms organized activities internally through authority and administration. States supplied legal, fiscal, infrastructural, and statistical conditions within which markets could operate at expanding scales. These arrangements did more than support production. They helped determine which economic subjects became visible, which claims could be administered, and which flows could be represented as movement among recognized units. The later entity-centered architecture of allocation emerged from this institutional history rather than from a purely methodological preference.

1.3.1 Industrial Production and Material Scarcity

Industrial production altered the relation between productive capacity and material organization by concentrating machinery, energy, fixed capital, labor, and technical knowledge within increasingly coordinated processes. Marx's analysis of machinery and modern industry treats mechanized production as a reorganization of the labor process rather than as a simple addition of more productive tools (Marx 1867). The relevant historical development for GRE is the growing dependence of output on combinations of assets and relations whose productive effect could not be assigned to isolated participants. Machinery required capital investment, maintenance, energy, technical knowledge, supplies, coordinated labor, and access to markets. Industrial capacity therefore existed in organized configurations before it appeared in the output attributed to a firm, sector, or worker.

Material scarcity remained a central organizing condition within this order. Productive assets, energy, land, raw materials, transport capacity, working time, and financial means could not be expanded without cost or delay. Industrial organization responded by concentrating resources, improving throughput, extending specialization, and creating institutions capable of coordinating larger stocks and flows. The economic importance of scarcity consequently attached not only to final consumption goods but also to access to the productive conditions through which future goods could be generated. A machine, factory site, transport connection, patent, or stock of working capital mattered because it conditioned later production. The industrial order therefore strengthened the distinction between current possession and productive capacity.

Scarcity also became increasingly mediated by infrastructure. Railways, ports, warehouses, energy systems, urban utilities, and communication networks changed the effective availability of resources by altering distance, timing, reliability, and scale. A material resource could be physically present while remaining economically

inaccessible because transport, finance, legal permission, or organizational coordination was absent. Conversely, infrastructure could transform distant resources into practically available inputs. The economic meaning of scarcity was therefore relational even where the scarce object was material: availability depended on the configuration connecting the resource to those capable of using it.

This configuration helps explain why industrial economics could profitably abstract toward measurable inputs and outputs. Factories, inventories, wages, investment, prices, and physical production created comparatively legible interfaces with economic activity. Accounting and administrative practices could attach costs and claims to recognizable organizational units. The resulting abstractions were historically powerful because they corresponded to material arrangements in which production was increasingly concentrated, standardized, and measured. Their usefulness did not require every generative relation to be visible. It required enough stability that productive resources, organizations, and recipients could be treated as identifiable over the horizon relevant to management and policy.

The GRE significance of industrial materiality lies in this combination of concentration and relational dependence. Industrial production made entities such as factories and firms appear increasingly substantial while simultaneously deepening the relations on which their productive capacities depended. The entity and the relational configuration developed together. Later economic analysis inherited the entity because it was legally and administratively useful; GRE returns to the surrounding configuration because the capacity attributed to the entity was generated through it.

1.3.2 Wage Labor and Social Reproduction

Wage labor supplied one of the principal institutional links between productive participation and access to livelihood in industrial economies. Marx's treatment of labor-power makes this relation explicit by distinguishing the worker's capacity to labor from the labor performed within production and by connecting the reproduction of labor-power to the means of subsistence required for its continued availability (Marx 1867). The analytical importance of the wage relation extends beyond the wage as a price. It connects production, income, household provisioning, time discipline, mobility, bargaining position, and future participation in the labor process.

Polanyi's account of market society provides a complementary historical perspective by emphasizing the institutional transformation involved when labor, land, and money are organized through market mechanisms (Polanyi 1944). Labor is especially revealing because the object transacted in a labor market remains inseparable from the person whose activity is organized. The labor relation therefore couples a market contract to domains that exceed the transaction itself: health, family life, residence, education, social membership, and the reproduction of the capacity to work. The wage can be recorded as a payment to an individual worker, while the conditions that make labor possible are distributed across households, communities, public institutions, and infrastructures.

This distributed structure gives social reproduction a direct economic relevance. Industrial production requires renewed labor capacity across days and generations. Food, housing, care, education, health, socialization, and periods outside paid em-

ployment contribute to that reproduction through arrangements whose institutional forms vary historically. Some are purchased in markets, some are provided within households, some are organized publicly, and some are sustained through community relations. The wage relation becomes especially consequential when access to these conditions depends heavily on employment income. Employment then mediates more than contribution to production; it mediates access to a wider field of material and institutional resources.

The resulting architecture stabilizes the worker as an economic subject. Payroll systems, labor contracts, taxation, social insurance, workplace rules, and household budgets can all identify the worker as a recipient of income and bearer of obligations. This administrative clarity is historically significant. It permits production costs to be assigned, income to be taxed, entitlements to be calculated, and labor supply to be represented in aggregate. The same clarity places part of the reproductive structure outside the immediate representation. A wage record identifies the payment while saying relatively little about the household relations, care arrangements, commuting infrastructure, education, health, or social supports through which the worker remains capable of participation.

For GRE, the wage relation is therefore an early example of a relation whose economic effects propagate beyond the nominal parties to a contract. Changes in wages, employment security, working time, or workplace authority can alter household dependence, geographic mobility, care burdens, learning opportunities, and future bargaining power. The historical prominence of wage labor helped make the individual worker a practical unit of economic administration, yet the capacity represented by that worker was reproduced through a wider relational system. The distinction between recipient and generative conditions later becomes central to GRE's treatment of allocation and value.

1.3.3 Property and Productive Assets

Property regimes stabilized claims over productive assets and thereby shaped who could decide how resources were used, who could exclude others, who could transfer claims, and who could receive returns generated through production. North's institutional analysis treats property rights and other formal rules as part of the institutional structure that shapes incentives and economic performance through time (North 1990). For the present argument, the important point is that a productive asset becomes economically consequential through a relation among material control, legal recognition, organizational use, and expectations concerning future returns.

Industrial production increased the importance of this stabilization because productive means became increasingly durable, specialized, and costly. Ownership of land, machinery, buildings, inventories, and financial claims could organize access to productive opportunities over horizons much longer than a single transaction. Property therefore linked present legal claims to future streams of decision-making and income. A productive asset was valuable partly because it could be inserted into an institutional environment in which exclusion, transfer, contracting, financing, and enforcement were sufficiently reliable for long-horizon organization.

The relation between property and productive capacity also clarifies why owner-

ship and control deserve analytical separation. A legal owner may delegate operational control to managers; creditors may constrain decisions without owning the asset; regulators may determine permissible uses; workers may possess indispensable knowledge; suppliers and platforms may control access to complementary infrastructure. Industrial organization consequently produces layered claims around the same productive object. Ownership remains economically important, yet its practical effect is mediated through a configuration of rights, contracts, dependencies, and institutional authorities.

Property also contributes to the stabilization of economic boundaries. Assets can be assigned to persons, firms, households, public bodies, or other recognized entities. Accounting systems can record acquisition, depreciation, liabilities, and returns relative to those boundaries. Contracts can specify transfers across them. This improves administrative legibility and supports the representation of economic activity as flows among units with identifiable holdings. The boundary is institutionally real because claims and obligations are enforced through it, even when the productive process itself crosses that boundary through labor relations, supply chains, shared infrastructure, finance, and knowledge flows.

GRE later develops a relational account of ownership, control, access, and allocation in greater detail. At this historical stage, the relevant result is that property helped make productive resources administratively attachable to entities while organizing the relations through which those resources could generate future value. The institutional success of entity-centered representation therefore rested on relational arrangements that stabilized entities sufficiently for ownership, contracting, and accounting to operate.

1.3.4 Firms and Administrative Organizations

The industrial firm gave durable organizational form to coordination that could not be represented adequately as a sequence of isolated exchanges. Coase's analysis begins from precisely this problem: market exchange coordinates through the price mechanism, while the firm coordinates some activities through internal direction because using markets also entails costs (Coase 1937). The firm therefore marks an institutional boundary across which the dominant mode of coordination changes. Prices remain important outside and within the organization, yet authority, routines, roles, schedules, budgets, technical standards, and managerial judgment coordinate many productive relations internally.

This organizational form expanded the range of activities that could be planned across time. Capital investment, hiring, training, procurement, production sequencing, inventory, quality control, sales, and finance could be coordinated through administrative structures that persisted beyond individual contracts. The firm became an economic subject because law, accounting, taxation, and organizational practice treated it as a continuing unit capable of owning assets, entering contracts, receiving revenue, bearing liabilities, and employing workers. Its continuity depended on relations among participants, yet the organization could be represented as an entity for many practical purposes.

Administrative organization also transformed information. A large productive pro-

cess requires knowledge to be distributed among specialized roles while remaining coordinated sufficiently for collective action. Records, reports, budgets, plans, technical specifications, and chains of authority create channels through which local knowledge is translated into organizational decisions. Some knowledge remains tacit or local, and some is formalized into procedures and indicators. The organization therefore does not merely collect information; it structures which information can travel, who is authorized to interpret it, and which representations acquire decision-making force.

This point links the firm to the broader genealogy of coordination developed in Section 1.2. Hayek's account of dispersed knowledge emphasizes the difficulty of concentrating the information relevant to economic adjustment (Hayek 1945). The firm does not eliminate that difficulty. It creates another coordination regime in which some decisions are internalized, some knowledge is formalized, and some authority is concentrated. Market and organization consequently represent different relational arrangements for coordinating heterogeneous knowledge and activity rather than mutually exclusive forms of economic order.

The GRE relevance of the firm lies in the contrast between entity manifestation and relational constitution. The firm is sufficiently stable to sign a contract, own an asset, receive funding, report profit, and appear as a node in administrative data. At the same time, its capacities depend on internal authority, cooperation, knowledge flows, routines, worker relations, supplier relations, finance, law, and infrastructure. Treating the firm as an entity is therefore often operationally indispensable. Treating the entity as analytically self-sufficient obscures the relations that generate its capacities and determine how an intervention directed at the firm propagates through the surrounding system.

1.3.5 State, Market, and Economic Coordination

Industrial modernity joined market coordination to a dense institutional architecture of law, public authority, infrastructure, taxation, monetary organization, administration, and social provision. Polanyi's historical account is especially useful here because it treats market society as institutionally produced and politically sustained rather than as the spontaneous expansion of exchange alone (Polanyi 1944). North similarly places economic performance within rule systems that structure incentives and reduce or redistribute uncertainty (North 1990). The modern market therefore operates through institutional conditions that define property, contract, legal personality, money, enforcement, and permissible forms of organization.

The state participates in this order through several analytically distinct roles. It recognizes and enforces legal claims; raises and spends revenue; builds and regulates infrastructure; produces statistics and administrative classifications; supplies or finances services; defines corporate, labor, and property forms; and intervenes in crises. These activities vary widely across historical periods and political systems. Their common relevance lies in the production of conditions under which dispersed economic actors can coordinate over scales that exceed local relations. State capacity can enlarge markets by standardizing rules, currencies, transport, information, and enforcement even when many transactions remain privately organized.

Market coordination addresses a different dimension of the same problem. Hayek's account of the price system emphasizes its capacity to communicate information relevant to adjustment when knowledge is dispersed among many participants (Hayek 1945). Prices can coordinate decentralized decisions without requiring a central actor to possess all local knowledge. Their informational economy is powerful precisely because the system does not attempt to reproduce every relation or circumstance in a complete representation. A price condenses some conditions into a signal useful for action while leaving many causal and normative dimensions outside the signal.

The institutional and market dimensions of coordination therefore operate through different forms of selective representation. Law translates heterogeneous social relations into rights, duties, persons, and claims. Administration translates them into categories, records, eligibility conditions, and budgets. Markets translate some dispersed conditions into prices and contracts. Firms translate productive complexity into organizational roles, accounts, and plans. Each representation supports coordination by reducing complexity, and each leaves a remainder that can become economically consequential when conditions change.

This selective structure is the historical bridge to the next sections of the chapter. Industrial modernity did not merely produce more goods or larger organizations. It stabilized a recognizable architecture of workers, owners, households, firms, assets, contracts, markets, and states. These units became economically real through persistent relations and institutionally powerful manifestations. They could be taxed, regulated, funded, counted, contracted with, and made responsible. The resulting order made entity-centered economic analysis unusually effective because economic administration itself increasingly operated through entities.

The same historical achievement establishes the limit that GRE later addresses. Coordination through entity boundaries and compressed representations works best when the relevant entities remain sufficiently stable and when the consequences of action can be assigned within the chosen boundary and horizon. As production becomes more networked, knowledge-intensive, transnational, platform-mediated, and dependent on distributed infrastructures, those conditions become less secure. The next sections therefore examine how modern economic subject positions and institutional entities were constructed before turning to the contemporary transformations that place increasing pressure on the architecture industrial modernity stabilized.

1.4 Historical Formation of the Economic Subject

This section reconstructs the historical formation of the principal subject positions through which modern economic life became administratively and analytically intelligible. Its role in the chapter is to move from the institutional architecture of industrial modernity to the actors that this architecture made recognizable as bearers of labor, property, consumption, responsibility, and organizational capacity. The objective is to show that the worker, owner, consumer, household, firm, and corporation acquired economic salience through historically specific relations and institutions rather than entering economic analysis as timeless units. The discussion proceeds

through five subject positions and uses a historical-institutional method: each position is examined through the relations that stabilize it, the capacities and claims attached to it, and the forms of representation through which economic analysis and administration can act upon it.

The term *economic subject* is used here in a deliberately limited sense. It designates a configuration that can occupy an economically consequential position, enter recognized relations, receive or control resources, bear claims or obligations, and become a unit of observation or intervention. The term therefore concerns economic standing rather than a complete theory of personhood or agency. A biological person may occupy several economic subject positions at once; an organization may acquire a durable economic position despite lacking the experiential properties associated with a human subject. This distinction permits the historical analysis to remain compatible with the ontological restraint developed later in the book.

Industrial modernity strengthened the stability of several such positions simultaneously. Wage labor made the worker legible through employment and remuneration. Property institutions stabilized the owner as a bearer of enforceable claims. Expanding markets and demand analysis strengthened the consumer as a distinct economic role. Household organization linked income, care, consumption, and reproduction. Firms and corporations condensed collective productive relations into continuing organizational subjects. These positions overlap in practice, yet their differentiation made modern economic administration unusually capable of assigning income, liability, entitlement, taxation, and policy effects to recognizable units.

1.4.1 Worker as Economic Subject

The worker became a particularly consequential economic subject through the institutional expansion of wage labor. Industrial production required large numbers of people to enter continuing relations in which labor capacity could be contracted, scheduled, supervised, remunerated, and coordinated with machinery and organizational routines. Marx's analysis of labor-power and the industrial labor process treats this relation as constitutive of capitalist production: productive activity is organized through a relation between those who control productive assets and those who sell labor-power within the production process (Marx 1867). Polanyi later emphasized the historical specificity of treating labor as if it were a commodity available to market coordination, while insisting that labor remains inseparable from the lives of the persons who supply it (Polanyi 1944).

The economic importance of the worker position lies partly in the forms of legibility that wage labor makes possible. Employment attaches income to a recognizable relation between worker and employer. Hours, wages, occupations, contracts, payrolls, and employment status create administrative traces through which labor can be taxed, regulated, insured, compared, and statistically represented. The worker can therefore appear as a unit of production and as a recipient of income at the same time. This dual standing helped make employment one of the principal institutional bridges between productive contribution and access to material resources in modern economies.

The worker position also illustrates the difference between an economic subject

and the whole person occupying that position. A person who appears in an employment relation as a worker may simultaneously be a household member, caregiver, consumer, owner of some assets, citizen, borrower, learner, or participant in a professional community. Economic administration frequently selects one of these relations because a particular institution requires a determinate role. Payroll systems need an employee; labor law needs parties to an employment relation; national accounts need categories of compensation and production. The resulting representation is operationally powerful precisely because it suppresses dimensions that are irrelevant to the immediate administrative task.

The relational constitution of the worker remains visible even within this compressed representation. Productive capacity depends on education, health, transport, care, technical infrastructure, organizational knowledge, legal protection, bargaining arrangements, and access to employment. The wage received by one worker may support several household members; a change in scheduling may alter care relations; a change in technology may reallocate authority and skill inside the workplace. The worker is therefore economically real as a stabilized subject position, while the capacities attached to that position remain generated and sustained through a wider relational field.

This historical construction becomes especially important later in the chapter when artificial intelligence and automation place pressure on employment as a mechanism of economic recognition and allocation. The relevant historical point at this stage is narrower. Modern institutions made the worker sufficiently stable and legible to serve simultaneously as a productive subject, an income recipient, a bearer of labor rights, and an object of economic measurement. That stabilization was an institutional achievement with a history, and its future transformation is therefore also historically possible.

1.4.2 Owner as Economic Subject

The owner position formed around enforceable claims to assets, resources, and the income streams associated with them. Property institutions identify a subject that may exclude others from specified uses, transfer claims, pledge assets, authorize access, and receive returns. The economic owner is therefore constituted through a relation among the claimant, the asset, other potential users, and the institutions that recognize and enforce the claim. North's institutional analysis is relevant because property and contracting arrangements operate within rule systems that structure incentives, expectations, and the security of exchange over time (North 1990).

Industrial development increased the importance of ownership by expanding the scale and durability of productive assets. Land, machinery, factories, inventories, financial claims, intellectual resources, and later corporate securities could be assigned to recognized owners and represented through accounts, registries, contracts, and balance sheets. Ownership thereby became a major interface between present control and future income. The owner could appear in economic analysis as a recipient of profit, rent, interest, or capital gain even when the productive process generating those returns depended on many relations extending beyond the owner's direct activity.

The position is internally differentiated. Ownership may coexist with delegated management, creditor claims, regulatory constraints, contractual commitments, shared ownership, or limited rights of use. An owner may hold legal title while another actor exercises operational control; several owners may share claims to the same organization; a lender may constrain asset use without becoming owner. These arrangements show that ownership is a structured bundle of recognized relations rather than a simple physical connection between a person and a thing. The stability of the subject position comes from institutional recognition of those relations and from the capacity to reproduce them through transfer, enforcement, and accounting.

Ownership also shapes the distribution of economic possibility. A productive asset can support future borrowing, investment, bargaining, mobility, inheritance, and access to organizational opportunities. The economic consequence of ownership therefore exceeds the current income attached to the asset. It can alter the owner's future action space while simultaneously affecting the access of others. This temporal dimension becomes central later in GRE's analysis of generative value, capability, and option-space transformation.

The owner position consequently provides an early example of a subject whose economic significance is relational and generative even when conventional representation is entity-centered. Administrative systems may record that an asset belongs to a named owner, and this representation is often indispensable. The effects of that ownership emerge through relations of exclusion, access, finance, production, inheritance, and institutional recognition. Section 1.5 develops the legal and administrative construction that makes these ownership relations increasingly durable and governable.

1.4.3 Consumer as Economic Subject

The consumer became a distinct economic subject as demand, choice, and the allocation of expenditure acquired greater analytical and practical importance. Marshall's synthesis of demand, supply, wants, value, and marginal adjustment gives consumption a systematic place within a unified account of economic coordination (Marshall 1890). The consumer in this framework appears as a chooser whose demands help organize production through market exchange. Later scarcity-centered formulations extended the same analytical grammar by representing economic conduct through the allocation of scarce means among alternative ends (Robbins 1932).

The rise of the consumer position depended on more than an intellectual shift in economic theory. Expanding markets, standardized goods, monetary income, retail infrastructures, advertising, credit, transport, and mass production increased the practical importance of household expenditure and individualized purchasing decisions. Prices and product categories made heterogeneous goods comparable within budgets, while market institutions provided repeated occasions on which persons could be addressed primarily through their role as buyers. The consumer therefore became both an analytical unit and an institutional target for pricing, marketing, credit, taxation, regulation, and later consumer protection.

This subject position has a characteristic abstraction. The consumer is represented through preferences, purchasing power, observed choices, or welfare effects, while

the relations that formed those preferences and capacities may remain outside the immediate model. Income depends on labor, ownership, transfers, or household relations; access depends on infrastructure and geography; preferences may be shaped by culture, information, social position, and commercial representation; credit can move purchasing power across time. A demand curve or expenditure record captures an economically meaningful manifestation while leaving many of its generative conditions unresolved.

The distinction becomes more important when consumption itself generates relations. Repeated purchases can sustain platforms and distribution networks; credit creates durable obligations; subscription models establish continuing dependence; digital consumption produces data that can be converted into commercial knowledge; consumption inside households redistributes access among members. These effects do not eliminate the analytical usefulness of the consumer. They reveal that the consumer is one stabilized position within a larger relational process.

The historical formation of the consumer therefore complements the formation of the worker and owner. Industrial modernity increasingly organized persons through several differentiated roles: contribution could be represented through work, command over assets through ownership, and access to goods through consumption. Economic policy could then act upon each role through different instruments. The later GRE problem is to understand how these roles interact, how they generate one another's capacities, and how contemporary digital systems increasingly blur their boundaries.

1.4.4 Household as Economic Subject

The household occupies a distinctive position because it joins economic subjecthood to a relational configuration that is visibly composite. Modern economic analysis and administration frequently treat the household as a unit of income, expenditure, taxation, welfare, housing, saving, and consumption. At the same time, the household contains relations of care, dependence, authority, contribution, inheritance, and intergenerational support that cannot be inferred from aggregate household resources alone. The household therefore makes especially clear how an entity-level representation can be administratively necessary while remaining analytically incomplete.

Industrial wage labor strengthened the household's role as a bridge between production and social reproduction. Wages and other incomes enter the household through members occupying positions in labor, ownership, or public transfer systems. Food, shelter, care, education, health, and time are then organized through internal relations that reproduce capacities required for future participation in economic and social life. Polanyi's account of market society is useful for locating this process historically because the expansion of labor markets placed livelihoods under market dependence while social institutions continued to mediate the reproduction of persons and communities (Polanyi 1944).

The household also became a practical allocation unit. Taxes, benefits, housing policy, social insurance, surveys, and consumption statistics frequently require a boundary within which income and need can be aggregated. Such aggregation enables public administration to compare households and distribute resources at scale.

The boundary acquires causal force because eligibility rules, tax liabilities, benefits, and legal responsibilities may depend on who counts as a member. Economic subjecthood is therefore partly generated by the institutional act of recognizing a household and assigning consequences to its boundary.

Aggregation can also conceal internal asymmetry. Two households with the same measured income may distribute time, care, consumption, control, and risk very differently. A transfer formally received by one member may alter bargaining relations among several members. Unpaid care can sustain labor-market participation without appearing as wage income. Intergenerational transfers can expand the future options of some members while limiting those of others. These possibilities show why the household is an important bridge from entity-centered economics toward relational analysis: the entity is empirically real and administratively consequential, yet the distribution of capacities within it depends on internal relations.

GRE therefore has reason to retain the household as an economic subject while refusing to treat its boundary as self-explanatory. Later chapters can ask how resources entering a household propagate through internal relations, how care and dependence generate capabilities, and how household membership affects access to institutions. The historical point developed here is that modern economic order increasingly stabilized the household as a recognizable recipient and decision unit, creating a powerful administrative representation whose relational interior remains economically consequential.

1.4.5 Firm and Corporation as Collective Economic Subjects

The firm and corporation extend economic subjecthood beyond the individual and household by stabilizing a collective organization as a continuing participant in economic relations. Coase's analysis of the firm identifies an organizational domain in which coordination occurs through internal direction rather than through a sequence of market exchanges, thereby making the boundary of the firm an economic problem in its own right (Coase 1937). The corporation adds greater durability through institutional forms that allow assets, contracts, liabilities, roles, and organizational purposes to persist across changes in particular participants.

Collective economic subjecthood depends on the capacity to act through organized relations. A firm can hire workers, purchase inputs, own or control assets, borrow, sell outputs, enter contracts, maintain accounts, and make investment decisions because internal authority and procedures translate distributed activity into organizational action. No single participant needs to perform every function. The organization acquires practical continuity through roles, routines, records, governance structures, and external recognition. Its economic agency is therefore generated through coordination among relations that are distributed across many persons and institutional interfaces.

The collective subject also produces a distinctive form of representation. Revenue, cost, profit, debt, assets, employment, and output can be assigned to the organizational boundary even though the processes generating these quantities extend through workers, suppliers, creditors, customers, regulators, infrastructures, and knowledge networks. Accounting and legal systems make the organization suffi-

ciently coherent to appear as one unit in contracts, taxation, statistics, investment, and policy. The unit is therefore neither fictitious in the ordinary sense nor analytically exhaustive. It is a stabilized manifestation of a more complex relational configuration.

This distinction matters for economic intervention. A subsidy paid to a firm may alter wages, investment, bargaining power, supplier dependence, market position, knowledge accumulation, or shareholder returns. A liability imposed on a corporation may be redistributed internally through prices, employment, contracting, or investment decisions. Entity-level policy therefore operates through collective subjects whose responses are generated by internal and external relations. The organization can remain the correct administrative recipient while relational analysis is required to understand the consequences of intervention.

The historical formation of the firm and corporation closes the section because it reveals the general pattern most clearly. Modern economic subject positions became powerful through institutional stabilization, selective representation, and repeated practical use. Worker, owner, consumer, household, firm, and corporation are heterogeneous kinds of subject, yet all can become economically real through durable participation in recognized relations. Section 1.5 turns from these subject positions to the legal, contractual, accounting, statistical, and administrative mechanisms that construct the entities through which modern economic systems allocate rights, resources, obligations, and responsibility.

1.5 Institutional Construction of Economic Entities

This section examines the institutional processes through which heterogeneous persons, organizations, assets, activities, and relations become recognizable as bounded economic entities. Its role in the chapter is to connect the historically formed economic subjects of Section 1.4 to the administrative architecture that permits modern economies to assign ownership, liability, income, taxation, reporting, eligibility, and responsibility to identifiable units. The objective is to show how legal personality, property, contract, accounting, statistics, and institutional membership jointly stabilize economic boundaries without requiring those boundaries to coincide with every relation that generates the entity's capacities. The discussion proceeds through five institutional mechanisms and uses a historical-institutional method: each mechanism is treated as a practical technology of recognition that makes an economic unit durable enough to enter law, administration, measurement, and allocation.

The term *economic entity* refers here to a recognized unit to which economically consequential attributes can be attached. Such attributes include assets, liabilities, income, expenses, rights, obligations, tax status, employment, eligibility, and reporting responsibility. This use remains compatible with the ontological restraint of GRE. Institutional recognition can make an entity relationally and causally consequential without settling the ultimate ontology of the person, organization, or collective concerned. The relevant historical achievement is the stabilization of boundaries that permit repeated action upon a unit across time.

Modern economic coordination depends heavily on this achievement. Credit re-

quires a debtor whose obligations can be identified. Property requires a bearer of enforceable claims. Contracts require parties. Accounting requires a reporting boundary. Statistics require units that can be classified and aggregated. Public administration requires membership criteria that determine who enters a category, receives a transfer, pays a tax, or bears a duty. These mechanisms operate differently, yet together they transform fluid relational configurations into operationally persistent entities. The resulting compression makes large-scale economic coordination possible while creating a recurrent analytical tension between the recognized unit and the wider relations on which its capacities depend.

1.5.1 Legal Personality

Legal personality supplies one of the strongest mechanisms for stabilizing an economic entity through time. Organizational law provides standardized legal forms through which an enterprise can hold assets, enter agreements, incur liabilities, and persist across changes in the individuals who participate in it. Hansmann and Kraakman's analysis of organizational law emphasizes asset partitioning: organizational law can separate a pool of entity assets from the personal assets and creditors of owners or managers, thereby creating a durable structure of claims around the organization (Hansmann and Kraakman 2000). This separation supports the practical treatment of the organization as a continuing economic unit.

The significance of legal personality for GRE lies in its capacity to stabilize relations into a recognizable locus of responsibility. A corporation, association, trust, or other recognized organization may contain many internal relations and participate in many external ones, yet legal form gives selected relations a persistent boundary. Contracts can be entered in the entity's name. Assets can remain attached to the organization when shareholders, managers, or employees change. Creditors can direct claims toward specified pools of assets. Courts and regulators can address the organization as a continuing bearer of rights and duties. Legal personality therefore creates temporal continuity that exceeds the continuity of any individual participant.

This institutional continuity should be distinguished from the complete economic reality of the organization. The legal entity identifies a powerful boundary, yet productive capacity may depend on workers, suppliers, public infrastructure, household reproduction, financing networks, tacit knowledge, and social trust that extend beyond that boundary. The legal unit nonetheless remains economically consequential because claims, liabilities, and decision authority are routed through it. Entity-centered analysis gains much of its practical force from this institutionally produced concentration of responsibility.

The relation between legal and economic entityhood becomes especially clear in systems of national accounting. The System of National Accounts treats institutional units as fundamental economic units capable of owning assets, incurring liabilities, engaging in transactions, and taking economic decisions on their own behalf (European Commission et al. 2009). Legal responsibility and accounting capacity are therefore closely connected to the statistical constitution of the economy. The institutional unit is useful because economic flows can be attributed to a recognizable bearer across repeated transactions and accounting periods.

Legal personality also demonstrates that economic entities can be generated through institutional recognition. The organization does not become economically significant merely because a legal label is attached to it; the label works because it reorganizes relations of ownership, liability, contracting, and enforcement. The legal form is therefore performative in an economically specific sense: recognition creates durable channels through which resources, obligations, and claims can circulate. Later GRE chapters return to this point when examining how institutional recognition can generate capacities and constraints beyond the immediate legal relation.

1.5.2 Property and Contract

Property and contract deepen entity stabilization by attaching resources and obligations to recognized parties. Property identifies who may hold, transfer, exclude, pledge, or authorize uses of an asset within a given institutional order. Contract identifies parties whose commitments can be interpreted, enforced, and carried forward through time. Together they create a relational grammar in which economically consequential claims can be assigned to named units and transferred among them.

North's account of institutions is relevant because property and contracting operate through rule systems that structure incentives, reduce some forms of uncertainty, and make repeated exchange more predictable (North 1990). Coase's analysis of the firm similarly shows that contracting and market exchange have costs and that organizations emerge partly through alternative arrangements for coordinating transactions (Coase 1937). These traditions differ in focus, yet both demonstrate that exchange depends on institutional structures that define who can commit, what can be transferred, and how agreements are sustained.

Property contributes to entity formation by linking a recognized subject to a durable set of claims over resources. Organizational law can intensify this effect through asset partitioning. Hansmann and Kraakman show that the ability to shield entity assets from the personal creditors of owners or managers supports the construction of large-scale organizations whose assets can remain committed to organizational purposes (Hansmann and Kraakman 2000). The entity thereby acquires a resource base whose continuity is institutionally protected.

Contract contributes a complementary temporal structure. An agreement links present recognition to future performance. Parties can coordinate production, finance, labor, supply, insurance, and exchange across time because obligations have a recognized bearer and an enforceable form. Contractual relations therefore extend the temporal horizon of economic organization. They also connect entity boundaries: one firm's obligation becomes another firm's claim, one household's debt becomes another unit's asset, and one employment contract links a worker position to an organizational position.

The resulting architecture makes economic activity legible as relations among entities with identifiable holdings and commitments. This architecture is analytically powerful because many economically important questions genuinely concern the distribution and transfer of recognized claims. The same architecture also compresses relational complexity. A contract records selected obligations, while leaving many

background relations unstated. Property identifies a recognized claimant, while the asset's productive use may depend on wider infrastructures and collective conditions. GRE therefore treats property and contract as real institutional structures whose analytical boundaries remain narrower than the total field of relations sustaining economic activity.

1.5.3 Accounting and Organizational Boundaries

Accounting converts organizational continuity into a recurring representational boundary. Assets, liabilities, equity, income, and expenses are recorded relative to a reporting entity, allowing economic activity to be summarized across a period and compared across periods. The reporting boundary makes a heterogeneous organizational process appear as a unit with a balance sheet, flows, and a calculable financial position. This representation supports investment, lending, taxation, governance, and organizational decision-making.

The IFRS Conceptual Framework makes the constructed character of this boundary explicit. A reporting entity can be a single legal entity, a portion of an entity, or a group of entities, and the reporting boundary therefore need not coincide exactly with legal personality (IFRS Foundation 2018). Consolidated accounts may represent a parent and its subsidiaries as one reporting entity even though several legal entities remain in existence. Combined accounts can likewise gather economic activities across organizational units when the reporting purpose requires a wider boundary.

This flexibility reveals an important methodological point for GRE. Economic entityhood is partly representation-dependent because different institutional practices may draw different boundaries around the same relational field. Legal personality, managerial control, accounting consolidation, statistical sectoring, and productive coordination can produce overlapping rather than identical entities. A corporate group may be many legal persons, one consolidated reporting entity, several operational divisions, and part of a wider supply network at the same time. Each boundary serves a different analytical and institutional function.

Accounting nonetheless gives these boundaries substantial causal force. Budgets, performance measures, debt covenants, taxation, capital allocation, and managerial evaluation are frequently organized around reported units. Once a reporting boundary is stabilized, internal and external flows are classified differently depending on which side of the boundary they occur. A transfer between subsidiaries may disappear in consolidated reporting while remaining visible in the accounts of each subsidiary. The representational boundary therefore changes which economic relations appear as internal coordination and which appear as transactions between entities.

This point provides a direct bridge from institutional construction to later GRE analysis. The entity represented in accounts is neither arbitrary nor exhaustive. It is selected for an institutional purpose, sustained through conventions and control relations, and capable of shaping decisions once established. Accounting therefore participates in the economic reality it represents: it defines operational units, channels attention, and conditions how resources are compared and allocated. The relational question is consequently not only whether the accounts are accurate, but also which relations become internal, external, visible, or invisible under the chosen boundary.

1.5.4 Statistics and Administrative Legibility

Statistical systems extend entity construction from organizations to entire economies. Large-scale administration requires units that can be counted, classified, compared, and aggregated. Households, corporations, government units, workers, industries, sectors, residents, and non-residents become statistical categories through definitions that specify membership and delimit the observations that belong together. Desrosières' history of statistical reasoning shows that classification and quantification depend on conventions that make heterogeneous cases comparable, while Porter emphasizes the institutional role of standardized quantitative procedures in producing impersonal forms of judgment (Desrosières 1998; Porter 1995).

The System of National Accounts provides a particularly clear example. It organizes macroeconomic accounting around institutional units and sectors, treating units capable of owning assets, incurring liabilities, entering transactions, and making economic decisions as the building blocks of the statistical economy (European Commission et al. 2009). Corporations, government units, nonprofit institutions, and households are then grouped according to functions and institutional characteristics. This statistical architecture permits production, income, saving, investment, assets, and liabilities to be aggregated into coherent macroeconomic accounts.

Administrative legibility emerges through similar operations. Tax systems define taxable persons and entities. Social insurance systems define contributors and beneficiaries. Business registries identify organizations. Labor statistics classify employment status and occupation. Regulatory systems distinguish sectors, ownership forms, and reporting obligations. Each system draws a boundary around the unit relevant to its task and attaches standardized attributes to that unit. The economy becomes governable partly because actors and activities can be entered into interoperable categories.

These classifications have practical consequences. Eligibility, tax liability, regulatory burden, access to programs, statistical visibility, and policy targeting can depend on categorical membership. The category therefore becomes more than a passive description. Hacking's historical work on classification is useful for understanding how classificatory practices can participate in the formation of the objects and kinds they describe (Hacking 2002). In economic administration, classification can alter incentives, organizational forms, reporting behavior, and the distribution of resources.

GRE inherits both the analytical power and the caution implied by this history. Statistical units are indispensable for large-scale comparison and coordination. Their usefulness depends on abstraction: a household or corporation must be treated as sufficiently coherent for aggregation to work. The same abstraction can conceal internal relations, boundary disputes, informal activity, cross-border structures, and dependencies that do not fit the chosen classification. Administrative legibility therefore increases the capacity to act on an economy while narrowing the dimensions that remain visible within the administrative representation.

1.5.5 Economic Recognition and Institutional Membership

Economic recognition completes the institutional construction of entities by determining which units count as members of particular economic categories and which rights, duties, resources, and observations follow from that membership. Recognition occurs through law, registration, accounting, statistics, employment, taxation, licensing, residence, organizational affiliation, and many other institutional practices. The resulting membership conditions establish who is visible to a system and in what capacity.

Institutional membership is rarely singular. A person may simultaneously be recognized as worker, taxpayer, household member, shareholder, resident, borrower, student, beneficiary, and citizen. An organization may be recognized as employer, corporation, taxpayer, regulated firm, statistical unit, contractor, or recipient of public support. Each membership relation attaches a different set of capacities and obligations. Economic subjecthood therefore becomes layered through multiple institutional recognitions rather than exhausted by one universal status.

The SNA illustrates this layering at the macroeconomic level. Institutional units are assigned to sectors and to residence categories so that transactions and balances can be organized consistently (European Commission et al. 2009). The classification is analytically productive because it creates stable membership rules for statistical aggregation. Similar logics operate in administrative systems: participation in a program or regulatory regime depends on satisfying criteria that determine whether an actor belongs inside or outside the relevant category.

Recognition can also generate economic capacity. Registration may permit an organization to open accounts, contract, receive funds, employ workers, or incur debt. Employment status may provide access to wages, insurance, pensions, or legal protections. Residence may determine access to public services or tax obligations. Institutional membership thus changes the action space available to an actor. North's treatment of institutions as constraints structuring interaction helps clarify why recognized status can alter incentives and feasible conduct (North 1990).

The generative consequence is important for the chapters that follow. An entity becomes economically durable when several systems converge in recognizing it: law attaches rights and duties, accounting draws a reporting boundary, statistics assign a category, administration records membership, and counterparties organize expectations around the resulting unit. These recognitions reinforce one another and make the entity increasingly actionable. Once stabilized, the entity can receive resources, bear obligations, appear in models, and become the target of intervention.

This institutional success explains much of the historical power of entity-centered economics. Modern economies created entities that were genuinely useful for administration because institutional systems repeatedly acted through them. The limit appears when economic effects propagate through relations that cross the recognized boundary or when important participants remain weakly recognized by the institutions that allocate resources and visibility. Section 1.6 therefore turns from the construction of entities to the historical logic of allocating scarce resources through administratively identifiable recipients.

1.6 Entity-Centered Allocation in Historical Context

The institutional construction of economic entities described in Section 1.5 made a particular form of allocation historically practical: resources could be assigned to recipients represented as persons, households, firms, corporations, public bodies, or other administratively identifiable units. This section examines the historical conditions under which recipient-centered allocation became a powerful economic abstraction. Its objective is explanatory rather than dismissive. The analysis proceeds through five steps: the formation of administrative recipients, the relation between material scarcity and allocation, the role of employment in connecting productive participation to livelihood, the relation between competitive capability and resource claims, and the limits that appear when evaluation stops at the recipient boundary. The method is historical and institutional. Entity-centered allocation is treated as an achievement of modern economic organization whose usefulness depends on the social and administrative conditions that made recipients sufficiently stable, comparable, and actionable.

1.6.1 Administrative Recipients

Allocation requires a recipient that an institution can identify before resources can be transferred, taxed, credited, insured, subsidized, or otherwise assigned. Modern economic administration therefore depends on categories that answer a practical question of membership: which unit is entitled, liable, responsible, or eligible within a given scheme. The institutional units of national accounting provide one explicit example. The System of National Accounts organizes production, income, saving, assets, and liabilities around units capable of owning assets, incurring liabilities, making economic decisions, and entering transactions (European Commission et al. 2009). Similar recipient structures appear in taxation, social insurance, business registration, corporate finance, public procurement, and household-based benefit systems.

The administrative recipient is consequently a historically stabilized interface between a relational economy and an allocative act. A transfer rarely reaches an abstract individual in isolation. It reaches a legally or administratively recognized unit whose identity has already been produced through registration, residence, employment, ownership, household membership, organizational status, or another institutional relation. The recipient category compresses these relations into a unit upon which an administrative system can act.

This compression yields substantial practical advantages. A public authority can assign a benefit to a registered household without reconstructing every relation within the household. A bank can lend to a corporation whose legal and accounting boundaries are sufficiently stabilized for liability and repayment to be represented. A firm can pay wages to an employee whose contractual status identifies both the payer and the recipient. Large-scale allocation becomes feasible because institutional systems can repeatedly address the same kinds of units without reproducing the whole relational history that makes those units possible.

Recipient construction also standardizes accountability. An allocation can be recorded as having been made to a defined unit; the unit can be asked to report,

repay, comply, or demonstrate eligibility. The reporting-entity logic discussed in the preceding section illustrates how flexible such units can be: accounting may construct a reporting boundary around one legal entity, part of an entity, or several entities when the institutional purpose requires it (IFRS Foundation 2018). Administrative allocation operates through the same general principle. The actionable unit is selected according to the function of the allocative regime.

The success of this architecture explains why recipient-centered analysis became deeply embedded in economic reasoning. Once recipients are legally, statistically, and administratively stabilized, flows among them can be represented with exceptional clarity. Income is received by households, investment is made by firms, taxes are collected from persons and organizations, transfers are paid to beneficiaries, and assets are assigned to owners. The representation is institutionally grounded rather than merely convenient. Modern economies developed extensive infrastructures that made entity-level flows real objects of administration.

1.6.2 Scarce Resources and Allocation

The allocative significance of stable recipients increased under conditions in which material resources, income, productive assets, public budgets, and institutional opportunities were treated as scarce relative to competing claims. Robbins' influential formulation of economics around scarce means with alternative uses made this structure explicit at the disciplinary level (Robbins 1932). Scarcity does not by itself determine who should receive a resource, yet it creates the recurrent requirement to choose among possible uses and recipients.

Entity-centered allocation offered a tractable response to that requirement. Once a scarce resource could be represented as a quantity and potential recipients could be represented as discrete units, allocation could be organized through prices, wages, contracts, budgets, eligibility rules, credit decisions, taxation, transfers, or administrative assignment. The institutional problem acquired a common form even when the mechanisms differed: a bounded quantity was distributed among recognized claimants.

This form was especially compatible with the accounting and statistical infrastructures of industrial modernity. Monetary valuation allowed heterogeneous resources to enter common ledgers; property rules identified control; organizational accounts located assets and liabilities; state statistics aggregated households and firms; and administrative records linked claims to recognized recipients. Allocation could therefore be represented through flows that entered and left defined units. The entity-centered picture matched the operational needs of institutions that had to decide, record, compare, and audit distributions at scale.

Welfare economics extended the allocative problem from private choice to the evaluation of social arrangements. Pigou's analysis of economic welfare connected economic assessment to the national dividend and to divergences between private and social net product (Pigou 1920). Whatever normative criterion is adopted, such analysis presupposes some representation of who receives benefits and bears costs. Recipient categories thereby enter both administrative action and evaluative reasoning.

The historical importance of this architecture should be separated from a claim

of timeless necessity. Material scarcity remains economically consequential, and recipient-level distribution remains indispensable in many domains. A household requires income, a firm requires financing, and a public program requires rules for eligibility. The historical question concerns the scope of the abstraction. Recipient-centered allocation became powerful because the major resources and institutions of modern economic organization could often be represented through identifiable owners, workers, firms, taxpayers, and beneficiaries. The same abstraction becomes less complete when the economic consequence of an allocation depends heavily on relations that extend beyond the recipient unit.

1.6.3 Employment and Access to Livelihood

Employment became one of the central institutional relations connecting productive participation to command over resources. Under wage labor, the employment relation links the contribution of labor to a monetary claim that can be exercised elsewhere in the economy. Marx's analysis of labor-power and the wage relation places this institutional form within the production process of capital, while Polanyi's historical account emphasizes the broader social transformation involved in organizing labor through markets (Marx 1867; Polanyi 1944). The importance of employment for allocation follows from its position between production and livelihood.

For the worker, wage income supplies purchasing power. For the firm, employment identifies a recognized contributor within an organizational structure and a recurrent claim on organizational resources. For public administration, employment status can become an observable category used in taxation, insurance, pensions, labor regulation, and statistical classification. The employment relation therefore coordinates several institutional systems at once. Productive participation, monetary income, legal status, organizational membership, and administrative recognition become partially coupled through one relation.

This coupling makes employment more than a production arrangement. In societies where access to many goods and services is mediated by monetary income, labor-market position influences the practical ability to secure housing, food, mobility, education, care, and other conditions of life. The strength of the connection varies across welfare regimes, households, property structures, and public institutions, yet the general architecture remains significant: employment often provides a major route through which productive participation is converted into an individual or household resource claim.

The recipient-centered form is visible in the way the wage appears. A complex production process involving technologies, infrastructures, organizations, legal protections, accumulated knowledge, suppliers, customers, and public institutions results in a payment assigned to an identifiable worker. The wage is real and materially consequential, while the recipient unit abstracts from the wider relations that made the productive process possible. The same structure appears on the employer side, where revenues and profits are attributed to legally and organizationally bounded entities even when production depends on extended relational networks.

Employment therefore helped stabilize a broader social expectation: participation in production could function as a legitimate basis for access to income. This expect-

tation never exhausted the sources of livelihood, since property income, household transfers, public provision, informal production, inheritance, and many other mechanisms also mattered. Its analytical significance lies in how effectively it connected three institutional domains that might otherwise remain separate: production, recognition, and allocation. Later sections will examine why artificial intelligence and changing productive capabilities place pressure on this historically powerful connection.

1.6.4 Competitive Capability and Resource Claims

Market allocation frequently connects resource claims to capabilities that can be exercised within recognized exchange and production relations. Labor earns income when a worker's capacities can be supplied under an employment relation; property generates claims when an owner controls an asset that can be leased, sold, financed, or used productively; firms obtain revenue when organizational capacities produce goods or services that counterparties purchase. Competitive capability therefore matters because institutions convert certain capacities into claims over resources.

The conversion is mediated rather than automatic. A capability has economic consequence only within a structure that recognizes, prices, contracts, or otherwise enables its exercise. The same skill may command a high wage in one labor market and little income in another. An asset may generate revenue under one property regime and remain economically inactive under another. Organizational knowledge may become valuable when complemented by finance, infrastructure, legal permission, and market access. Resource claims consequently depend on the relation among capability, institution, and opportunity rather than on capability considered in isolation.

Sen's entitlement analysis offers a particularly clear formulation of the distinction between aggregate availability and a person's command over resources. His analysis of famine shifts attention from total food supply toward the legally available means through which persons can command food, including production possibilities, trade opportunities, and state entitlements (Sen 1981). The argument is narrower than a general theory of allocation, yet its structural lesson is important here: the existence of a resource in an economy does not establish an actor's effective claim upon it. Claims are mediated by ownership, exchange, production, public provision, and other institutional relations.

This perspective clarifies the historical connection between competitive capability and allocation. Markets can translate productive or exchange capabilities into monetary claims when institutions recognize the relevant relations. The resulting claims can then be exercised against other resources. A worker's labor capacity, a firm's organizational capacity, an owner's asset, or a creditor's contractual right enters an allocative system through a different institutional channel, yet each channel links a recognized position to a claim.

The architecture also generates asymmetry. Differences in productive capability, asset ownership, bargaining position, institutional recognition, and access to markets can become differences in resource command. Once resource command changes, it can alter future capabilities by affecting education, health, mobility, technology, finance, networks, and institutional access. Recipient-centered allocation therefore

contains a dynamic dimension even when the immediate transaction is represented as a transfer between fixed entities. Present claims influence the conditions from which future claims can be made.

The historical relevance of this mechanism becomes especially important when capabilities change rapidly. If productive capacity is increasingly concentrated in artificial systems, platforms, capital-intensive infrastructures, or organizations that control them, the inherited connection between individual productive contribution and resource claims may weaken. Chapter 1 develops this problem later. The present point is preparatory: modern allocation systems often rely on institutional pathways that translate recognized capabilities and positions into claims, and those pathways have histories rather than universal form.

1.6.5 Historical Limits of Recipient-Centered Evaluation

Recipient-centered evaluation is strongest when the principal question concerns the immediate distribution of a resource among identifiable units. It can show who received income, credit, property, benefits, or public expenditure; it can compare quantities across households or firms; and it can support accountability for decisions made under scarcity. These capacities remain indispensable. The limitation emerges when the evaluative object extends from receipt to the relational and generative consequences of receipt.

The first limit concerns internal structure. Two households receiving the same transfer may contain different dependency relations, care obligations, decision structures, and access conditions. Two firms receiving the same investment may organize labor, knowledge, ownership, and external suppliers differently. A recipient-level measure records the transfer accurately while leaving the internal relational configuration unresolved. The resulting abstraction is often appropriate for administration, although it cannot by itself establish how the resource changes relations within the recipient unit.

The second limit concerns external structure. A resource assigned to one entity may alter other actors through prices, employment, bargaining, supply relations, debt, care obligations, network access, or institutional precedent. These effects cross the administrative boundary used to record the allocation. The same nominal transfer can therefore generate different economic consequences depending on the recipient's position within wider relational configurations.

The third limit concerns effective command. Sen's entitlement analysis demonstrates, in an especially stark domain, that aggregate resource availability does not determine who can command the resource (Sen 1981). Recipient-centered evaluation can likewise become incomplete when formal receipt and effective access diverge. A formally allocated resource may be unusable without complementary infrastructure, knowledge, legal status, time, mobility, or relational support. The allocative event and the realized capability can therefore differ substantially.

The fourth limit concerns temporal extension. Immediate receipt is an event; its consequences may unfold through a trajectory. An educational grant can alter later capabilities and networks. Credit can enable production while also creating dependence. Infrastructure can change future market access. A transfer to an organization

can reshape internal authority or external bargaining power. Evaluating only the first recipient and the first period captures a real part of the process while omitting the later relations generated by the allocation.

These limits establish the historical transition that motivates GRE without retroactively invalidating entity-centered economics. Modern law, accounting, statistics, markets, and public administration created stable recipients because large-scale allocation required them. Economic theory developed abstractions that corresponded closely to this institutional architecture. The contemporary challenge arises as increasingly important economic effects depend on networks, platforms, knowledge infrastructures, institutional access, heterogeneous capabilities, and long-horizon transformations that cross recipient boundaries.

Recipient-centered allocation therefore remains a valid analytical layer within a wider relational account. GRE's later allocation theory will retain the administrative recipient while asking additional questions about the relations through which resources become usable, the configurations altered by transfer, and the future capacities and constraints generated afterward. Before that theory can be developed, Chapter 1 must first reconstruct how capital, labor, and knowledge became historically differentiated sources of economic power. Section 1.7 takes up that task.

1.7 Capital, Labor, Knowledge, and Economic Power

The preceding sections reconstructed the institutional conditions that made economic subjects legible and recipient-centered allocation administratively tractable. This section turns to the historically differentiated capacities through which economic subjects could command resources, participate in production, and influence the conditions of later action. Its objective is to clarify how capital, labor, and knowledge became distinct yet convertible sources of economic advantage and how their institutional conversion contributed to economic power. The discussion proceeds through five steps: capital as productive control, labor as the material basis of social reproduction, knowledge as a productive resource whose scarcity differs from that of material goods, expertise as an institutionally organized hierarchy of epistemic capability, and the conversion among capability, resources, and power. The method is historical and relational. Each source of advantage is treated through the relations that make it effective rather than as an isolated substance possessed independently of institutions, organizations, and other economic subjects.

1.7.1 Capital and Productive Control

Capital acquired historical significance through more than the accumulation of valuable objects. Productive assets become economically consequential when ownership, finance, organization, and law permit their holders to determine how production is initiated, combined, expanded, or withheld. The distinction matters because the same physical object can occupy very different economic positions according to who controls it, what claims are attached to it, and which organizational relations it enables. A machine incorporated into a firm, a parcel of land pledged as collateral, and

a financial claim that can be mobilized for investment all participate in productive control through different institutional arrangements.

Marx's analysis of capital emphasized this relational dimension by locating capital within a production process in which money, means of production, and purchased labor-power are organized toward the production of surplus value (Marx 1867). Whatever theoretical commitments are adopted regarding value, the historical insight is broader: control over productive assets can organize the labor of others, determine access to productive opportunities, and shape the technical form of production. Capital therefore carries a capacity to structure relations in addition to representing stored wealth.

Modern organizational law strengthened this capacity by creating durable arrangements through which assets could be partitioned, committed to organizational purposes, and insulated from some claims against owners or managers (Hansmann and Kraakman 2000). Productive control could thereby persist beyond the lifespan or immediate decisions of particular individuals. Corporations, banks, partnerships, and other organizational forms stabilized pools of assets and claims that could hire labor, purchase technology, enter contracts, and accumulate further resources. Capital increasingly operated through organizations rather than through isolated proprietors alone.

Financial institutions added another layer. Access to credit allows claims on future income to be converted into present productive capacity, while collateral and creditworthiness determine which subjects can perform that conversion on favorable terms. Productive control can consequently expand without prior possession of the full material resources required for production. The institutional ability to mobilize claims, pledge assets, and coordinate investment becomes part of the effective content of capital.

This historical structure created a recursive advantage. Control over productive assets can generate income; income can be retained or invested; investment can expand productive capacity; expanded capacity can strengthen bargaining position and access to finance. The recursion is neither automatic nor uniform, because markets, crises, regulation, competition, technological change, and collective organization can interrupt it. The important point for the present chapter is structural: capital can reproduce its economic effectiveness through institutional channels that connect present control to future control.

Economic power begins to appear within this recursion. At this stage, the term refers provisionally to an actor's capacity to influence access to productive conditions, the organization of production, or the feasible economic actions of other participants. A firm controlling essential infrastructure, a lender controlling access to finance, or an owner controlling a productive asset can affect choices available to subjects who depend on those conditions. Later chapters develop a fuller theory of power. The historical task here is to identify the material and institutional channels through which productive control became one of its recurrent economic sources.

1.7.2 Labor and Material Reproduction

Labor occupies a different position because productive activity is inseparable from the reproduction of the persons and households that perform it. Wage labor links these domains by converting participation in production into income through which food, shelter, care, mobility, education, and other conditions of continued life are acquired. The employment relation therefore connects the organization of production to the material reproduction of economic subjects.

Marx's treatment of labor-power made this connection explicit by distinguishing labor from the capacity to labor and by locating the reproduction of labor-power within the material conditions required for its continued availability (Marx 1867). Polanyi's historical account likewise emphasized the social transformation involved in subjecting labor to market organization, since labor is inseparable from the human beings whose lives are affected by the conditions under which it is bought and sold (Polanyi 1944). These traditions differ in theory and purpose, yet both illuminate the historical dependence of wage labor on arrangements extending beyond the immediate labor contract.

The worker's productive capacity depends on relations that precede the employment encounter. Health, education, care, household support, transportation, legal status, social networks, and public infrastructure all condition whether labor can be offered and under what terms. The wage relation makes only part of this supporting structure visible in monetary form. A firm pays for labor according to institutional and market conventions, while many conditions that reproduce labor are provided by households, communities, public institutions, or earlier investments whose contribution is not represented in the wage itself.

Labor also generates capacities that can outlast particular employment relations. Skills are learned through practice; workers acquire knowledge of technologies, organizations, clients, procedures, and collective routines; occupational communities transmit standards and techniques. Employment can therefore reproduce and transform productive capability at the same time that it uses it. The distribution of learning opportunities becomes economically important because work positions differ in the degree to which they expand future capability.

The relation between labor and power follows from this dual character. A worker may depend on employment for livelihood while an organization depends on workers for productive continuity. The relative strength of these dependencies varies with labor-market conditions, legal protections, skill scarcity, collective organization, ownership structure, mobility, and the availability of alternatives. Labor power in the ordinary political-economic sense and productive capability in the present historical reconstruction meet at this point: the capacity to work becomes economically consequential through institutions that determine who can withhold, replace, coordinate, or reorganize that capacity.

This perspective also clarifies why labor cannot be reduced to a quantity of effort. Labor is situated within a relational infrastructure of reproduction and learning. Its economic effect depends on technical systems, organizational coordination, access to tools and information, household conditions, and institutional recognition. The material reproduction of labor and the productive use of labor are therefore connected

moments of one economic architecture. That architecture becomes especially important once knowledge and expertise begin to account for a larger share of differentiated productive capability.

1.7.3 Knowledge as a Scarce Productive Resource

Knowledge entered modern economic analysis through several routes that gradually made its productive importance more explicit. Industrial production had long depended on skill, scientific knowledge, technical design, and accumulated experience. During the twentieth century, economic theory increasingly represented education, training, information, and innovation as objects of investment and coordination in their own right. This development expanded the concept of productive resources beyond physical assets while also revealing that knowledge obeys a different scarcity structure from many material goods.

Human-capital analysis provided one influential formulation. Becker treated education and on-the-job training as investments that affect earnings, employment, and productivity, thereby representing acquired knowledge and skill as economically consequential capacities rather than as external cultural attributes (Becker 1964). This framework made differences in training and education analytically comparable with other forms of investment, while leaving open broader questions about the institutions through which learning is provided and recognized.

Hayek approached knowledge from the problem of coordination. His argument centered on the dispersion of incomplete and frequently local knowledge across separate individuals, emphasizing that economically relevant information is rarely concentrated in one mind or institution (Hayek 1945). Knowledge scarcity in this sense concerns location and access as much as aggregate quantity. A system may contain the relevant knowledge while no participant possesses a complete representation of it.

Arrow's analysis of invention introduced another important distinction. Information generated through inventive activity has economic properties that complicate ordinary commodity treatment, including uncertainty about its production and difficulties surrounding appropriation once information becomes available (Arrow 1962). The result is a productive resource whose cost of creation, cost of transmission, and cost of repeated use can differ sharply. A technical insight may require substantial investment to produce while becoming inexpensive to reproduce after codification. Knowledge can therefore be scarce at one stage and abundant at another.

These observations support a more differentiated historical account of knowledge scarcity. Scarcity may arise in discovery, acquisition, verification, interpretation, certification, access, or the ability to apply knowledge effectively. Codified information can circulate widely while the capacity to understand or operationalize it remains concentrated. Scientific knowledge can be publicly available while access to laboratories, data, compute, professional networks, or institutional authority remains restricted. The economically relevant unit is consequently broader than a stock of information.

Knowledge also has cumulative properties. Earlier discoveries provide inputs to later discoveries; education increases the ability to acquire further knowledge; or-

ganizations develop routines that preserve experience; scientific communities build infrastructures through which findings can be criticized and extended. These cumulative relations can generate increasing advantages for subjects already positioned within rich knowledge environments. The historical scarcity of knowledge therefore concerns both the content known and the relations through which further knowing becomes possible.

This distinction prepares an important transition in Chapter 1. Industrial modernity treated many forms of expertise as scarce because advanced knowledge required long training, specialized institutions, and limited access to technical infrastructures. Contemporary artificial intelligence alters parts of this scarcity structure by reducing the cost of producing certain symbolic and cognitive outputs. Section 1.11 develops that transformation. The present section first establishes the earlier architecture in which knowledge scarcity supported differentiated expertise and economic hierarchy.

1.7.4 Expertise and Epistemic Hierarchy

Expertise is knowledge placed within an institutional relation of recognized competence. An individual may know something without occupying an expert position, while an expert position usually depends on certification, organizational role, professional reputation, demonstrated performance, or another mechanism through which others are authorized to rely on that knowledge. Epistemic hierarchy arises when these recognition structures distribute unequal authority to interpret information, define problems, evaluate evidence, or make decisions whose consequences extend to other participants.

The economic significance of expertise follows from its connection to scarce productive and coordinative tasks. When training is costly and specialist knowledge is unevenly distributed, expertise can affect wages, occupational access, organizational authority, and bargaining position. Human-capital analysis captures part of this structure by linking education and training to productivity and earnings (Becker 1964). The wider institutional structure includes professional licensing, educational credentials, technical standards, internal promotion systems, and reputational mechanisms that determine which capabilities receive economic recognition.

Expertise also creates organizational dependence. Complex production requires decisions about design, maintenance, finance, medicine, law, logistics, software, research, and many other specialized domains. Organizations therefore allocate decision rights partly according to recognized competence. The resulting hierarchy can be productive because it concentrates responsibility where relevant knowledge is expected to be available. At the same time, dependence on specialized judgment creates asymmetry between those who possess or are recognized as possessing relevant knowledge and those who must act on their conclusions.

Hayek's account of dispersed knowledge places an important limit on epistemic hierarchy. Local participants often possess situational knowledge unavailable to centralized experts, and coordination problems arise when institutions fail to transmit or respond to such distributed information (Hayek 1945). The tension is structural. Specialized expertise may be indispensable for tasks requiring deep technical knowl-

edge, while economic systems also depend on dispersed, contextual, and practical knowledge that cannot be fully centralized.

The hierarchy of expertise is consequently shaped by both knowledge and recognition. Credentials can signal capability, yet they also determine access to occupations and decision arenas. Organizational roles can place experts close to resources and strategic information, which in turn expands their opportunities for further learning. Research institutions can accumulate instruments, data, and networks that make future knowledge production easier. Epistemic advantage can therefore become institutionally self-reinforcing even when the underlying knowledge remains contestable or distributed.

This structure matters for GRE because economic power can arise through control over interpretation as well as control over material resources. Subjects able to classify risk, certify quality, authorize technical claims, design information systems, or define admissible evidence can shape which actions appear feasible to others. Later chapters examine epistemic power more directly. The historical point here is that modern economies increasingly organized productive differentiation around recognized knowledge, thereby adding an epistemic dimension to the older relations of capital and labor.

1.7.5 Conversion among Capability, Resources, and Power

Capital, labor, and knowledge become most consequential when their economic effects are examined through conversion. Productive capability can generate income; income can purchase education, tools, mobility, or access; knowledge can attract finance; ownership can purchase expertise; organizational position can provide information; and institutional recognition can convert a capability into an authorized claim on resources. Economic advantage therefore develops through pathways that translate one form of capacity or resource into another.

The conversion from capability to resources is institutionally mediated. Becker's human-capital framework links acquired education and training to earnings and productivity, yet the strength of that link depends on labor markets, credentials, organizational demand, discrimination, bargaining arrangements, and other institutions (Becker 1964). A capability that lacks recognition, market demand, legal permission, or access to complementary resources may generate little income. Productive capacity and resource command therefore remain analytically distinct.

The reverse conversion is equally important. Income and wealth can finance training, relocation, health, experimentation, equipment, and time for learning. Capital can purchase access to expertise and technical infrastructure. Organizational resources can support research whose results create further capability. Resource inequality can consequently become capability inequality through cumulative investment, while capability inequality can become resource inequality through differentiated returns. These conversions create feedback across time.

Sen's entitlement analysis provides a related insight from a different problem-space: aggregate availability of resources does not determine what particular persons can command, because access depends on the entitlement relations through which resources are obtained (Sen 1981). The same principle applies more broadly

to capability conversion. Possessing a useful skill does not by itself establish access to the assets, institutions, markets, or rights required to exercise it. Resources become effective through relations, and capabilities become economically consequential through conversion structures.

Power enters when these conversions affect the feasible actions of others or the rules governing access. Control over capital can determine who receives finance; recognized expertise can determine which claims are accepted; organizational authority can allocate productive opportunities; ownership can restrict access to infrastructure; labor scarcity can strengthen the capacity to bargain or withhold participation. Economic power therefore often emerges through control of conversion points rather than through possession of a single resource considered in isolation.

The conversion structure also helps explain persistence. An actor with capital may purchase knowledge, hire labor, and expand organizational capacity. An actor with recognized expertise may gain income and authority, then use those resources to acquire further credentials, networks, or productive assets. A worker in a position rich in learning opportunities may accumulate capabilities that widen future options. Conversely, subjects lacking access to key conversion interfaces may experience cumulative disadvantage even when they possess valuable capacities. Historical inequality can therefore reproduce itself through unequal conversion conditions.

This synthesis provides the bridge to the next stage of Chapter 1. Capital, labor, and knowledge are distinct sources of productive capacity, yet their economic effects depend on relations that connect subjects across organizations, infrastructures, institutions, and markets. As production becomes more interdependent, the relevant unit of analysis increasingly extends beyond the isolated owner, worker, or expert. Section 1.8 develops this expansion by examining relational production through division of labor, organizations, production networks, knowledge infrastructures, and transnational economic relations.

1.8 Expansion of Relational Production

The preceding sections reconstructed the modern economic subject and the institutional mechanisms through which capital, labor, and knowledge became recognizable sources of productive capability and economic power. This section changes the scale of analysis. Its role is to show how production increasingly depended on relational configurations that extended beyond any single worker, owner, firm, or administrative recipient. The objective is to identify the historical forms through which productive activity became distributed across interdependent tasks, organizations, supply relations, knowledge infrastructures, and multiple territorial scales. The discussion proceeds from division of labor to organizations, production networks, scientific and knowledge infrastructures, and transnational relations. The method is historical and relational: each form of production is examined through the dependencies, coordination mechanisms, and infrastructures that make distributed activity productive as a configuration.

1.8.1 Division of Labor and Interdependence

The division of labor offers one of the clearest historical routes from entity-centered description toward relational production. Smith treated specialization as a source of increased productive power because repeated tasks permit greater dexterity, reduce time lost in switching activities, and encourage the development of specialized machinery (Smith 1776). The resulting productivity, however, belongs to a configuration of differentiated activities rather than to any isolated worker. Specialization increases output by making each participant more dependent on the complementary work of others.

This dependence changes the economic meaning of productive capability. A highly specialized worker may possess a narrow and valuable skill while being unable to produce a finished good independently. A specialized machine may be extremely productive within one production sequence while having little use outside it. A firm may excel at one stage of production while relying on logistics, suppliers, finance, standards, infrastructure, and downstream distribution. Productive capability therefore becomes increasingly positional: its value depends on how a capacity is connected to other capacities within an organized sequence.

The relation between specialization and interdependence is historically important because it expands the coordination problem at the same time that it expands productive capacity. Tasks must be sequenced, inputs must arrive at appropriate times, specifications must be compatible, and failures in one component can propagate to others. Market prices can coordinate some of these relations, while organizations, contracts, standards, routines, and long-term exchange relationships coordinate others. Hayek's account of dispersed knowledge captures one dimension of this problem by emphasizing that economically relevant knowledge is distributed across participants and cannot be fully concentrated in a single mind or office (Hayek 1945).

Interdependence also makes attribution more difficult. An observed output can often be assigned legally or administratively to a firm, yet the productive conditions of that output may include workers trained elsewhere, infrastructure financed publicly, suppliers operating under long-standing technical standards, scientific knowledge produced in universities, and demand coordinated through institutions beyond the firm. The administrative unit remains useful for accounting and responsibility, while the productive process exceeds it.

Granovetter's analysis of embeddedness sharpened this point by arguing that economic action is situated within ongoing structures of social relations rather than adequately represented by atomized actors alone (Granovetter 1985). For the historical argument developed here, the significance lies in the unit of explanation. Productive outcomes increasingly depend on the pattern of relations among specialized actors, and the stability, trust, information, and expectations carried by those relations can affect what the participants are able to accomplish.

Relational production therefore begins with a simple historical observation: specialization makes productive capacity increasingly distributed. The more extensive the division of labor, the less adequate it becomes to identify production solely with the attributes of a nominal producer. The relevant productive object begins to include the configuration through which differentiated capacities are connected.

1.8.2 Organizations as Relational Structures

Organizations provide one institutional response to the coordination demands generated by specialization. Section 1.5 examined the legal and accounting mechanisms that stabilize organizations as recognizable entities. The present subsection examines their internal productive structure. An organization can act as an economic unit because relations among roles, authority, information, assets, routines, and responsibilities are sufficiently stabilized to sustain coordinated action over time.

Coase's analysis of the firm framed the organizational question through the costs of using the price mechanism and the circumstances under which coordination is brought inside the firm (Coase 1937). That formulation remains important because it makes organizational boundaries economically consequential. Yet the boundary does not eliminate relational complexity. It reorganizes it. Activities that might otherwise be coordinated through repeated market transactions become connected through employment, managerial authority, internal procedures, budgeting, technical standards, and shared infrastructure.

The internal productive capability of a firm is consequently distributed across relations among positions. A design team depends on manufacturing knowledge; manufacturing depends on procurement and maintenance; sales depends on production and logistics; finance depends on information generated throughout the organization. The formal organization chart captures part of this architecture, while informal relations, professional communities, accumulated routines, trust, and tacit knowledge often carry coordination that the formal chart does not specify.

Network forms of organization make the relational character of production even more visible. Powell describes network forms as organized through reciprocal patterns of communication and exchange that differ from both discrete market transactions and hierarchical command (Powell 1990). Such arrangements can support collaboration where participants remain organizationally distinct yet require continuing coordination, mutual adjustment, and information exchange. Productive organization therefore admits more than a binary choice between market and hierarchy.

This plurality matters for economic analysis because the same nominal set of assets and workers can perform differently under different relational arrangements. A tightly centralized organization, a federated organization, a long-term alliance, and a project network distribute authority, information, learning, and risk differently. Productive capability is consequently partly a property of organizational architecture. Reorganization can alter output without changing the physical assets or formal skills initially available, because the relations through which those resources are combined have changed.

Organizations also accumulate relational memory. Routines preserve solutions to recurrent problems; reputations shape expectations; shared standards reduce coordination costs; informal communication channels accelerate problem solving; and prior collaboration changes the ease with which new collaboration can occur. These features make organizational history productive. Present capability depends partly on relations generated through earlier activity.

The firm remains a legitimate unit of economic analysis, especially where legal responsibility, financial reporting, and contractual control are concerned. Relational

analysis adds a second description: the firm is also a stabilized configuration of internal and external relations whose productive capacity depends on how those relations are maintained and transformed. This dual description becomes increasingly necessary once production extends across organizational boundaries.

1.8.3 Production Networks and Supply Chains

Production networks extend relational production beyond the boundaries of a single organization. Modern goods and services are frequently produced through sequences in which design, components, assembly, logistics, finance, certification, distribution, and after-sales support are distributed among legally distinct organizations. The resulting system cannot be understood only by adding the attributes of participating firms, because the pattern of dependence among them affects coordination, bargaining, learning, vulnerability, and the distribution of returns.

Supply relations differ in depth. Some transactions are relatively standardized and can be shifted among suppliers with limited reorganization. Others require dedicated equipment, jointly developed specifications, long-term learning, quality assurance, shared information systems, or repeated adaptation. The economic relation between buyer and supplier therefore ranges from relatively impersonal exchange to durable interdependence. These differences affect how productive knowledge moves and how risks are distributed across the network.

The global value-chain literature provides a developed account of such variation. Gereffi, Humphrey, and Sturgeon identify governance patterns ranging from market relations to modular, relational, captive, and hierarchical forms, with differences shaped by transaction complexity, codifiability, and supplier capabilities (Gereffi et al. 2005). Their typology is valuable here because it shows that formally separate firms can stand in relations with very different degrees of coordination and power asymmetry. A contractual boundary therefore provides limited information about the substantive structure of production.

Network position can also affect access to opportunity. A firm connected to technically demanding customers may receive information, standards, and incentives that support learning, while a supplier confined to narrowly specified tasks may have fewer opportunities to develop independent capabilities. The productive consequences of a relation can therefore extend beyond the immediate exchange. Repeated transactions may generate new competencies, lock in specialized investments, deepen dependence, or alter the set of future partners available to each participant.

This perspective clarifies why disruptions propagate. A production network can appear decentralized because ownership is distributed among many firms, while remaining operationally concentrated around a small number of critical nodes, standards, logistics routes, or technological interfaces. Failure at one point can interrupt production elsewhere even when downstream firms remain financially solvent and legally intact. The vulnerability belongs to the relational configuration through which production is organized.

Relational production also complicates the attribution of value creation. Final prices and accounting systems assign revenues to identifiable organizations, but

productive contribution may be dispersed across suppliers, infrastructure providers, workers, knowledge institutions, and complementary technologies. The question of how value generated through a network is recognized, converted, and distributed will become central in later GRE chapters. At this historical stage, the important development is the expansion of production beyond the boundaries used to identify economic recipients and owners.

Supply chains therefore provide a concrete transition from entity-centered production to relational architecture. Firms remain real and consequential economic subjects, yet their productive capacities are conditioned by positions within networks of dependence, learning, standards, and exchange. The unit of production increasingly spans relations among organizations.

1.8.4 Scientific and Knowledge Infrastructures

Knowledge-intensive production makes the relational basis of productive capacity especially visible because usable knowledge depends on infrastructures that no single organization normally creates by itself. Universities, laboratories, professional associations, journals, standards bodies, databases, patent institutions, training systems, public research agencies, and technical communities contribute to the production, verification, transmission, and preservation of knowledge. Firms and workers draw on these infrastructures even when the resulting commercial output is attributed to a particular organization.

Arrow's analysis of invention emphasized uncertainty and the distinctive appropriation problems surrounding information production (Arrow 1962). These features help explain why knowledge production has repeatedly relied on institutional forms that differ from ordinary commodity exchange. Public funding, professional norms, publication systems, intellectual-property regimes, collaborative research, and organizational secrecy create different arrangements for balancing dissemination, appropriation, and continued investment.

The productive importance of knowledge infrastructures extends beyond the stock of published information. Scientific and technical work depends on instruments, methods, repositories, standards, peer communities, credibility systems, and trained personnel able to interpret results. A published finding may be widely accessible while the capability to reproduce, evaluate, or apply it remains concentrated in institutions possessing complementary infrastructure. Knowledge abundance at the level of symbols can therefore coexist with scarcity at the level of effective participation.

Interorganizational collaboration provides one mechanism through which such capabilities are distributed. Powell, Koput, and Smith-Doerr's study of biotechnology located important innovation processes within networks of learning that crossed organizational boundaries (Powell et al. 1996). The broader lesson is not specific to biotechnology. Where relevant knowledge is dispersed and rapidly developing, productive capability can depend on access to relations through which organizations learn from one another, recruit expertise, interpret developments, and combine complementary resources.

Knowledge infrastructures are also cumulative. Training institutions reproduce specialist communities; standards allow artifacts produced in different places to in-

teroperate; archives and databases preserve earlier results; professional criticism filters claims; and shared technical vocabularies reduce the cost of future coordination. These infrastructures are products of earlier relations and conditions for later production. Their economic contribution is therefore temporally extended and difficult to assign to a single transaction.

This structure matters for the historical emergence of GRE because it weakens the correspondence between ownership of an output and generation of the conditions that made the output possible. A firm may own a patent or sell a product while depending on publicly financed science, professional standards, prior open knowledge, supplier learning, and collective technical infrastructures. The administrative attribution of revenue can remain precise even when the generative history of capability is distributed.

Knowledge-intensive production thus expands the economic object from information possessed by an actor to an infrastructure of relations through which knowing becomes possible, credible, applicable, and renewable. Later chapters will examine the implications for knowledge capital, epistemic economy, and generative return. The present section establishes the historical condition: productive knowledge increasingly operates through relational infrastructures whose boundaries exceed the firms that commercialize their outputs.

1.8.5 Transnational and Multiscalar Economic Relations

The expansion of relational production eventually exceeds both organizational and territorial boundaries. Production can be coordinated through relations among workers, firms, financial institutions, public authorities, standards bodies, logistics systems, and knowledge institutions located in different jurisdictions. The resulting architecture is multiscalar: local workplaces operate within organizations; organizations operate within production networks; networks depend on national institutions and transnational infrastructures; and changes at one scale can alter feasible actions at another.

This multiscalar structure changes the meaning of economic location. A plant is physically situated in one place while depending on financing, intellectual property, technical standards, intermediate inputs, software, demand, and managerial decisions distributed across other places. A worker's employment conditions can be affected by procurement decisions made elsewhere. A supplier's upgrading opportunities can depend on standards imposed by lead firms or regulatory regimes outside its jurisdiction. Territorial location therefore remains important while becoming only one coordinate of economic position.

Global value-chain analysis again illustrates the point. Governance patterns within transnational production can distribute coordination and power asymmetrically even where production is legally fragmented across independent firms (Gereffi et al. 2005). The economic significance of a firm consequently depends partly on its relation to lead firms, standards, technologies, logistics channels, and markets that operate at larger scales.

Multiscalar production also connects different institutional tempos. Employment law, industrial capacity, infrastructure, financial conditions, and technical standards

change at different rates. A rapid shift in procurement or finance can therefore encounter much slower changes in worker skills, transport systems, regulatory institutions, or regional productive structures. The historical consequences of production cannot always be understood on the timescale of the transaction that initiated them. This observation prepares the later GRE treatment of generative spacetime and fast-slow economic dynamics.

The expansion across scales also changes the distribution of responsibility. Economic benefits and harms may be generated through chains of relations in which no single participant controls the whole configuration. Lead firms may influence specifications without directly employing workers who implement them; investors may alter financing conditions without managing production; states may regulate entities within their jurisdiction while key dependencies lie elsewhere. Legal and administrative responsibility remains essential, yet causal and generative responsibility can extend across a wider relational field.

Section 1.9 develops globalization as a distinct historical transformation of these relations, including transnational production, capital flows, distributed labor, institutional interdependence, and economic relations beyond territorial boundaries. The present section establishes the prior analytical shift. Once productive activity is distributed across interdependent tasks, organizations, networks, knowledge infrastructures, and scales, the economy becomes increasingly difficult to represent as a collection of independently productive entities connected only by exchange. Relational production becomes part of the historical architecture of economic life.

1.9 Globalization and the Extension of Economic Relations

The preceding section showed how productive activity expanded from specialized tasks and organizations into interorganizational networks and knowledge infrastructures. This section extends the historical scale once more. Its role is to reconstruct globalization as an enlargement and densification of economic relations across territorial, organizational, and jurisdictional boundaries. The objective is to identify how production, capital, labor, institutions, and claims upon value became increasingly transnational while remaining embedded in unequal capacities, legal regimes, and historically specific forms of coordination. The discussion proceeds through transnational production, global capital flows, distributed labor and unequal exchange, global institutional interdependence, and economic relations whose operative structure exceeds a single territorial container. The method is historical and relational: globalization is treated as a reorganization of connections among economic actors and institutions rather than as a simple increase in the volume of international transactions.

1.9.1 Transnational Production

Transnational production extends the relational structure of production across territorial jurisdictions. A finished good, service, or technological system may combine design in one location, components from several others, assembly elsewhere, financing through another jurisdiction, and distribution through globally coordinated logis-

tics. The productive object presented to a final buyer can therefore be the terminal manifestation of a geographically dispersed system of relations among firms, workers, infrastructures, standards, and institutions.

Global value-chain analysis made this architecture analytically visible by distinguishing different governance forms through which lead firms and suppliers coordinate complex transactions. Gereffi, Humphrey, and Sturgeon show that transaction complexity, the codifiability of information, and supplier capabilities help shape governance arrangements ranging from relatively market-like relations to more captive and hierarchical forms (Gereffi et al. 2005). Their framework is useful here because it shows that geographic dispersion does not imply organizational dispersion in the same degree. Production can be territorially distributed while coordination, specification, intellectual property, market access, or strategic decision-making remains highly concentrated.

This distinction changes the meaning of location. A factory is located in a territory, yet its productive role may be defined by relationships extending far beyond that territory. Its output may depend on imported components, standards specified by a distant lead firm, credit provided through international financial networks, software maintained elsewhere, and demand originating in multiple markets. The economic significance of the local unit therefore cannot be inferred from territorial location alone. Position within a transnational production structure affects bargaining power, learning opportunities, vulnerability to substitution, and the share of downstream value that can be retained.

Transnational production also separates several forms of economic control that industrial organization had often joined within the same firm or national economy. Ownership may be located in one jurisdiction, production in another, technical authority in a third organizational center, and consumption across many territories. Contractual relations can coordinate these distributed elements without bringing them under a single legal person. The resulting architecture combines entity-level responsibility with relational dependence across entities.

The historical importance of this arrangement lies in the enlargement of the relevant economic system. A national economy remains a meaningful institutional and statistical object, yet many productive processes acquire conditions of reproduction that extend beyond national boundaries. Input availability, logistics, exchange rates, standards, foreign demand, and decisions by transnational firms can alter domestic production without originating within the domestic institutional field. Globalization therefore expands the set of relations that must be considered when explaining how a nominally local economic capability is generated and sustained.

This expansion does not erase territory. Infrastructure, labor law, taxation, education, energy systems, political authority, and physical logistics continue to be territorially organized to substantial degrees. Transnational production instead layers cross-border coordination upon territorial institutions. The result is a multiscalar productive architecture in which local, national, regional, and global relations jointly condition what any single economic actor can do.

1.9.2 Global Capital Flows

Global capital flows extend economic relations through claims on assets, liabilities, income streams, and future payments. Cross-border finance allows savings generated in one jurisdiction to support investment in another, firms to raise funds from foreign investors, states to issue debt internationally, and households or institutions to hold claims on assets located elsewhere. The relevant relation is therefore partly financial and partly temporal: present resources are connected to expectations about future returns across institutional and territorial boundaries.

The expansion of gross external positions illustrates how strongly these relations can exceed the scale suggested by net flows alone. Lane and Milesi-Ferretti constructed estimates of foreign assets and liabilities for a large set of countries and documented the growing scale and changing composition of international financial positions during the late twentieth century (Lane and Milesi-Ferretti 2006). For relational analysis, the importance of such data lies less in a single aggregate measure than in the multiplication of cross-border creditor, debtor, equity, reserve, and direct-investment relations that bind economic actors to developments outside their immediate jurisdiction.

Financial integration can redistribute both opportunity and exposure. Access to external capital may relax domestic financing constraints, support investment, and diversify risk. The same relational extension can transmit shocks through balance sheets, refinancing conditions, exchange-rate movements, and changing investor expectations. A financial relation that appears as an asset to one actor is simultaneously a liability or obligation within another balance sheet. Changes in one part of the system can therefore alter the conditions faced by actors connected through portfolios, credit lines, collateral structures, or payment systems.

Capital mobility also changes the relation between economic activity and institutional jurisdiction. Productive facilities remain materially situated, while claims on their earnings can circulate through financial centers and legal vehicles far from the site of production. Ownership, control, risk bearing, and income receipt can thus become geographically separated. These separations complicate any analysis that equates the location of productive activity with the location of economic benefit or decision power.

The same architecture creates asymmetry among participants. Actors with access to deep financial markets, reserve currencies, diversified portfolios, sophisticated intermediaries, or favorable legal jurisdictions can often manage cross-border exposure differently from actors dependent on a narrow set of external creditors or volatile financing channels. Global capital flows consequently create a relational field in which mobility itself becomes unequally distributed capability. The ability to enter, exit, hedge, refinance, or relocate a claim may matter as much as the nominal size of the resource being moved.

For GRE, the historical lesson is preparatory. Capital becomes increasingly intelligible as a set of mobile claims embedded in institutional relations, rather than only as a stock attached to a territorial owner. The later analysis of power, value circulation, and conversion will require this distinction because financial mobility can transform resources, obligations, and bargaining positions across scales without any

corresponding movement of the underlying productive activity.

1.9.3 Distributed Labor and Unequal Exchange

The globalization of production distributes labor across economies characterized by different wage structures, productivity levels, bargaining institutions, currencies, social protections, and positions within international production networks. This distribution generates a long-standing problem in political economy: similar productive interdependence can coexist with markedly unequal compensation, bargaining power, and capacity to appropriate the gains associated with international exchange.

The structuralist tradition formulated an early version of this problem through the center–periphery distinction. Prebisch’s analysis of Latin American development emphasized asymmetries in productivity, industrial structure, and terms of trade between industrial centers and primary-exporting peripheral economies (Prebisch 1950). The analytical significance of this tradition lies in its insistence that participation in international exchange occurs from historically differentiated productive positions. Exchange connects economies whose capacities to innovate, set prices, absorb shocks, and alter the composition of production may differ substantially.

Theories of unequal exchange developed the argument in a more explicitly distributive direction. Emmanuel linked international exchange to persistent differences in wages and to the international organization of production, arguing that trade can reproduce transfers of value between differently positioned economies (Emmanuel 1972). This is a contested theoretical tradition, and the present chapter does not adopt its full value-theoretic apparatus. Its relevance to the historical argument is narrower: globalization made it increasingly necessary to ask how the distribution of labor across jurisdictions interacts with price formation, wage differentials, ownership, and the allocation of returns across the production process.

Global production networks sharpen the question because the physical contribution of labor and the capacity to command downstream returns can be institutionally separated. Workers and suppliers may perform indispensable stages of production while lead firms control design, standards, branding, finance, intellectual property, market access, or final distribution. The governance patterns identified in global value-chain research provide one way to describe these asymmetries without presupposing a single theory of value (Gereffi et al. 2005). Different positions in the chain carry different opportunities for learning, upgrading, substitution, and bargaining.

The term *unequal exchange* therefore names more than one possible mechanism and should be used carefully. Terms-of-trade dynamics, wage differentials, ownership structures, market concentration, technological dependence, currency hierarchy, and contractual governance can each contribute to unequal distributions of gains. Their relative importance is an empirical and theoretical question. A relational framework gains little by compressing them prematurely into one explanation.

The more durable insight is that labor mobility, capital mobility, and value appropriation need not move together. Capital and contractual authority can cross borders with relative ease in some settings, while workers remain subject to migration law, local labor institutions, family obligations, language, and other forms of territorial

embeddedness. The resulting asymmetry can alter bargaining positions across the global production system. Globalization therefore expands productive interdependence while also creating new sites at which differences in mobility, institutional protection, and control shape distribution.

This historical problem prepares a later GRE question concerning heterogeneous value and conversion. Labor performed in different institutional settings becomes comparable through prices, contracts, accounting systems, and exchange rates, yet those mechanisms of commensuration may also conceal relations that produced the observed distribution. Section 1.9 remains at the historical level; later chapters will return to the normative and power-theoretic consequences of such conversion structures.

1.9.4 Global Institutional Interdependence

Globalization depends on institutions that make cross-border relations sufficiently stable to support repeated economic activity. Trade agreements, customs rules, financial regulation, payment systems, accounting standards, investment regimes, central-bank arrangements, dispute procedures, technical standards, and private certification systems all contribute to the conditions under which transnational exchange can occur. Economic integration is therefore institutionally produced as well as commercially enacted.

Institutional interdependence creates a layered structure of authority. A firm operating transnationally may face domestic corporate law, foreign labor and environmental regulation, international trade rules, sectoral standards, contractual obligations, financial-market requirements, and private platform or certification rules at the same time. These rule systems differ in source and enforceability, yet jointly shape the feasible actions of the firm. The economic relation is consequently embedded in overlapping jurisdictions rather than contained by one institutional order.

This layering also reveals a persistent tension between economic reach and political jurisdiction. Rodrik emphasizes that deep international economic integration encounters constraints created by national jurisdiction and political organization (Rodrik 2000). The point is especially useful for relational analysis because the relevant boundaries do not coincide: production networks, capital claims, information flows, and market relations may extend globally while taxation, social protection, electoral accountability, and many regulatory capacities remain organized primarily through territorial states.

Institutional interdependence therefore changes the consequences of domestic decisions. A regulatory change in one major market can alter production standards elsewhere; a monetary or financial decision can affect external borrowers; a change in customs treatment can reorganize sourcing relations; a shift in certification requirements can redistribute access to markets. The causal chain often crosses several jurisdictions and organizations before reaching the affected actor.

The resulting structure can also redistribute institutional power. Actors capable of setting standards, providing settlement infrastructure, controlling market access, or defining widely adopted contractual forms can shape behavior beyond their formal territorial authority. Smaller jurisdictions and firms may retain legal autonomy while

adapting to rules generated elsewhere because access to a larger network depends on compliance. Such influence operates through relational position and dependence rather than through direct command alone.

Global institutional interdependence thus extends the economic field beyond trade volumes. It includes the rule-producing and rule-transmitting relations that allow transnational production and finance to remain coordinated. These institutional layers will become important later when GRE distinguishes resource ownership from access, capability from effective control, and formal sovereignty from relational dependence.

1.9.5 Economic Relations beyond Territorial Boundaries

The combined development of transnational production, global finance, distributed labor, and institutional interdependence produces economic relations whose operative boundaries differ from territorial boundaries. A production network may span several states; a financial position may connect balance sheets located on different continents; a technical standard may organize firms across many jurisdictions; and an ownership structure may separate the place of production from the place in which claims on its income are legally held.

Territory remains economically consequential because infrastructure, coercive authority, taxation, public services, labor regulation, and political membership are still extensively territorialized. Globalization adds relational structures that cut across these territorial containers. The same actor can therefore occupy several economic spaces simultaneously: a local labor market, a national legal order, a transnational supply chain, an international financial network, and a global market for knowledge or standards.

This multiplicity creates analytical problems for any unit of analysis treated as self-contained. National accounts can describe production and income according to carefully defined territorial and residence conventions, firms can report consolidated entities, and trade statistics can record cross-border transactions. These representations remain indispensable. Their boundaries, however, answer specific administrative and analytical purposes. They do not exhaust the relations through which productive capability, vulnerability, knowledge, risk, and bargaining power are generated.

A relational description therefore supplements territorial accounting with attention to connection, position, dependence, and cross-scale transmission. The relevant questions concern which actors are connected, which infrastructures mediate those connections, which institutions govern them, how easily relations can be substituted, and how effects propagate when one node or jurisdiction changes. Such questions become increasingly important as the same production process is divided among actors whose legal, financial, and political environments differ.

Globalization also changes the meaning of economic distance. Geographic separation can coexist with tightly synchronized production, while neighboring actors can remain weakly connected economically. Contractual standards, transport infrastructure, communication systems, legal compatibility, financial access, and organizational integration can reduce some forms of distance while preserving or enlarging

others. Economic proximity becomes partly relational rather than purely spatial.

The historical transition examined in this section therefore prepares the move to digital modernity. Once economic coordination already extends beyond territorial and organizational boundaries, digital infrastructures can intensify mediation, observation, matching, and algorithmic coordination across those relations. Section 1.10 develops that next stage. The globalization considered here establishes the preceding transformation: economic activity becomes increasingly organized through multiscalar relations whose effective boundaries exceed the entities and territories through which they are administratively represented.

1.10 Digital Modernity and Relational Intermediation

This section locates digital modernity within the historical expansion of relational production developed in Sections 1.8 and 1.9. Its role is to identify how digital infrastructures alter the conditions under which economic relations are formed, observed, coordinated, and concentrated. The objective is to show that digitalization changes more than the speed or scale of existing transactions: it creates new infrastructures of information, inserts platforms between previously more direct relations, increases the frequency and granularity of observation, delegates parts of coordination to algorithmic systems, and changes the returns associated with network position. The discussion proceeds through information as productive infrastructure, platforms as mediators, data-intensive observation, algorithmic coordination, and the concentration dynamics associated with network effects. The method is historical and relational. Digital technologies are treated as institutional and technical arrangements that reorganize connections among economic actors rather than as a separate economy detached from earlier productive structures.

1.10.1 Information as Productive Infrastructure

Digitalization changes the economic role of information by making its storage, processing, replication, transmission, and recombination available at scales and speeds that earlier organizational forms could not sustain. Information has always mattered to production, exchange, administration, and coordination. Digital infrastructures deepen that dependence by embedding information-processing capacities directly into productive systems, logistics, finance, communication, and organizational decision-making.

Goldfarb and Tucker characterize digital economics through reductions in several costs associated with digital activity, including search, replication, transportation of information, tracking, and verification (Goldfarb and Tucker 2019). This formulation is useful for the present historical argument because each cost reduction modifies a relation. Lower search costs alter how buyers, sellers, workers, firms, and information sources find one another. Lower replication and transmission costs change the circulation of symbolic and informational goods. Lower tracking costs expand the observability of interactions. Lower verification costs can reorganize the institutional arrangements through which claims are authenticated.

Information therefore becomes infrastructural in a stronger sense than simply being an input possessed by an organization. Digital production depends on shared protocols, databases, cloud services, communication networks, identifiers, software standards, application interfaces, payment rails, and security systems. These infrastructures establish conditions under which geographically distributed actors can coordinate actions that remain intelligible to one another. A transaction recorded in one system can trigger inventory updates, logistics instructions, payment processes, compliance checks, and subsequent recommendations across several organizational boundaries.

The productive significance of such infrastructure often appears indirectly. A database or protocol may generate little value as an isolated object while enabling a large number of downstream relations. Its economic importance can therefore reside in the range of interactions it makes possible, the compatibility it establishes, and the future actions it renders easier or harder. This is an early instance of a distinction developed later in GRE between possession of a resource and the generative conditions supported by that resource.

Digital infrastructure also changes the temporal structure of coordination. Information that previously arrived through periodic reports can be transmitted at much shorter intervals. Inventory, demand, prices, locations, payments, and user interactions may be updated in near real time. Decisions can consequently respond to changing conditions before slower administrative routines would have registered them. The historical consequence is a tighter coupling between observation and action, a theme that becomes central in the later discussion of data and algorithmic coordination.

The expansion of information infrastructure does not remove the material basis of economic activity. Data centers require energy and equipment; communication networks require physical infrastructure; software systems depend on organizations and trained labor; digital records rely on institutional conventions that make identifiers, permissions, and categories operational. Digital modernity therefore adds a dense informational layer to material production while also making many productive relations increasingly dependent on that layer.

1.10.2 Platforms and Mediated Relations

Platforms reorganize economic relations by becoming durable intermediaries between groups whose interactions were previously arranged through more fragmented markets, organizations, or bilateral channels. Search platforms connect users and information providers; marketplaces connect buyers and sellers; payment systems connect merchants, consumers, and financial institutions; labor platforms connect clients and workers; application ecosystems connect developers, device owners, advertisers, and infrastructure providers. The platform's economic object is therefore frequently a configuration of relations among several participant groups.

The economics of two-sided markets provides an established account of this structure. Rochet and Tirole analyze platforms whose viability depends on bringing multiple sides of a market into participation and on choosing a price structure that reflects interactions across those sides (Rochet and Tirole 2003). Their formulation shows

that the platform cannot be understood adequately by examining a single bilateral transaction in isolation. Participation on one side changes the attractiveness and economic possibilities available on another.

Relational mediation gives the platform several forms of influence. It can determine conditions of entry, specify contractual terms, rank or recommend participants, standardize interfaces, collect fees, define permitted forms of interaction, and alter which information is visible to whom. These functions need not be exercised uniformly, and their institutional significance varies across platform types. The common feature is positional: the platform occupies an interface through which other actors reach one another.

This interface position can reorganize the distinction between internal and external economic relations. A seller may remain legally independent while depending on a platform for discovery, payment, reputation, and access to demand. A worker may formally contract as an independent provider while task allocation, evaluation, and visibility are mediated by platform rules. A developer may own an application while access to users depends on an operating system, store, or technical interface controlled elsewhere. Legal independence can therefore coexist with substantial relational dependence.

Platforms also transform market boundaries. The relevant market may become partially constituted by technical standards, ranking systems, identity protocols, and participation rules maintained by the intermediary. Changing an interface or recommendation policy can alter demand allocation without changing the underlying legal status of participants. Economic coordination increasingly occurs through modifications to the architecture of mediation itself.

For GRE, the historically significant development is the emergence of intermediaries whose productive role consists partly in organizing relations among others. This does not make the platform external to the economy it mediates. The platform becomes one participant within the relational configuration, while its position gives it unusual capacity to alter the conditions under which other relations form and persist.

1.10.3 Data and Continuous Observation

Digital mediation produces data as a recurrent by-product of economic interaction. Searches, purchases, clicks, payments, movements, communications, ratings, inventories, delivery events, and work activities can generate machine-readable traces as they occur. Economic observation consequently shifts from periodic description toward increasingly granular and high-frequency recording in domains where digital infrastructures are present.

Einav and Levin describe the growing use of large-scale administrative and private-sector data in economics and emphasize that such data can provide granular and increasingly timely views of economic activity (Einav and Levin 2014). Their discussion concerns economic research, yet the same historical transformation extends beyond research. Firms, platforms, governments, and other organizations can use transaction-level or interaction-level data as operational inputs to pricing, logistics, risk assessment, advertising, fraud detection, performance evaluation, and service

design.

The reduction of tracking costs identified by Goldfarb and Tucker is especially consequential here (Goldfarb and Tucker 2019). Observation becomes woven into the transaction itself. A purchase records demand; a delivery records location and timing; a recommendation records exposure and response; a digital workflow can record sequence, duration, completion, and evaluation. The distinction between conducting an activity and producing a record of that activity narrows in systems designed around continuous data capture.

This increased observability changes economic power because visibility is distributed unevenly. A platform may observe interactions among many participants while each participant sees only a narrow local portion of the system. A firm operating a digital marketplace can possess information about aggregate demand, search behavior, conversion rates, and seller performance that individual sellers cannot reconstruct independently. Data concentration can therefore produce epistemic asymmetry even where the underlying transactions involve formally independent parties.

Data also depend on categories. A recorded event becomes usable only after a system defines what counts as a user, transaction, impression, conversion, task, defect, risk, or successful outcome. Continuous observation therefore increases the importance of classification rather than eliminating it. What can be optimized depends on what has been rendered observable, and what is observable depends on technical and institutional decisions embedded in the data infrastructure.

The resulting expansion of evidence has an ambivalent historical character. Granular observation can reveal heterogeneity, improve coordination, detect failures, and make previously opaque processes available for analysis. The same infrastructure can intensify surveillance, concentrate informational advantage, and encourage action toward what is easiest to record. Later chapters examine these epistemic and normative consequences directly. At this stage, the central transformation is that economic relations increasingly generate traces that can feed back into the organization of those relations at short temporal intervals.

1.10.4 Algorithmic Coordination

The accumulation of machine-readable data creates conditions for algorithmic coordination. Rules encoded in software can match participants, rank alternatives, set prices, route deliveries, detect anomalies, recommend actions, schedule labor, allocate attention, and update decisions in response to incoming information. Economic coordination thereby acquires a computational layer operating between observation and action.

Algorithmic coordination differs from simple automation of a fixed task when the system continually mediates relations among multiple actors. A recommendation system affects which producers become visible to which users. A dispatch system allocates work across drivers and locations. A pricing system changes incentives and demand across a marketplace. A ranking system redistributes attention among competing participants. In each case, the algorithm operates upon a relational configuration and can alter that configuration through repeated decisions.

Organizational research has documented this development in the workplace. Kellogg, Valentine, and Christin synthesize research on algorithmic control and identify mechanisms through which employers can use algorithms to direct, evaluate, and discipline work (Kellogg et al. 2020). Their account is relevant beyond labor control because it demonstrates a broader institutional change: observation, classification, recommendation, evaluation, and sanction can be integrated into computational systems that operate repeatedly and at scale.

The economic significance of algorithmic coordination depends on both the rule and the information available to it. A technically sophisticated allocation rule produces little if the relevant state of the system is poorly observed. Conversely, extensive data can have limited operational effect until a decision process translates them into rankings, predictions, or actions. Digital infrastructures therefore join observation and coordination through feedback: records generated by interaction become inputs to later decisions, and later decisions generate new records.

Such feedback can make coordination highly adaptive while also producing path dependence. Ranking based on prior activity can direct attention toward already visible participants; recommendation based on accumulated behavior can reinforce established patterns; performance metrics can induce workers or firms to adapt toward measured criteria. The resulting economic order is partly generated through repeated interaction between participants and the computational procedures that mediate them.

Algorithmic coordination remains institutionally situated. Objectives, permissible actions, data access, error tolerances, review procedures, and accountability structures are supplied by organizations and legal systems. The algorithm is therefore neither an autonomous economic subject by default nor a neutral technical layer. It is part of an institutional relation through which some actors obtain the capacity to coordinate the actions of others at increased scale and frequency. The later treatment of artificial intelligence will examine what changes when computational systems acquire substantially greater productive and epistemic capabilities.

1.10.5 Network Effects and Digital Concentration

Digital intermediation can generate concentration through the interaction of network effects, accumulated participation, data, standards, and control of interfaces. The mechanism begins from a relational property: the usefulness of some systems depends partly on how many other participants, complementary products, or compatible services are connected to them.

Katz and Shapiro analyze systems markets in which compatibility and network effects make popular products more valuable and influence adoption dynamics, competition, and standardization (Katz and Shapiro 1994). Platform markets add cross-group effects: participation by one side can increase the value of the platform to another side, creating feedback across participant populations (Rochet and Tirole 2003). These mechanisms do not guarantee monopoly, yet they can make installed participation and interface control economically consequential assets.

Concentration can arise through several mutually reinforcing channels. A larger participant base may attract more complementary suppliers; greater activity may

generate more data; more data may improve matching or prediction; improved coordination may attract further participation; and wider adoption may increase incentives for complementary products to conform to the platform's standards. The result is a feedback structure in which relational position contributes to future capacity.

This process changes the meaning of capital and competitive advantage. Productive assets remain important, yet control of a widely used interface, identity system, communication network, data repository, technical standard, or application ecosystem can become a durable source of economic power. The value of such an asset may depend less on its isolated technical properties than on the relations already organized through it and the costs participants face when moving elsewhere.

Digital concentration also creates asymmetry between scale and jurisdiction. A platform can mediate relations across many territories while regulation, labor law, taxation, consumer protection, and public accountability remain divided among jurisdictions. The multiscale tension developed in Section 1.9 therefore persists within digital modernity, now combined with technical mediation and high-frequency observation. Concentrated intermediaries can coordinate relations at a scale larger than many institutions responsible for governing their effects.

The historical importance of digital concentration lies in the coupling of relational centrality with productive and epistemic capability. A central intermediary can influence access, visibility, standards, data collection, and coordination simultaneously. Economic power consequently becomes increasingly difficult to infer from ownership of conventional productive assets alone. Position within a digital relational architecture can itself generate capabilities that reinforce that position.

Digital modernity thus intensifies several tendencies developed across this chapter: production becomes more relational, observation becomes more continuous, coordination becomes more mediated, and economic power becomes increasingly connected to control over interfaces and infrastructures. Section 1.11 turns to artificial intelligence, where these digital conditions support a further transformation in cognitive production, symbolic generation, knowledge production, and comparative productive capability.

1.11 Artificial Intelligence and Productive Transformation

This section identifies the transformation that artificial intelligence introduces into the productive conditions established by digital modernity. Its role is to distinguish the economic significance of contemporary AI from the preceding expansion of information infrastructure and algorithmic coordination. The objective is to show how artificial systems increasingly participate in tasks historically performed through human cognitive labor, generate symbolic outputs at very low marginal cost, enter organized processes of knowledge production, alter comparative productive capability across workers and organizations, and concentrate important forms of productive capacity in the infrastructures that make advanced models possible. The discussion proceeds from cognitive-task automation to symbolic generation, machine-mediated knowledge production, comparative capability, and concentration. The method is

historical and economic. Artificial intelligence is treated here through the productive relations it changes rather than through a claim about machine consciousness, intelligence in the philosophical sense, or the ontological status of artificial systems. Those questions remain open and are addressed only where their economic consequences require them.

1.11.1 Automation of Cognitive Production

Earlier waves of mechanization transferred portions of physical production from human labor to machines. Contemporary artificial intelligence extends automation into tasks whose outputs were historically associated with human cognitive labor: drafting, classification, prediction, translation, coding, summarization, pattern recognition, search, recommendation, and parts of analytical work. The economic novelty does not lie in the appearance of automation itself. It lies in the widening range of tasks for which computational systems can produce outputs that enter ordinary organizational workflows.

A task-based perspective provides a useful starting point. Acemoglu and Restrepo distinguish automation from productivity improvement in already assigned tasks by emphasizing changes in the allocation of tasks between labor and capital (Acemoglu and Restrepo 2019). In their framework, automation creates a displacement effect when capital substitutes for labor in tasks previously performed by workers, while the creation of new tasks can generate a reinstatement effect that expands domains in which labor retains comparative advantage. The framework is deliberately more precise than a binary contrast between “jobs lost” and “jobs created.” Production is decomposed into tasks, and technological change alters who or what can perform them.

Generative AI extends this task reallocation into forms of work whose intermediate products are symbolic. A worker drafting a memorandum, preparing a customer response, writing code, summarizing a document, or producing a first-pass analysis can now combine human judgment with machine-generated candidate outputs. This arrangement does not imply complete substitution. It alters the sequence of production. Human effort can move away from some acts of initial generation toward specification, selection, correction, verification, integration, and responsibility for the final result.

Experimental evidence already demonstrates that this reorganization can be economically consequential in bounded tasks. Noy and Zhang randomly assigned college-educated professionals to complete occupation-specific writing tasks with or without access to ChatGPT and found substantial reductions in completion time together with improvements in evaluated output quality (Noy and Zhang 2023). Brynjolfsson, Li, and Raymond study the staggered introduction of a generative-AI assistant among customer-support workers and report an average productivity increase, with especially large gains among less experienced and lower-skilled workers (Brynjolfsson et al. 2025). The studies concern particular tasks and organizational settings, yet they establish a point needed for the historical argument: systems generating language can already change the productive performance of human workers rather than merely serving as experimental demonstrations outside production.

The resulting transformation is therefore relational. Productive capability no longer resides only in the worker or only in the model. It can arise from a configuration involving the worker, the model, the interface through which prompts and outputs are exchanged, the organization's proprietary information, evaluation procedures, and the institutional rules determining which outputs may be used. A model that performs impressively in isolation may contribute little where data access, workflow integration, or verification is absent. Conversely, a modest model embedded in a well-designed process may alter productivity substantially. The productive object is increasingly the human-machine arrangement rather than either component taken alone.

This reorganization also changes the meaning of skill. Some skills remain directly productive because the worker performs the task. Others become supervisory or evaluative because the worker judges machine-generated candidates. Still others become more valuable because AI lowers the cost of complementary activities. The historical transition is therefore better understood as a redistribution of productive functions than as the simple removal of human cognition from production. Section 1.13 later examines the allocation consequences when such redistribution creates persistent asymmetries in comparative capability.

1.11.2 Machine-Generated Symbols

Generative models also alter the material conditions of symbolic production. Text, software code, images, audio, structured summaries, translations, synthetic examples, and other representational outputs can be produced at scales and speeds that differ sharply from earlier human-only processes. Digitalization had already lowered the cost of storing, copying, transmitting, and searching symbols. Generative AI lowers the cost of producing new symbolic arrangements in response to instructions and context.

This distinction matters because symbolic production had historically remained constrained by human time even after reproduction became cheap. A digital document can be copied almost without limit, yet drafting the first document still required someone to produce it. Generative systems weaken that bottleneck. Once a model is available, additional candidate texts, images, code fragments, classifications, or reformulations can be generated at low marginal cost relative to the human labor that would have been required to produce each candidate independently.

The resulting abundance should be interpreted carefully. Symbolic output is not identical with meaning, truth, or knowledge. Bender, Gebru, McMillan-Major, and Mitchell emphasize that large language models operate over linguistic form and warn against attributing understanding merely from the fluency of generated text (Bender et al. 2021). Their argument is particularly useful for GRE because it separates an economic fact from an epistemic conclusion. The economic fact is that systems can generate symbols that are usable within many production processes. The epistemic standing of those symbols depends on their relation to evidence, context, interpretation, verification, and subsequent use.

Machine-generated symbols therefore create a new selection problem. When candidate production becomes cheap, the scarce activity can migrate toward speci-

fication, evaluation, authentication, and attention. A writer facing one blank page confronts a production problem. A writer facing hundreds of plausible machine-generated drafts confronts a different problem: selecting, interpreting, checking, and integrating among abundant candidates. The productive constraint shifts from generation alone toward orientation and judgment.

The same shift occurs in organizations. Marketing teams can generate many variations of copy; software teams can generate multiple code suggestions; researchers can generate candidate summaries or hypotheses; administrative systems can generate draft classifications and reports. In each case, the volume of symbolic material can expand more quickly than the institutional capacity to evaluate its accuracy, relevance, provenance, or consequences. Symbolic abundance can therefore increase the demand for trustworthy filters even while reducing the direct cost of producing symbols.

This transformation provides the immediate historical background for the representational crisis examined in Section 1.14. Economic institutions depend on symbols: prices, reports, contracts, classifications, credentials, ratings, applications, records, and claims about states of the world. When symbol generation becomes abundant, the old inference from costly production to evidential significance becomes less reliable. The problem moves from whether a representation can be produced toward how its relation to a referent is established and maintained.

1.11.3 Machine-Mediated Knowledge Production

Artificial intelligence enters knowledge production through a more demanding pathway than symbolic generation. Knowledge is an institutional and epistemic achievement: claims must be related to evidence, methods, prior work, standards of justification, and communities capable of criticism and revision. A fluent output becomes economically useful knowledge only when it is incorporated into relations that establish why it should be relied upon.

AI can nevertheless alter those relations materially. Systems can search large information spaces, retrieve and reorganize documents, generate code for analysis, propose candidate explanations, identify patterns, produce simulations, and translate material across technical or linguistic boundaries. These functions can shorten the path between a question and a set of candidate answers. They can also make forms of expertise available to workers who would previously have required longer training, specialized assistance, or access to a more experienced colleague.

The productivity evidence from generative AI illustrates this knowledge-transfer function without establishing that the model itself possesses knowledge in the same sense as a person or institution. Brynjolfsson, Li, and Raymond find especially large gains among less experienced customer-support workers and present evidence consistent with the system disseminating practices associated with stronger workers (Brynjolfsson et al. 2025). The economically important mechanism is relational: information distilled from prior interactions becomes available through the model to workers who did not personally accumulate the same experience.

This possibility changes the historical scarcity of expertise. Expertise traditionally combined knowledge with the time required to learn, practice, encounter rare cases,

and gain entry to communities and institutions where such learning occurs. AI can reduce some of these access costs by placing codified or statistically reconstructed regularities into an interface available during performance. The result may narrow capability differences in some tasks, as the heterogeneous productivity effects in experimental and workplace studies suggest (Noy and Zhang 2023; Brynjolfsson et al. 2025).

The transformation remains partial. Tacit knowledge, situated judgment, responsibility, domain-specific interpretation, and the recognition of novel or misleading circumstances may remain difficult to externalize. AI outputs can also be incorrect, weakly grounded, or inappropriate to the local context. The epistemic problem therefore changes rather than disappears. A worker may gain access to a larger candidate knowledge space while simultaneously acquiring a new burden of verification.

Knowledge production also becomes more recursive. Human-generated records train or condition models; models generate outputs that enter human workflows; those workflows produce new records; and later systems may be trained, evaluated, or adapted using traces of the resulting activity. The boundary between knowledge infrastructure and productive activity therefore becomes increasingly porous. What an organization knows can be partly embodied in databases, models, prompts, retrieval systems, evaluation procedures, and the accumulated interaction between artificial systems and human participants.

For GRE, the important historical consequence is that knowledge can become both more abundant and more infrastructurally concentrated at the same time. Access to a general-purpose model can diffuse capabilities widely, while the production, training, adaptation, and control of the most capable systems may remain concentrated. Section 1.12 examines the corresponding transformation of scarcity, while Section 1.17 later returns to the economic organization of knowledge and epistemic capacity.

1.11.4 Transformation of Comparative Productive Capability

The economic significance of AI becomes sharper when productive capability is treated comparatively rather than absolutely. A worker can become more productive after adopting AI and simultaneously lose relative scarcity if similar or greater capability becomes available to many others at low cost. Conversely, a worker whose domain expertise complements AI effectively may become more productive and more valuable even though the technology performs parts of the underlying task. The relevant transformation concerns the distribution of comparative capability across participants.

Task-based automation already provides the structure for this analysis. When technology performs tasks previously assigned to labor, the task content of production shifts. New tasks, complementary skills, or organizational roles can partly counterbalance displacement (Acemoglu and Restrepo 2019). Generative AI makes this reallocation unusually visible in cognitive and symbolic tasks because the same system can assist drafting, coding, classification, summarization, translation, and other activities across occupations rather than being confined to one narrowly engineered physical process.

Evidence from current deployments suggests that the effect can be heterogeneous

across prior capability levels. Noy and Zhang report larger gains among initially weaker performers in professional writing tasks (Noy and Zhang 2023). Brynjolfsson, Li, and Raymond similarly find larger improvements among less experienced and lower-skilled customer-support workers, while effects for the most experienced workers are smaller (Brynjolfsson et al. 2025). These findings do not establish a universal equalizing effect. They demonstrate that AI can change the shape of a productivity distribution rather than simply shifting every participant upward by the same amount.

This transformation can alter the economic value of accumulated experience. Some experience is encoded in routines that a model can reproduce or make available to newcomers. Other experience becomes more valuable because it supports verification, problem framing, exception handling, or judgment in cases where machine output is unreliable. Comparative advantage therefore moves within the task structure. A capability that once differentiated an expert may become common, while a previously secondary capability becomes the new bottleneck.

The same logic applies at the organizational level. Firms differ in proprietary data, complementary software, workflow design, managerial capacity, domain expertise, computing infrastructure, and the ability to redesign production around new tools. Access to the same general-purpose model can therefore yield different productive effects. AI capability is not a scalar resource added uniformly to every organization; it enters a pre-existing relational architecture and combines with other capacities.

Comparative productive capability also has distributive consequences because modern institutions connect productive contribution to wages, profit, bargaining position, investment, access, and organizational authority. A technology that changes comparative capability can therefore alter resource claims even before aggregate output changes dramatically. The later sections on scarcity, allocation, and subjecthood examine the resulting tension: an economic order historically organized around comparative productive necessity encounters artificial systems capable of performing an expanding range of tasks at potentially much lower replication cost.

1.11.5 Concentration of Artificial Productive Capacity

The diffusion of AI-generated outputs coexists with concentration in the infrastructure that produces frontier artificial capability. This combination is historically important. A service can be accessible to millions of users while the training, deployment, and strategic control of the underlying systems remain located in a relatively small number of organizations and technical infrastructures.

The Stanford AI Index documents this concentration at the frontier. Its 2026 report states that industry produced more than 90 percent of notable AI models in 2025 and records rapid growth in global AI compute capacity together with substantial concentration in advanced hardware and hyperscale infrastructure (Stanford Institute for Human-Centered Artificial Intelligence 2026). These observations do not imply that useful AI is available only from a few firms. Open-weight models, smaller systems, local adaptation, and falling inference costs can diffuse access widely. The relevant distinction is between diffusion of use and control over the resources required to create or operate the most capable systems.

Artificial productive capacity depends on several complementary layers: specialized chips, data centers, energy, network infrastructure, model architectures, training data, engineering talent, capital, and organizational experience in large-scale deployment. Bottlenecks at any layer can shape who can develop frontier systems. The productive asset is therefore not the model file alone. It is a relation among computational, financial, informational, organizational, and infrastructural capacities.

Concentration at these interfaces can create economic leverage beyond conventional ownership of a single productive asset. An organization controlling model access can influence prices, usage conditions, permissible applications, data retention, technical compatibility, and the pace at which downstream users receive new capabilities. Cloud providers and hardware suppliers can shape the cost and feasibility of model development. Organizations possessing proprietary data or distribution networks can combine those assets with models in ways that new entrants cannot easily reproduce.

The structure is therefore double-sided. AI can decentralize capability by allowing a small organization or individual worker to perform tasks that previously required a larger team or specialized expert. The same technological system can centralize upstream productive power by increasing dependence on compute, model providers, data infrastructure, and technical standards. Decentralized use and centralized capacity are compatible because they occur at different layers of the productive architecture.

This layered concentration complicates conventional measures of economic power. Ownership of factories, machinery, or financial assets remains relevant, yet control of artificial productive capacity may depend on infrastructure positions that organize access for many downstream actors. A user can possess no share of the model provider while becoming deeply dependent on the provider's interface. A firm can own its data while lacking the compute or model capability needed to extract comparable productive value from them. Relational position becomes increasingly important to effective command.

Artificial intelligence therefore intensifies the historical tendencies developed in Sections 1.7 through 1.10: knowledge becomes a productive resource, production becomes distributed across networks, digital interfaces mediate access, and informational advantage can become economic power. AI adds the capacity to generate cognitive and symbolic outputs at scale and to reorganize comparative capability. The next section turns to the resulting transformation of scarcity: abundance in some forms of production can coexist with new scarcities in attention, trust, institutional access, infrastructure, and meaningful opportunities to participate in generative economic relations.

1.12 Transformation of Economic Scarcity

This section consolidates the productive changes described in Sections 1.10 and 1.11 into a historical account of changing scarcity. Its role is to clarify that technical abundance in one domain can redistribute scarcity across the wider economic system. The objective is to distinguish continuing material scarcity from the rapid expansion of

symbolic and codified informational output, then identify the growing importance of attention, trust, institutional access, and opportunities for consequential participation. The discussion proceeds from material constraints to symbolic abundance, qualified knowledge abundance, emergent relational scarcities, and generative opportunities. The method is historical and economic. Scarcity is treated as a relation between available means, institutional arrangements, technological capacities, and the uses through which those means become consequential; no claim is made that contemporary economies have entered a general condition of abundance.

1.12.1 Material Scarcity

Material scarcity remains a constitutive condition of economic organization. The digitalization of information and the automation of cognitive tasks alter the cost structure of production, yet they continue to depend on energy, land, physical infrastructure, computing equipment, logistics, human time, and access to productive assets. The historical transformation therefore concerns the composition and location of scarcity as much as its aggregate level. A task can become cheap in one dimension while becoming more dependent on scarce complementary resources in another.

The classical economic language of scarce means retains its relevance here. Robbins's influential formulation treated economics as concerned with human behavior under conditions in which means are scarce relative to alternative uses (Robbins 1932). Contemporary digital production changes which means are scarce and which alternatives compete for them. Storage and replication costs can fall sharply while advanced computation, energy, specialized hardware, skilled maintenance, and infrastructure access become more economically consequential. The relevant scarcity therefore shifts across the production system rather than simply receding.

Artificial intelligence makes this redistribution especially visible. Frontier systems can generate symbolic outputs at very low marginal cost once a model has been trained and deployed, yet the capacity to build and operate such systems remains materially intensive. The 2026 AI Index documents continued growth in the scale and concentration of advanced AI compute and infrastructure, alongside broader diffusion of downstream use (Stanford Institute for Human-Centered Artificial Intelligence 2026). The coexistence of cheap generated output and concentrated upstream capacity illustrates the central point: abundance at the output interface can rest on scarcity and concentration in the infrastructure that makes the output possible.

Material scarcity also persists through bottlenecks that lie outside the immediate production process. A digitally abundant service may still depend on limited transmission capacity, electricity supply, physical distribution, regulatory permission, or geographic access. Similarly, an organization may possess abundant information while lacking the personnel, time, or institutional authority required to act upon it. The economic significance of scarcity therefore depends on the relations linking resources to use. A resource can be physically present and economically unavailable when the institutional or relational conditions of access are absent.

This historical persistence of material constraint prevents a simple narrative in which digital technology replaces scarcity with abundance. The more precise development is a differentiation of scarcity. Some forms of production become increas-

ingly reproducible and automatable, while other inputs become relatively more decisive because they remain difficult to reproduce, difficult to access, or concentrated in a small number of institutional locations. This differentiation prepares the analysis of symbolic abundance, where the cost of producing representations falls much faster than the cost of interpreting, validating, and acting upon them.

1.12.2 Symbolic Abundance

Symbolic production has undergone one of the sharpest scarcity transformations in the contemporary economy. Digital technologies reduced the costs of copying, transmitting, searching, and storing information, while generative AI extends the cost reduction to the production of new symbolic arrangements. Goldfarb and Tucker characterize digital technologies through reductions in several economic costs associated with search, replication, transportation, tracking, and verification (Goldfarb and Tucker 2019). Generative systems add another layer by lowering the cost of producing candidate texts, images, software code, classifications, and other representations.

The economic consequence is a rapid expansion in the supply of symbols relative to the human time required to inspect them. Section 1.11 distinguished this expansion from knowledge production in the stronger epistemic sense. That distinction remains essential. Bender and colleagues caution against inferring understanding from fluent language output and thereby provide a useful boundary between symbolic competence and epistemic warrant (Bender et al. 2021). Symbolic abundance therefore describes an observable productive condition: representations can be generated at scales that no longer require proportional human authorship. It does not establish that truth, justification, or understanding have become correspondingly abundant.

This shift alters the location of economic effort. When symbolic generation is expensive, production itself absorbs time and skill. When candidate production becomes cheap, selection, comparison, verification, interpretation, provenance assessment, and integration become relatively more important. The scarcity problem moves downstream. An organization may be able to generate hundreds of reports, designs, or strategic options while lacking the attention required to evaluate them. A researcher may generate many plausible hypotheses while still facing the harder task of determining which ones merit sustained inquiry. A consumer may face abundant recommendations while remaining uncertain which sources deserve trust.

Symbolic abundance also changes the strategic value of visibility. When publication or generation costs fall, the ability to enter an informational environment no longer guarantees that an output will be noticed. Search ranking, platform position, reputational signals, institutional endorsement, and network location become increasingly important in determining which symbols reach consequential audiences. The problem of scarcity therefore reappears through channels of selection and recognition.

The expansion of machine-generated symbols further increases the importance of provenance. When many outputs can be produced rapidly and with similar stylistic plausibility, the economic value of knowing where a representation came from, which process produced it, and what evidential relations support it can rise. Section 1.14 develops this issue as a representational crisis. The present section records

the prior economic condition: production of symbols can become abundant faster than the institutional capacity to authenticate, rank, and absorb them.

1.12.3 Knowledge Abundance

Knowledge abundance requires a more qualified interpretation than symbolic abundance. Modern economies possess unprecedented capacities to store, reproduce, retrieve, and recombine codified knowledge. Digital networks make scientific articles, technical documentation, educational materials, software, data, and institutional records available across distances at costs far below those of earlier print and archival systems. Artificial intelligence can further reduce the effort required to search, summarize, translate, and reorganize such materials. These developments expand the effective availability of codified knowledge even when the production of justified knowledge remains demanding.

The economic properties of information help explain this transformation. Arrow's analysis of invention emphasized that information differs from ordinary commodities because its production involves uncertainty and its use can be difficult to appropriate exclusively once the information exists (Arrow 1962). Digital reproduction strengthens the practical significance of this distinction by making many informational objects cheap to copy and distribute. The marginal cost of another reader, user, or software instance can become very small relative to the original cost of discovery, design, or verification.

Expanded availability, however, does not eliminate the dispersed and situated character of economically relevant knowledge. Hayek's analysis of knowledge in society emphasized that knowledge of particular circumstances remains distributed among many actors and cannot be treated as if it were centrally given (Hayek 1945). Contemporary data systems and AI can aggregate and mediate some of this knowledge, yet they do so through representations, classifications, and institutional access arrangements. Tacit knowledge, local judgment, practical familiarity, and context-sensitive interpretation remain difficult to detach from the relations in which they are formed.

The phrase *knowledge abundance* therefore refers here to an expansion in the accessible stock and recombining ability of codified knowledge rather than to a disappearance of epistemic scarcity. Several complementary scarcities can intensify as codified material proliferates. Verification remains costly. Provenance must be established. Contradictory sources require adjudication. Questions must be selected. Relevance must be judged relative to a concrete problem. The scarce activity increasingly concerns orientation within a large possibility space of available information and candidate explanations.

AI-mediated work illustrates this redistribution. Brynjolfsson, Li, and Raymond find evidence consistent with generative AI helping less experienced workers access and apply patterns embodied in the practices of more experienced workers (Brynjolfsson et al. 2025). Such diffusion can reduce certain knowledge asymmetries inside an organization. At the same time, the organizational value of the system depends on who can access the tool, what proprietary knowledge it can retrieve, how its outputs are checked, and which institutional processes accept those outputs as actionable.

The scarcity of codified know-how can decline while the scarcity of trustworthy interpretation, authorized access, and responsibility for judgment becomes more salient.

This distinction is important for GRE because an economy with abundant informational artifacts can still distribute epistemic capability very unequally. Access to a large corpus differs from the capacity to formulate worthwhile questions, discriminate among explanations, connect knowledge to a local situation, or persuade an institution to act. The later analysis of epistemic economy will treat these differences directly. At the historical level, the relevant transition is already visible: economic scarcity moves from possession of information alone toward the relations that make information interpretable, credible, and practically consequential.

1.12.4 Attention, Trust, and Institutional Access

As symbolic and informational supply expands, several relational conditions become more economically scarce. Attention is the clearest case. Simon's analysis of information-rich organizations argued that an abundance of information creates a scarcity of the attention required to receive and process it (Simon 1971). The proposition becomes increasingly important in digital environments where both human and machine systems can produce outputs continuously. The cost of producing another message may approach zero while the time available to read, interpret, compare, and act upon messages remains bounded.

Attention scarcity is relational because visibility depends on competition among claims upon finite recipients. A document can exist without being read, a recommendation can be generated without being considered, and an opportunity can be technically available without entering the decision horizon of the relevant actor. Search systems, ranking algorithms, professional reputation, institutional position, and social networks therefore influence which signals obtain scarce attention. Abundance in production can increase the economic value of selection mechanisms.

Trust becomes consequential through a related mechanism. Economic relations frequently involve commitments whose quality cannot be fully established before action. Arrow's discussion of gifts and exchanges emphasized the pervasive role of trust in commercial and social transactions (Arrow 1972). In information-rich environments, the number of potential counterparties, claims, and representations can expand faster than the capacity to verify each one independently. Trust, reputation, certification, and institutional assurance then perform a filtering function. Their economic importance grows when direct verification is expensive relative to the abundance of available claims.

Trust should not be treated here as a homogeneous stock. It is generated and sustained within particular relations, histories, institutions, and repeated practices. Trust in a person, trust in a profession, trust in a platform, and trust in a public institution arise under different conditions and can fail in different ways. The scarcity relevant to economic coordination is therefore the limited availability of credible relations through which actors can rely on information, promises, identities, and standards without reproducing the entire verification process themselves.

Institutional access forms a third scarcity. Institutions organize entry to resources, credentials, markets, finance, scientific communities, legal protections, and decision

procedures. North's institutional analysis emphasizes that rules and organizations structure incentives and transaction possibilities rather than merely recording choices after they occur (North 1990). In a digital economy, information about an opportunity can be abundant while actual access to the institution controlling that opportunity remains constrained. Knowing that a funding programme exists differs from being eligible to apply; knowing that a model exists differs from having compute access; knowing that a scientific field exists differs from possessing the credentials, language, networks, or institutional affiliation through which participation becomes possible.

The capability perspective provides a useful bridge from resource possession to effective opportunity. Sen's account of development as freedom emphasizes substantive opportunities and the institutional conditions through which people are able to pursue valued ways of living (Sen 1999). For the present historical argument, the important point is analytical: equal information or nominal resources can coexist with unequal effective access because conversion depends on social, institutional, and personal conditions. Scarcity can therefore reside in the relation between a subject and an institution even when the informational description of the opportunity is widely available.

Attention, trust, and institutional access share a common structural feature. They mediate the conversion of abundant possibilities into consequential action. Symbolic abundance produces many candidate claims; attention determines which are considered; trust influences which are treated as credible; institutional access affects which can be acted upon. Their growing importance marks a transformation in scarcity from possession alone toward relational gateways.

1.12.5 Generative Opportunities and Meaningful Participation

The final scarcity considered in this section concerns opportunities to participate in relations that can generate further capacities, knowledge, projects, recognition, and future options. The term *generative opportunity* is used here in a preliminary sense. It denotes access to a relational configuration in which participation can contribute to outcomes that extend beyond immediate consumption or transfer. Examples include entry to a research community, participation in an organization where skills can develop, access to collaborators and infrastructure, involvement in a public institution, or membership in a network through which new projects become possible. The full theory of generative value and generative opportunity is developed later in the book; the present section identifies the historical conditions that make the category increasingly relevant.

This perspective extends the analysis of scarcity beyond the ownership of goods. A person may possess material resources and still lack access to relations in which those resources can be converted into capability. A worker may possess skill yet lack organizational entry. A researcher may possess knowledge yet lack collaborators, infrastructure, or recognition. A community may possess local knowledge yet lack institutional channels through which that knowledge can alter decisions. In each case, scarcity resides in a missing relation or a constrained pathway of conversion.

Sen's capability approach provides an important precedent for shifting attention

from resources to substantive opportunity (Sen 1999). GRE carries the question further into the structure of relations that generate later possibilities. The distinction is analytical rather than competitive: capabilities concern what a person is effectively able to do or be, while generative opportunities direct attention to the relational processes through which new capacities and future option spaces may emerge. The two can overlap, and their exact relation remains an open theoretical problem for later chapters.

Artificial intelligence intensifies the relevance of this issue because productive abundance can weaken the historical link between human participation and material necessity. If machines can perform a growing range of cognitive and symbolic tasks, economic inclusion cannot be understood only through the scarcity of labor required for production. Access to livelihood remains crucial, yet another problem becomes visible: whether people possess opportunities to participate in relations where their judgment, curiosity, care, creativity, or responsibility can become consequential. Section 1.15 examines the resulting pressure on the modern economic subject, and Section 1.17 develops relational and generative conditions of subjecthood more directly.

Meaningful participation is therefore introduced here as an economic object whose scarcity cannot be inferred from unemployment statistics or income measures alone. Its empirical specification requires caution. Meaningfulness varies across persons and contexts, and GRE does not assume a single universal content. The economically relevant issue is whether institutional arrangements provide plural pathways through which participants can enter, sustain, and transform consequential relations. Scarcity appears where such pathways are narrow, heavily gated, or concentrated among actors already possessing relational advantage.

The transformation of scarcity described across this section can now be stated in historical terms. Material scarcity persists, while digitalization and AI produce relative abundance in selected symbolic and informational domains. This abundance raises the importance of attention, trust, institutional access, and generative opportunity as mediating scarcities. Economic analysis organized only around the possession and allocation of material goods therefore captures a declining share of the constraints that determine who can convert resources into consequential participation. Section 1.13 turns next to allocation under radical capability asymmetry, where these new scarcities interact with increasingly unequal productive capacities.

1.13 Allocation under Radical Capability Asymmetry

This section examines a historical allocation problem created when productive capability becomes highly unequal across human and artificial participants. Its role in the chapter is to connect the changing scarcity structure of Section 1.12 to the institutional mechanisms through which income, access, and economic participation are distributed. The objective is to distinguish five elements that are often conflated: employment as an allocation channel, comparative capability as a condition of competitive access, automation as a transformation of task demand, productive abundance as distinct from broad resource access, and capability asymmetry as a possible source of

economic power. The discussion proceeds historically and institutionally rather than as a forecast of aggregate employment. Artificial intelligence is treated as a contemporary case in which productive capability may become radically asymmetric while the allocation of livelihood and participation remains organized through institutions formed under earlier productive conditions.

1.13.1 Employment-Based Allocation

Wage employment became one of the central allocation mechanisms of industrial economies because it linked productive participation to command over resources. A worker supplied labor through an employment relation and received wages or salary that could be converted into food, housing, services, savings, and other goods. The arrangement also frequently mediated access to benefits, credentials, professional networks, legal status, and forms of institutional recognition. Employment therefore functioned as more than a contract for a discrete productive task. It connected a position within production to a wider bundle of claims upon economic resources.

The historical character of this arrangement is important. Polanyi's account of market society emphasizes the institutional construction required to treat labor as something organized through markets and contracts (Polanyi 1944). Sen's entitlement analysis makes a related point from the side of command over resources: access to goods depends on the entitlement relations through which a person can transform endowments, production possibilities, trade, and public transfers into effective command (Sen 1981). Wage employment is one such entitlement relation. Its importance arises from the way institutions connect labor contribution to purchasing power rather than from any timeless identity between work and livelihood.

Employment-based allocation also packages several relations together. The employer receives access to labor under organizational authority; the worker receives income; public institutions may attach taxation, insurance, pension rights, immigration status, or social protection to the employment relation; and households organize consumption and care around the resulting income stream. The economic significance of losing a job therefore depends on the institutional bundle attached to employment. Two economies with similar productive technologies can generate very different consequences from the same reduction in labor demand when healthcare, housing, social insurance, or legal membership are connected to employment in different ways.

This bundling made employment an efficient administrative recipient for allocation in periods when productive systems required extensive and continuing human labor. Income could be distributed through organizations already responsible for coordinating production, while participation in production supplied a socially legible basis for claims on output. The arrangement also created vulnerability because access to livelihood became dependent on continued inclusion in productive organizations. Historical labor politics, social insurance, and public provision can be read partly as responses to this vulnerability rather than as departures from an otherwise self-sufficient labor market.

Artificial intelligence makes the contingency of the employment-allocation linkage more visible. If a growing range of tasks can be performed with less human labor, the

economic question extends beyond the number of jobs created or destroyed. It concerns whether institutions continue to use employment as the primary route through which people receive income, institutional access, and opportunities for participation. The issue is therefore one of allocation architecture: a productive system can change faster than the institutions that connect productive contribution to resource claims.

1.13.2 Comparative Capability and Competitive Access

Competitive allocation depends on relative as well as absolute capability. Where access to jobs, contracts, finance, education, or professional positions is limited, applicants are compared against one another under institutionally defined criteria. Improvements in one person's productive capacity can therefore alter the competitive position of others even when their own abilities remain unchanged. The scarcity lies in the number and quality of positions, while capability influences who can enter them.

Human-capital theory formalized one important part of this structure by treating education and training as investments that can raise productivity and earnings (Becker 1964). The approach helps explain why individuals invest in capabilities that improve access to scarce economic positions. A relational analysis adds the competitive environment in which those capabilities are evaluated. The return to a skill depends on the distribution of comparable skills, the technologies with which the skill is combined, the institutions controlling entry, and the criteria by which organizations recognize productive contribution.

The capability perspective further distinguishes possession of resources from effective opportunity. Sen emphasizes that people differ in their ability to convert resources into substantive freedoms because personal, social, and institutional conditions differ (Sen 1999). In labor and professional markets, the corresponding conversion problem includes credentials, networks, geographic access, legal status, organizational membership, and technological complements. A person can possess knowledge without possessing a recognized pathway through which that knowledge becomes economically consequential.

Artificial intelligence alters comparative capability in a complex way. Experimental and workplace studies have found that generative AI can raise performance in selected writing and customer-support tasks, with especially large gains for some lower-performing or less-experienced workers (Noy and Zhang 2023; Brynjolfsson et al. 2025). In such settings, AI can compress certain within-task performance differences by diffusing practices or supplying assistance. This equalizing effect at the task level can coexist with wider inequalities in access to frontier systems, proprietary data, compute, organizational authority, and the ability to redesign work around the technology.

Comparative capability therefore remains institutionally mediated. The relevant economic unit is rarely a human capability considered in isolation or an artificial capability considered in isolation. Competitive access depends on combinations of persons, tools, organizations, infrastructures, and recognized claims. Radical capability asymmetry becomes economically significant when some actors can assemble these combinations more effectively than others or control the institutional inter-

faces through which combinations are admitted into competition.

1.13.3 Artificial Intelligence and Employment Scarcity

Artificial intelligence introduces a new form of uncertainty into employment-based allocation because it can substitute for selected cognitive and symbolic tasks while also complementing workers and generating new tasks. The task-based literature on automation distinguishes displacement effects from reinstatement effects: technologies can reduce labor demand in tasks they automate while new tasks and activities can restore labor demand elsewhere (Acemoglu and Restrepo 2019). The historical record therefore offers no basis for treating task automation as mechanically equivalent to aggregate technological unemployment.

The relevant problem for GRE is narrower and more structural. Employment scarcity can increase for particular groups, occupations, or periods when the number of positions capable of providing stable income and institutional membership grows more slowly than the number of people seeking such positions. AI can contribute to this condition when organizations reorganize production around machine performance and require fewer workers for tasks that previously justified large employment structures. New tasks, new industries, or increased demand can offset the effect, yet the offset depends on the speed, location, accessibility, and institutional form of the new opportunities.

Korinek and Stiglitz analyze AI and worker-replacing technological progress through distributional channels that include innovation rents, changes in factor prices, and possible forms of technological unemployment under specified conditions (Korinek and Stiglitz 2019). Their analysis is useful here because it separates greater productive capacity from the distribution of the resulting gains. A technology can increase aggregate output while changing bargaining positions, income shares, and the allocation of surplus. The distributive consequences depend on ownership, substitution patterns, policy, and institutional design rather than on productivity alone.

Current evidence also counsels against a simple capability hierarchy in which AI uniformly replaces weaker workers first. Brynjolfsson, Li, and Raymond find substantial productivity gains from an AI assistant in customer support, with larger gains among less-experienced and lower-skilled workers in their setting (Brynjolfsson et al. 2025). Such evidence suggests that AI can operate as a capability-diffusion mechanism inside organizations even while reducing the scarcity of some human task-specific skills. The same technology can therefore complement workers at one margin and weaken the market value of previously scarce expertise at another.

Employment scarcity should consequently be understood as a relational outcome rather than a direct property of AI capability. It depends on how quickly tasks are reorganized, which new roles appear, how income is connected to those roles, who can access complementary tools, and whether institutions create alternative routes to livelihood and participation. The historical significance of AI lies in its potential to weaken the inherited coupling between human productive necessity and resource allocation. Whether that potential is realized, and in which sectors or populations, remains an empirical question.

1.13.4 Productive Abundance and Resource Concentration

Greater productive abundance does not determine how control over productive assets or resulting income is distributed. A production system can generate rapidly expanding quantities of output while ownership of capital, infrastructure, intellectual property, market access, or organizational decision rights remains concentrated. The distinction becomes especially important when the marginal cost of selected outputs falls faster than the cost of building or controlling the systems that generate them.

Digital and AI production display this asymmetry clearly. Section 1.11 noted that frontier AI development remains concentrated in large organizations with access to substantial compute, data, engineering capacity, and financing, even as downstream use diffuses more widely (Stanford Institute for Human-Centered Artificial Intelligence 2026). The result is a layered productive structure: symbolic and cognitive services can become cheaper and more abundant at the user interface while the infrastructure capable of producing frontier systems remains scarce and institutionally concentrated.

A broader empirical precedent is provided by the literature on superstar firms. Autor and colleagues document rising sales concentration in several industries and find that industries with larger increases in concentration experienced larger declines in labor's share, consistent with reallocation toward highly productive firms with relatively low labor shares (Autor et al. 2020). The analysis does not establish an AI-specific causal mechanism. Its relevance here is structural: productivity growth, concentration, and changes in the distribution of income can coexist when technological and market conditions shift activity toward a small number of highly productive organizations.

Resource concentration can also persist through control of complementary assets. Access to an AI model may be widespread while valuable deployment requires proprietary data, customer relationships, legal permissions, computing infrastructure, distribution channels, or organizational authority. The productive capability of the model becomes economically consequential through these relations. Actors controlling scarce complements can therefore capture a larger share of value even when the immediate technological output is reproducible at low marginal cost.

This distinction complicates abundance narratives. Productive abundance can reduce prices, widen access to services, and create new opportunities, while simultaneously increasing the strategic importance of the remaining scarce complements. Economic analysis therefore needs to track both output abundance and the distribution of control over the conditions that make abundance possible. Section 1.18 later identifies this divergence as one reason entity-level measures of possession can miss changes in relational and generative structure.

1.13.5 Capability Asymmetry and Economic Power

Capability asymmetry becomes economic power through institutional relations that convert superior capacity into control over resources, options, or the conduct of others. A technically superior system does not automatically possess economic power. Power emerges when capability is connected to ownership, organizational authority,

infrastructure control, bargaining position, institutional recognition, or the ability to determine access to scarce complements. The distinction is central to GRE because it separates differences in what actors can do from differences in what actors can compel, withhold, authorize, or capture.

The institutional mediation can be seen in ordinary firms. Organizational law and asset partitioning help establish entities capable of owning assets, entering contracts, and separating pools of claims (Hansmann and Kraakman 2000). Institutions therefore determine which actors can legally command productive resources and which claims are recognized when conflicts arise. Similar mediation applies to artificial capability. An AI system's productive capacity is exercised through the organization that owns, operates, licenses, deploys, or integrates it. The economically relevant power may reside in the relation among model capability, organizational control, and institutional recognition rather than in the model considered by itself.

Radical human–AI capability asymmetry makes this mediation especially important. The economic value of advanced AI partly arises because its performance can exceed human performance in selected tasks. Eliminating every capability difference would also eliminate much of the productive reason for deploying the technology. The practical problem is therefore not the existence of asymmetry alone. It is the structure through which asymmetry is converted into dependency, bargaining power, exclusion, or concentrated control over future opportunities.

Several conversion pathways are possible. An organization that controls scarce AI infrastructure can convert technical capability into lower production costs, then into competitive advantage, market access, revenue, and further investment. A platform can convert informational and computational capability into control over ranking, matching, or access. A worker equipped with an AI assistant can convert machine capability into higher individual productivity when organizational rules permit use and recognize the resulting output. Different institutional arrangements can therefore produce different power distributions from similar underlying technologies.

The same logic applies in the opposite direction. Humans and institutions can retain substantial power over technically superior systems through ownership, legal authority, access controls, energy supply, deployment permissions, and the design of organizational objectives. Capability and power can therefore diverge in both directions. The degree of divergence is a property of the relational architecture connecting subjects, institutions, and productive systems.

This distinction completes the section's historical argument. Employment-based allocation, competitive access, AI-mediated task transformation, resource concentration, and institutional conversion together create a setting in which productive capability can become radically asymmetric without determining a single distributive outcome. Economic consequences depend on the relations that connect capability to claims, resources, institutions, and participation. Section 1.14 turns next to symbolic abundance and representational crisis, where expanding productive capacity changes the reliability of the symbolic interfaces through which those relations are recognized and coordinated.

1.14 Symbolic Abundance and Representational Crisis

This section examines a second consequence of contemporary productive transformation: the expansion of symbolic production changes the conditions under which economic actors rely on representations of persons, organizations, transactions, risks, assets, claims, and states of the world. Its role in the chapter is to connect the symbolic abundance introduced by digitalization and artificial intelligence to a problem of economic coordination. The objective is to distinguish five layers of that problem: the scale of symbolic production, the relation between symbols and referents, the institutional effects of classification, the growing importance of provenance and epistemic trust, and the consequences of representational instability for coordination. The discussion is historical and institutional. It treats representations as economically consequential interfaces whose reliability depends on practices of classification, verification, authorization, and interpretation.

1.14.1 Expansion of Symbolic Production

Modern economies have long depended on symbolic production. Contracts, accounts, prices, reports, ratings, certificates, legal identities, statistical indicators, inventories, forecasts, and organizational records allow dispersed actors to coordinate without direct access to every underlying event or relation. Digital infrastructures greatly expanded the volume, speed, and granularity of these representations. Section 1.10 described how platforms, continuous observation, and algorithmic systems intensified the production and use of data. Section 1.11 added a further change: generative systems can now produce linguistically and visually plausible symbolic artifacts at very low marginal cost.

The economic significance of this expansion lies in the changing ratio between symbolic production and the resources required to evaluate it. A report, explanation, image, recommendation, summary, classification, or synthetic record can be generated quickly. The work required to determine provenance, contextual adequacy, factual support, relevance, and institutional standing may remain costly. Symbolic abundance therefore enlarges the set of representations available to actors while also increasing the burden placed on selection and verification.

The distinction between symbolic output and knowledge is essential. Language models can generate fluent text whose surface form resembles ordinary human communication, yet fluency alone does not establish understanding, reference, evidential grounding, or truth. Bender and colleagues emphasize this gap in their critique of large language models, warning against treating statistical success in language generation as equivalent to semantic understanding (Bender et al. 2021). The relevance for economics extends beyond debates about artificial intelligence. Economic coordination increasingly proceeds through representations whose production cost can fall much faster than the cost of establishing their epistemic status.

This asymmetry changes the practical meaning of abundance. A larger supply of symbols can improve access to explanation, translation, drafting, visualization, and administrative communication. The same expansion can also increase duplication, low-quality synthesis, strategic fabrication, and competition for attention. The result-

ing problem concerns the institutional relation among production, filtering, verification, and use. Symbolic production becomes economically productive only when actors can establish which representations deserve reliance within a particular decision context.

The contemporary transformation therefore adds a new layer to scarcity. Section 1.12 argued that abundant information can increase the relative scarcity of attention. Symbolic abundance also increases the relative importance of infrastructures that authenticate sources, preserve context, establish responsibility, and connect representations to the relations they purport to describe. These infrastructures become part of the productive conditions of knowledge-intensive economic coordination.

1.14.2 Symbols and Referents

A representation becomes economically useful through a relation to something beyond its own symbolic form. A balance sheet refers to an organization's financial position; a credential refers to an institutional status; a price refers to a transactionally recognized valuation; a risk score refers to an inferred probability or classification; a written claim refers to an event, object, or relation that may be investigated independently. The relevant economic question therefore concerns the stability of the relation between symbolic form and referent.

That relation has never been automatic. Documentary practices, accounting rules, legal procedures, statistical classifications, professional standards, audits, and institutional routines have historically supplied the conditions under which a representation acquires standing. A monetary figure in an audited statement, for example, carries a different institutional status from the same numeral written in an unauthenticated message. The visible symbol may be identical while its relation to evidence, authority, and responsibility differs.

Artificial symbolic production increases the importance of this distinction because surface plausibility can be generated independently of a corresponding referent. A text can describe an event that did not occur; an image can depict a scene that was never captured; a synthetic profile can imitate the ordinary markers of an institutional identity; a generated citation can possess the form of scholarship without referring to an existing publication. The representational problem therefore concerns the mapping from sign to referent and the institutional conditions under which that mapping can be trusted.

The problem is broader than deception. A representation can be sincere and still detach from the structure it is meant to summarize. Statistical categories may lag behind changing economic relations. Administrative records may preserve a legal identity while control has shifted elsewhere. Organizational reporting may describe formally bounded entities whose productive processes depend on external networks. A model can continue to produce internally coherent outputs after the relations that once justified its parameters have changed. Representational instability therefore includes both fabricated symbols and historically outdated or structurally incomplete mappings.

GRE later develops a more explicit distinction between relational reality and manifestational reality. The historical purpose of the present section is narrower. It estab-

lishes why the distinction becomes economically urgent under contemporary productive conditions. When institutions coordinate through increasingly abundant representations, the quality of economic action depends increasingly on the reliability of the relations connecting those representations to the subjects, processes, assets, and events they mediate.

1.14.3 Classification and Administrative Reality

Economic representations frequently take classificatory form. Persons are classified as employed, unemployed, self-employed, insured, resident, eligible, creditworthy, skilled, dependent, or vulnerable. Organizations are classified by legal form, sector, industry, size, ownership, reporting status, and jurisdiction. Assets, liabilities, expenditures, incomes, transactions, and risks are likewise organized through standardized categories. Classification makes complex relations administratively tractable by assigning them to recognized types.

The historical sociology of quantification shows that these categories are institutional achievements. Desrosières traces how statistical conventions construct comparability by defining categories, coding rules, and procedures of aggregation (Desrosières 1998). Hacking's work on historical ontology likewise emphasizes that classifications participate in the historical formation of the objects and kinds through which institutions organize knowledge (Hacking 2002). The relevant implication for GRE is that administrative representation can participate in constituting the economic reality upon which later decisions operate.

The System of National Accounts provides a concrete example. Its concept of an institutional unit identifies entities capable of owning assets, incurring liabilities, making economic decisions, and entering contracts, and uses those units to organize sectoral accounts (European Commission et al. 2009). Such classifications are indispensable for macroeconomic observation. They also establish a particular resolution at which economic activity becomes visible. Relations that cross recognized units or operate through informal, distributed, or hybrid arrangements can be less visible within the same framework.

Classification acquires performative force when eligibility, taxation, reporting duties, access, benefits, or regulatory treatment depend on category membership. An employment classification can affect social insurance and labor protections. A credit category can alter borrowing conditions. A platform's ranking category can affect discoverability. A statistical category can shape which population is recognized as a policy target. The representation therefore enters the relation it describes by altering the conditions under which actors are treated.

This feedback makes classification economically generative in a limited but important sense. A category can stabilize expectations, coordinate institutions, and support large-scale administration. It can also redirect behavior toward the criteria used for classification. As categories become integrated into automated systems, the speed of this feedback can increase. Classification, decision, and subsequent data generation can occur in a repeated loop, allowing an administrative representation to shape the future observations from which it is later updated.

The resulting issue is neither a rejection of classification nor a preference for un-

structured description. Complex economies require abstractions. The analytical task is to identify the relations that an abstraction preserves, the relations it suppresses, and the institutional consequences produced when the abstraction becomes an operational category. Representational crisis emerges when categories remain administratively powerful while their fit with changing relational structures becomes increasingly uncertain.

1.14.4 Authenticity, Provenance, and Epistemic Trust

Symbolic abundance increases the economic importance of authenticity and provenance. Authenticity concerns the standing of an artifact or claim as the artifact or claim it purports to be. Provenance concerns the history of its production, modification, custody, and transmission. Epistemic trust concerns the willingness to rely on a representation within a decision process. These dimensions interact, yet each answers a different question.

The distinction is visible in contemporary provenance standards. The C2PA Content Credentials specification defines technical mechanisms for binding signed provenance assertions to digital assets, recording creation and editing information, and validating whether those assertions remain associated with the underlying content (Coalition for Content Provenance and Authenticity 2026). The specification also draws an important boundary: validation of provenance data does not by itself determine whether the represented claim is true or normatively acceptable. Provenance supplies evidence about source and transformation history; interpretation remains necessary.

This separation has direct economic significance. A cryptographically verified document can still contain a mistaken forecast. An authenticated identity can make a false statement. A valid model output can be based on inappropriate assumptions. Conversely, an artifact without machine-verifiable provenance can still be accurate. Reliable coordination therefore depends on several layers of trust: confidence in source identity, confidence in chain of custody, confidence in the procedure that generated the claim, and confidence in the claim's adequacy to the decision at hand.

Porter's history of quantitative objectivity shows that impersonal procedures and standardized measures became important partly because institutions required forms of judgment that could travel across distance and survive disputes over personal authority (Porter 1995). Digital provenance infrastructures can be understood as a contemporary extension of the same organizational problem. Where direct familiarity is unavailable, institutions construct portable signals through which strangers can coordinate. The contemporary difference lies in the scale and speed of symbolic production, which increase the number of artifacts for which such signals may be demanded.

Provenance itself can therefore become an economic resource. Actors able to establish trusted identity, verified histories, recognized credentials, and auditable production chains may obtain advantages in markets where surface form is easy to imitate. Verification infrastructures can also become sites of dependency when a small number of institutions control identity, certification, reputation, or access to trusted channels. The problem of epistemic trust thus connects symbolic abundance to the

power questions developed later in the book.

The historical shift is consequently double. Artificial systems reduce the cost of producing persuasive symbolic form, while institutions respond by increasing investment in provenance, authentication, audit, reputation, and trusted intermediation. Economic coordination moves toward a regime in which the ability to produce symbols is widespread, while the ability to establish their standing remains institutionally differentiated.

1.14.5 Economic Coordination under Representational Instability

Economic coordination depends on representations because actors routinely decide in the absence of direct access to the underlying relations they seek to govern. Firms rely on accounts and forecasts; creditors rely on identities, histories, and risk measures; consumers rely on labels, reviews, and reputational signals; governments rely on administrative records and statistical indicators; platforms rely on profiles, rankings, and behavioral traces. Representational instability therefore becomes an economic problem when actors can no longer assume that the cost and reliability of these interfaces remain roughly stable.

Several consequences follow. Verification becomes a larger component of transaction and coordination costs. Organizations may devote more resources to authentication, audit, provenance, identity management, model validation, and source assessment. Trusted intermediaries can gain strategic importance because they reduce uncertainty for participants who cannot verify every claim independently. At the same time, concentration around trusted intermediaries can create new dependencies and bottlenecks. The infrastructure that restores trust can itself become a source of economic power.

Representational instability also affects temporal coordination. Decisions are often made before verification is complete. A false or poorly grounded representation can influence prices, hiring, credit, procurement, or public response before correction arrives. Later correction may fail to reverse the relational consequences already produced. The economic effect of a representation therefore depends partly on the speed with which it is produced, circulated, believed, acted upon, challenged, and revised.

Classification feedback adds a further layer. Once an institution acts on a representation, the action changes subsequent data. A risk classification can alter access to credit; altered access changes later repayment histories; those histories then become inputs to subsequent classification. A ranking can redirect attention; redirected attention changes observed popularity; popularity then reinforces the ranking. These loops make representational systems participants in the processes they measure.

The resulting crisis is best understood as a problem of alignment among symbolic production, referential reliability, institutional classification, and practical trust. The term *crisis* refers here to intensified instability in that alignment rather than to universal collapse of representation. Many representations remain highly reliable, and digital systems can improve traceability, consistency, and access. The historical change lies in the simultaneous expansion of symbolic productive capacity and the growing economic importance of institutions that determine which representations

can be trusted, acted upon, and converted into recognized claims.

This shift prepares the next problem in the chapter. Economic subjects themselves are partly stabilized through representational systems: occupational titles, employment records, credentials, income categories, organizational roles, legal identities, and measures of contribution all participate in defining how persons and organizations appear within economic institutions. When productive relations and representational systems change together, established subject positions can lose part of the institutional support that previously made them stable. Section 1.15 therefore turns from representational instability to the destabilization of the economic subject.

1.15 Destabilization of the Economic Subject

This section examines how contemporary productive transformation unsettles the institutional relations through which modern economic subjects have been stabilized. Its role in the chapter is to connect the preceding transformations of production, scarcity, allocation, and representation to a problem of subject formation. The objective is to identify the historical relations among economic roles, work, social recognition, productive necessity, and participation, and then to clarify why artificial intelligence places pressure on arrangements that have tied material access and social standing to comparative productive contribution. The discussion proceeds historically and institutionally. It first reconstructs economic roles as socially organized positions, then examines work as a source of recognition and membership, distinguishes productive contribution from productive necessity, analyzes the possibility of productive redundancy, and closes by identifying forms of economic participation that need not depend on productive superiority.

1.15.1 Economic Roles and Subject Formation

Modern economic institutions do more than allocate resources among pre-existing persons and organizations. They also assign recognizable positions through which participants become legible as workers, employers, owners, consumers, creditors, debtors, taxpayers, dependants, professionals, and members of organizations. Section 1.4 reconstructed these positions as historically formed economic subjects, while Section 1.5 showed how law, accounting, statistics, and administrative classifications stabilize the entities to which those positions attach. The present subsection develops the further implication that repeated participation in such roles can become part of the practical continuity through which persons understand their place in economic life.

An economic role carries a bundle of expectations, rights, obligations, resources, and relations. Employment connects a worker to wages, working time, organizational authority, occupational identity, social insurance, and often a trajectory of promotion or skill formation. Ownership connects an actor to control, exclusion, residual claims, and asset-based security. Professional status connects expertise to credentials, jurisdiction, reputation, and access to specialized work. Household positions connect provision, care, dependence, and intergenerational transfer. These arrangements differ across legal systems and historical periods, yet they share a general structure:

economic subject positions are reproduced through institutional relations that make some forms of participation recognizable and consequential.

This relational formation does not imply that persons are exhausted by economic roles. A person can remain a parent, friend, citizen, learner, creator, believer, neighbor, or participant in many other relations whose meaning exceeds economic classification. The analytical point concerns the historical importance of economic roles for access and recognition. In industrial and post-industrial societies, employment status, occupation, income, property, organizational affiliation, and professional credentials have often affected where a person can live, which institutions become accessible, how risk is borne, and which forms of contribution become publicly visible. Economic roles therefore participate in the stabilization of subject positions even though they never constitute the whole of subjectivity.

Polanyi's historical account of market society is useful here because labor, land, and money became increasingly organized through market institutions that restructured the relation between livelihood and social organization (Polanyi 1944). Marx's analysis of wage labor likewise places labor-power inside a historically specific relation in which access to the means of livelihood is mediated through the sale of labor capacity and the organization of production under capital (Marx 1867). The theoretical traditions differ substantially, yet both make visible a common historical fact: participation in production became deeply connected to the institutions through which subjects acquired income and occupied recognizable social positions.

The contemporary problem arises when the relations that stabilize those roles change faster than the categories through which subjects remain institutionally recognized. A person can retain knowledge, experience, responsibility, and the capacity for meaningful action while the labor market value of a previously recognized role declines. An organization can retain formal identity while its productive function is redistributed across platforms, contractors, automated systems, and external infrastructures. A credential can continue to certify a competence even as the task structure associated with that competence changes. Subject destabilization therefore begins as a mismatch between historically stabilized roles and changing productive relations.

The economic subject is consequently vulnerable to transformations that leave the underlying person or organization intact. The worker can persist while the job disappears; the professional can persist while the exclusive jurisdiction of a profession erodes; the firm can persist while productive control migrates toward infrastructure providers; the household can persist while the employment structure supporting its income becomes less stable. This distinction prepares the analysis of recognition in the next subsection. What becomes unstable is often the relation through which a subject is economically recognized, rather than the existence of the subject itself.

1.15.2 Work and Social Recognition

Work has historically supplied more than output. It can connect persons to income, routine, organization, skill formation, collective membership, reciprocal dependence, and publicly recognizable contribution. The relative importance of these dimensions varies widely across occupations and societies, and many forms of socially necessary activity remain unpaid or weakly recognized. Even so, the institutional centrality

of employment means that changes in work can affect both material access and the social recognition attached to participation.

The material relation is direct. Section 1.6 described employment as a major channel through which households receive claims on resources. Wages can determine access to housing, credit, insurance, savings, mobility, and many forms of consumption. Employment can also serve as an administrative gateway to benefits and legal protections. Where these arrangements are strong, work becomes one of the principal institutional bridges between productive participation and livelihood.

The recognitional relation operates differently. A role becomes socially recognizable when institutions and communities possess a vocabulary for identifying its contribution. Occupational titles, credentials, organizational positions, portfolios, rankings, and professional standards make contribution visible in standardized ways. Recognition can therefore depend on representation as well as performance. Section 1.14 showed that representational systems participate in economic coordination; they also participate in subject formation by stabilizing the categories through which contribution is noticed and attributed.

This dependence creates asymmetries among forms of work. Market-mediated labor often leaves records of hours, wages, outputs, contracts, and titles. Care, informal coordination, emotional support, household maintenance, community work, mentoring, and many forms of relational labor may generate substantial capacities while remaining weakly represented in conventional accounts. The economic subject produced through employment institutions therefore reflects a selective architecture of recognition. Activities become more visible when they enter established categories of paid contribution, while other relations can remain economically consequential without receiving equivalent institutional standing.

Sen's capability approach helps clarify the stakes because material resources matter through the substantive opportunities they enable rather than through possession alone (Sen 1999). Participation in work can be valuable partly because it supports income, but also because it can expand capabilities through learning, association, mobility, voice, and institutional membership. The same perspective also permits a distinction between work and the broader capabilities work sometimes supports. If technological change weakens the employment channel, the relevant loss cannot be assessed adequately by asking only whether income can be replaced. The consequences may concern access to relations, routines, opportunities for contribution, and forms of recognition previously carried by employment.

This point gives subject destabilization a wider scope than unemployment. A person may remain employed while experiencing reduced autonomy, weaker occupational jurisdiction, diminished bargaining power, or increasing dependence on algorithmic systems. Conversely, a person outside paid employment may possess rich capacities and participate in relations of care, learning, creation, and community life. Economic subjecthood therefore becomes unstable when inherited institutions continue to use employment as a privileged signal of contribution while productive and socially valuable activity becomes more heterogeneous.

Artificial intelligence intensifies this problem because it can alter the informational basis on which professional recognition rests. When generative systems assist writ-

ing, coding, analysis, design, diagnosis, or customer interaction, observed output becomes increasingly co-produced by persons, models, organizations, datasets, and infrastructures. Brynjolfsson, Li, and Raymond find that generative-AI assistance can increase worker productivity while producing heterogeneous gains and forms of knowledge diffusion (Brynjolfsson et al. 2025). The broader implication is that contribution becomes more relational: the performance attributed to an individual may depend increasingly on access to shared artificial capabilities. Recognition systems organized around isolated individual productivity may therefore become less informative about the relations that actually generate performance.

The historical role of work in social recognition should consequently be separated into its component relations. Employment can provide income, status, routine, membership, learning, bargaining position, and visibility of contribution. Technological transformation can affect these components differently. A future institution may preserve income while weakening membership, preserve work while changing autonomy, or expand productive capacity while concentrating recognition in those who control infrastructures. The economic problem concerns the reorganization of these relations rather than a single binary distinction between employment and unemployment.

1.15.3 Production and Social Necessity

The relation between production and subjecthood becomes especially important when productive capability grows faster than the need for particular forms of human contribution. This subsection distinguishes productive contribution from productive necessity. A person can contribute to production without being indispensable to it, and a society can possess the technical capacity to produce more output while reducing the number of human tasks required for a given production process. The distinction matters because modern allocation and recognition systems have often tied access to resources to the continuing demand for labor.

Industrial production already contained this tension. Mechanization increased output while reorganizing the demand for particular skills and tasks. The task-based framework developed by Acemoglu and Restrepo describes technological change through a balance between displacement, in which capital or automation performs tasks previously assigned to labor, and reinstatement, in which new tasks create new domains for labor demand (Acemoglu and Restrepo 2019). This framework is useful for GRE because it avoids treating technical capability as a direct forecast of employment. Productive transformation changes the task structure, while institutions, demand, complementary innovations, and new activities shape the resulting labor relation.

Artificial intelligence extends the same problem into domains of cognitive and symbolic production discussed in Section 1.11. The relevant change is the widening set of tasks for which machine systems can provide useful substitutes or complements. Writing, translation, coding, search, classification, planning, image production, and analytic support can increasingly be performed through human-AI configurations. The resulting productivity gains can expand output, lower costs, generate new demand, and create new roles. They can also reduce the amount of human labor re-

quired for some existing outputs. Both directions are compatible with the same technological advance.

The concept of social necessity adds another layer. A task may cease to be technically necessary for producing a given output while remaining socially meaningful. Human teaching, care, artistic performance, deliberation, companionship, craftsmanship, or face-to-face service can retain value even when some associated informational or productive functions are automatable. The persistence of such activity depends on institutions of valuation and participation as much as on comparative technical efficiency. Productive necessity and social desirability therefore belong to different analytical dimensions.

This distinction prevents a narrow interpretation of technological progress. If machine systems can produce a report faster than a person, the technical relation concerns the production of the report. The economic and social relation also concerns who learns through producing it, who is recognized for judgment, who bears responsibility for its consequences, who retains access to the relevant institution, and which capabilities are reproduced through repeated practice. An output-equivalent substitution can therefore change the relations surrounding production even when the immediate product appears unchanged.

The implications for subject formation are substantial. Modern societies have often connected the worth of economic participation to productive necessity: a worker receives income because an organization requires the worker's labor, and the need for that labor supports bargaining position and recognition. As artificial productive capacity expands, that connection can weaken in some domains. Technical abundance may make a person's participation less necessary for output without making the person's experience, judgment, relationships, or potential contribution meaningless. A tension then emerges between the organization of production and the organization of subject recognition.

Korinek and Stiglitz emphasize that the distributional effects of artificial intelligence depend on how technological change affects factor returns, innovation rents, ownership, and labor demand (Korinek and Stiglitz 2019). The implication for the present argument is institutional. Productive capability alone does not determine who receives resources or recognition. The connection between production and livelihood is mediated through property, employment, bargaining institutions, taxation, public provision, and other mechanisms of allocation. Productive necessity can therefore decline before the institutions built around it have been replaced or reconfigured.

The resulting subject problem is historically specific. Industrial modernity stabilized a powerful chain linking work, contribution, income, and social position. Contemporary artificial productive capacity can loosen individual links in that chain at different rates. Economic theory therefore needs concepts capable of distinguishing output production from capability formation, necessity from participation, and technical substitution from the broader relational functions historically carried by work.

1.15.4 Productive Redundancy and Subjective Instability

Productive redundancy names a condition in which an actor's participation becomes less necessary for achieving an established productive objective because comparable or superior output can be generated through other configurations. The term refers to a relation between an actor and a production system, rather than to a judgment about the actor's intrinsic worth. This distinction is central to the section. An economic system can treat a contribution as redundant while the person remains fully capable of experience, learning, care, judgment, creation, and participation in other relations.

Redundancy can arise gradually. A task may first receive machine assistance, then become standardized, then shift toward supervision, and eventually require little direct human production. It can also arise unevenly across workers. Access to artificial tools may allow some employees to perform a wider range of tasks, reducing demand for specialized positions elsewhere. Organizational redesign can redistribute tasks among workers, models, platforms, and external contractors. The resulting instability therefore concerns the configuration of productive relations rather than a uniform technological shock.

The subjective consequences depend on how strongly a person's economic continuity has been organized around a productive role. Where income, professional identity, daily routine, social network, institutional membership, and perceived contribution are concentrated in employment, the weakening of that role can produce several forms of instability at once. Material insecurity may be accompanied by a loss of recognized position, reduced opportunities for skill use, and uncertainty about the relation between personal capability and social need. These effects can occur even when aggregate production rises.

The concept should remain analytically restrained. Productive redundancy is a possibility generated by changing task relations; it is not a forecast that human labor as a whole will become unnecessary. Acemoglu and Restrepo's displacement–reinstatement framework explicitly leaves room for new tasks and new forms of human comparative advantage (Acemoglu and Restrepo 2019). Artificial intelligence may also create complementary roles whose content cannot yet be specified. GRE therefore treats redundancy as a local and historically variable relation that can expand, contract, or transform as institutions and production systems evolve.

Even a local form of redundancy can expose a deeper problem in allocation. If access to income and social resources depends heavily on being competitively necessary to production, a decline in necessity can become a decline in material command. Section 1.13 described the institutional channels through which capability asymmetry can affect allocation. Subjective instability adds the consequence that the same institutions also influence how persons understand their place in economic life. A system can therefore preserve productive abundance while generating insecurity among participants whose recognized roles weaken.

Income replacement addresses only part of this structure. Transfers, social insurance, public provision, or asset ownership can protect material access when employment falls. Such arrangements may be indispensable, yet the relations previously carried by work can remain unresolved. A person can possess income while lacking a recognized field of contribution; possess leisure while lacking access to commu-

nities of practice; possess consumption opportunities while lacking participation in decisions that shape shared institutions. The difference between material security and relational participation is therefore analytically important.

This distinction also constrains simplistic claims about a post-work economy. Reduced labor requirements can expand freedom from undesirable work, create time for care and learning, and weaken forms of domination tied to economic necessity. The same reduction can intensify exclusion if institutions continue to allocate recognition, voice, and access through employment. The consequences depend on which relations replace the employment nexus and how those relations distribute opportunities for meaningful participation.

The historical significance of artificial intelligence therefore lies partly in the possibility that productive abundance and subjective insecurity can coexist. A society may become technically capable of producing more with less human labor while retaining institutions that interpret reduced labor demand as reduced social claim. The tension arises from the lag between productive transformation and the institutional forms through which subject positions are recognized. This lag motivates the final subsection's focus on participation beyond productive superiority.

1.15.5 Economic Participation beyond Productive Superiority

The destabilization of inherited subject positions creates a positive analytical task: economic participation needs to be distinguished from productive superiority. Modern institutions have often rewarded actors who possess scarce capabilities, control productive assets, or occupy roles that organizations require. These mechanisms remain important, yet they provide an incomplete basis for subjecthood when artificial systems can exceed human performance across expanding sets of tasks. A relational economy therefore requires concepts of participation that remain intelligible even when contribution cannot be justified by superior output alone.

Sen's capability perspective offers one foundation because it evaluates substantive opportunities rather than reducing well-being to resource possession or commodity command (Sen 1999). Applied to participation, the relevant question concerns which relations persons are effectively able to enter, sustain, and shape. Access to education, public institutions, collaborative work, culture, care, research, community life, and collective decision-making can matter because these relations expand what persons are able to become and do. Their value need not depend on proving that human performance exceeds the best artificial alternative.

Economic participation can therefore be analyzed through several distinct dimensions. Material participation concerns access to resources and infrastructures required for life. Productive participation concerns contribution to the production and maintenance of goods, services, organizations, and shared capacities. Epistemic participation concerns access to inquiry, learning, interpretation, and the production of knowledge. Institutional participation concerns membership, voice, and the ability to influence rules. Relational participation concerns access to durable relations through which trust, care, identity, cooperation, and future opportunities can be generated. These dimensions can overlap while remaining analytically separable.

The distinction matters because technological systems affect them differently. Arti-

ficial intelligence may reduce the labor required for a productive task while increasing opportunities for learning, experimentation, or collaborative creation. It may improve access to knowledge while concentrating control over computational infrastructure. It may enable small organizations to perform functions previously requiring large staffs while increasing dependence on a few platform providers. Evaluation therefore requires tracing which forms of participation expand, which contract, and which become conditional on new intermediaries.

Participation also has a temporal dimension. A relation can be valuable because it generates later capabilities that were unavailable at the moment of entry. Apprenticeship, education, research collaboration, caregiving, community membership, and participation in institutions can produce skills, trust, confidence, reputation, and future opportunities. A purely output-centered account can undervalue such relations when their immediate productivity is modest. GRE later develops this problem through the concept of generativity. The historical point here is that the weakening of productive necessity makes these longer-horizon relational effects more visible as economic questions.

The same perspective changes the interpretation of artificial capability. If machine systems outperform humans on a task, the comparison answers a question about task performance under specified conditions. It does not determine whether humans should cease to participate in every relation organized around that task. Education offers the clearest example: a system may know more than a learner while the learner's engagement with a subject still generates human capability. Similar distinctions may arise in art, research, care, deliberation, and craftsmanship, where participation itself can transform the participant and the surrounding relational system.

This approach preserves the role of productive efficiency while limiting its jurisdiction. Efficiency remains relevant when a goal is well specified and output can be compared meaningfully. Subject formation, capability development, institutional membership, and relational continuity introduce additional objects of economic evaluation. A society facing radical artificial capability therefore confronts a problem of institutional design: how to organize access, contribution, recognition, and participation when comparative productive superiority no longer provides a stable general foundation for each of them.

The next section develops the relational basis required for that problem. Section 1.16 examines economic subjecthood as a relatively stable configuration generated and sustained through social and institutional relations. The move does not assign final ontology to the subject. It identifies the relational conditions under which subjects acquire sufficient persistence, recognition, and causal efficacy to become objects of economic analysis.

1.16 Relational Constitution of Economic Subjecthood

Section 1.15 identified a historical strain in the modern alignment among productive contribution, income, recognition, and subject position. The present section provides the conceptual bridge required by that diagnosis. Its role is to reconstruct economic subjecthood through the relations that generate, stabilize, and recognize an actor as

a participant in economic life. The objective is deliberately limited: the section develops a relational account sufficient for economic analysis while reserving stronger metaphysical questions for the ontological discussion in Chapter 4. The argument proceeds through five steps. It first locates subject formation within social relations, then examines the mechanisms that stabilize economic actors, turns to institutional recognition, states a minimal account of relational reality under ontological suspension, and finally considers how the framework applies to human and artificial economic subjects. The method is historical-institutional and analytical. It asks what practical conditions allow an actor to persist as a recognizable bearer of roles, resources, obligations, capacities, and consequences across time.

1.16.1 Social Relations in Subject Formation

Economic subjecthood becomes intelligible once the subject is approached through the relations in which economic capacities acquire determinate form. A worker is constituted economically through relations of employment, remuneration, authority, skill recognition, and social reproduction. An owner occupies relations of title, exclusion, transfer, inheritance, and control. A consumer appears through monetary access, market exchange, product classification, credit, and infrastructures of distribution. A household becomes actionable through membership, care, income pooling, taxation, welfare rules, and statistical recognition. These positions differ sharply in content, yet they share a structural feature: the economic subject is recognizable because a network of relations gives the position continuity and practical consequences.

This relational reconstruction extends the historical argument developed earlier in the chapter. Economic action has long been embedded in ongoing structures of social relations, and Granovetter's critique of atomized economic action remains useful precisely because it directs attention to the social structures through which apparently individual conduct becomes possible and consequential (Granovetter 1985). The relational point can be taken further without assuming that every feature of a person is socially produced. Economic subjecthood concerns a domain-specific position. The person who occupies that position exceeds it, while the position itself is formed through relations that make certain actions, claims, obligations, and resources available.

The distinction between bearer and position prevents two forms of analytical compression. One compression treats the subject as a self-contained unit whose economically relevant capacities are already present before interaction. The other treats the subject as fully exhausted by the relations currently observed. A relational account can preserve a practical identity while recognizing that the capacities attached to that identity are historically conditioned. Skills are acquired through education and work; creditworthiness is constituted through records and institutional judgment; ownership depends on recognized title; occupational identity depends on social classification and organizational demand; household responsibility depends on membership rules. These capacities become attributes of actors in practice even though their production is relational.

Historical categories participate in this formation. Hacking's historical-ontological work emphasizes that classifications, descriptions, and styles of reasoning can par-

ticipate in bringing into view kinds of persons and objects that become available for thought and action (Hacking 2002). In economic life, categories such as employee, unemployed person, shareholder, debtor, beneficiary, incorporated firm, or dependent household member do more than summarize pre-existing properties. They organize eligibility, obligation, comparison, and institutional response. Once repeatedly used, a category can become part of the relational environment through which actors understand their position and through which others act upon them.

Subject formation therefore has a recursive structure. Relations generate positions; positions organize expectations and conduct; conduct reproduces or modifies the relations that generated the positions. This recursion explains why economic subjects can appear stable while remaining historically transformable. A worker can persist as a recognizable subject across many transactions because employment law, payroll systems, occupational classifications, organizational routines, and social expectations stabilize the position. Changes in technology, law, family structure, or organizational form can later alter the same position without requiring the person occupying it to disappear.

The relational constitution of subjecthood also clarifies why new economic actors can emerge historically. Platform sellers, data workers, content creators, gig workers, cooperative members, algorithmic service providers, and other contemporary positions become economically salient when repeated relations stabilize distinctive expectations, flows, rights, risks, and forms of recognition around them. The emergence of a subject position can therefore be studied historically through the relations that make it durable enough to enter allocation, accounting, regulation, and collective interpretation.

1.16.2 Relational Stabilization of Economic Actors

Subject formation alone does not explain persistence. Economic analysis ordinarily requires actors that can be followed across transactions, held responsible for obligations, assigned resources, and recognized by counterparties. Relational stabilization names the process through which a changing configuration becomes sufficiently continuous to support those practices. The relevant stability is practical and historical. It can be strong enough for accounting, contracting, policy, and statistical analysis while remaining open to transformation.

Several mechanisms contribute to stabilization. Repetition is the most elementary. Recurrent interaction allows counterparties to form expectations about conduct, reliability, skill, payment, or role performance. Records extend repetition across time by making prior actions institutionally available. Contracts and property titles preserve obligations beyond the moment of agreement. Organizational routines preserve roles across personnel changes. Membership rules define who remains inside a household, association, cooperative, or firm. Trust lowers the need to renegotiate every relation from the beginning. Infrastructures such as payment systems, registries, identity systems, and digital platforms provide technical continuity across otherwise separate encounters.

Stabilization also depends on complementarity. A position becomes durable when other positions are organized around it. The worker position is sustained by employ-

ers, payroll systems, labor law, training institutions, tax systems, and household expectations. The firm is stabilized by workers, suppliers, customers, creditors, regulators, accounting practices, and legal recognition. The consumer position depends on sellers, money, distribution, standards, information, and rights of redress. The durability of an actor is therefore partly a property of the surrounding configuration. A subject can lose economic continuity when complementary relations dissolve even if the bearer remains materially unchanged.

This point is especially important for collective actors. A corporation may survive the departure of founders, managers, workers, investors, and customers because the relations composing the organization are reproduced through roles, records, procedures, assets, contractual continuity, and external recognition. Section 1.5 showed how organizational law and accounting provide institutional mechanisms for such persistence. Relational stabilization adds the wider economic condition: counterparties continue to orient their conduct toward the organization as one participant. The collective actor acquires continuity because many relations converge on a sufficiently stable reference point.

Stabilization admits degrees. Some subject positions are highly standardized and portable across institutions; others depend on a narrow local relation. An incorporated firm with registered assets, audited accounts, recurring contracts, and regulatory standing possesses stronger institutional continuity than an informal collaboration assembled for a single project. A household can be stable for purposes of taxation and unstable for purposes of care or resource sharing. A digital identity may be persistent within one platform and irrelevant outside it. Economic subjecthood therefore varies with the domain, scale, and relation under examination.

The degree structure matters for contemporary transformation. New technologies can produce economically consequential actors faster than legal and statistical systems can stabilize them. A software agent can repeatedly perform purchasing, pricing, recommendation, or coordination functions while formal responsibility remains assigned to a human or organization. A decentralized collective can coordinate resources before established law provides a clear organizational category. Relational stabilization can therefore precede full institutional recognition, creating a period in which practical economic agency and formal subject status diverge.

1.16.3 Institutional Recognition of Subject Positions

Institutional recognition converts relational stabilization into forms that large-scale economic systems can administer. Law, accounting, taxation, statistics, welfare systems, financial institutions, and organizational procedures require identifiable units to which rights, obligations, resources, liabilities, and observations can be attached. Recognition gives a subject position an address within these systems. It also changes the position by creating consequences that follow from being classified in one way rather than another.

Modern economic administration offers clear examples. The System of National Accounts defines institutional units through capacities such as owning assets, incurring liabilities, making economic decisions, and engaging in transactions in their own right (European Commission et al. 2009). Financial reporting similarly constructs re-

porting entities whose boundaries can include a legal entity, part of an entity, or several entities, depending on the informational purpose of the report (IFRS Foundation 2018). Organizational law stabilizes further distinctions by partitioning assets and creditor claims around recognized organizational forms (Hansmann and Kraakman 2000). These frameworks do not merely discover a universal natural boundary. They create standardized interfaces through which heterogeneous relational configurations become comparable and actionable.

Recognition has distributive consequences. Employment status can determine access to wages, insurance, leave, taxation, and legal protection. Household classification can alter benefit eligibility and tax liability. Corporate status can affect asset ownership, limited liability, financing, and continuity. Residency, citizenship, professional certification, and organizational membership can determine access to institutions and resources. Recognition therefore participates in the production of economic capacity. A subject position that enters an institutional category can acquire options unavailable to an otherwise similar actor outside it.

The same process can exclude. Relations that do not fit established categories may remain economically consequential while receiving weak institutional visibility. Informal care, unpaid collaboration, platform-mediated labor, collective knowledge production, or cross-border digital activity can generate value and obligations without aligning neatly with inherited administrative units. The mismatch does not make these relations unreal. It affects whether economic systems can see them, allocate to them, regulate them, or assign responsibility through existing procedures.

Institutional recognition is therefore performative in a bounded sense. A classification can change access, incentives, responsibility, and behavior because institutions act through it. Its efficacy still depends on relations beyond the classification itself. Declaring an organization independent does not produce suppliers, trust, productive capacity, or internal coordination. Recognizing a worker category does not determine the actual distribution of bargaining power or care burdens. Recognition becomes economically effective when classificatory rules connect to infrastructures, resources, expectations, and enforcement.

This bounded performativity provides a useful bridge to the later discussion of manifestation. Economic systems often encounter subjects through institutional representations: accounts, legal statuses, employment records, credit files, identifiers, statistical categories, or platform profiles. These manifestations can stabilize action while preserving only part of the underlying relational structure. The analytical task is therefore twofold: to take institutional recognition seriously as a causal component of economic subjecthood and to preserve the possibility that the recognized manifestation leaves economically relevant relations unresolved.

1.16.4 Relational Reality under Ontological Suspension

The preceding analysis permits a minimal ontological stance suitable for the historical argument of this chapter. An economic subject can be treated as relationally real when a configuration is generated and sustained through relations, persists sufficiently to support recurrent interaction, and exercises causal effects within the economic system. This criterion concerns analytical standing rather than ultimate meta-

physical substance. Chapter 4 develops the distinction more fully through ontological suspension, relational reality, and manifestational reality.

Ontological suspension is useful because the economic subjects considered in this chapter are heterogeneous. A person, household, corporation, cooperative, state agency, platform account, and software agent do not need to share one ultimate mode of being in order to participate in economically consequential relations. Economic analysis can ask how each is generated, recognized, stabilized, and causally effective while leaving deeper questions about consciousness, personhood, identity, or metaphysical composition open where the economic argument does not require their resolution.

Relational reality also guards against premature reification. A stabilized subject can be treated as a unit for a particular analytical purpose while its boundary, capacities, and continuity remain dependent on relations. The firm can be a real contracting and accounting subject even though its productive capacity depends on workers, suppliers, law, finance, infrastructure, and knowledge networks. The household can be a real administrative unit even though its internal distribution cannot be inferred from aggregate resources. A worker can be a real employment subject while occupational identity and earning capacity remain historically generated. Reality at the relational level therefore supports unit-level analysis without converting provisional boundaries into timeless essences.

The same stance preserves practical identity. Relational constitution does not require economic analysis to dissolve every subject into an unbounded network. Contracts, liabilities, rights, welfare claims, preferences, plans, and histories frequently attach to identifiable actors. The question is how those identities are sustained and what relations make them economically effective. A relational account can therefore move between levels: it can treat the subject as a unit where the unit is causally and institutionally stabilized, and it can reopen the unit when internal or external relations become necessary for explaining an outcome.

This level-sensitive stance is especially important for value and responsibility. Economic systems often need to know who receives a transfer, who owes a debt, who controls an asset, who bears a risk, or who is accountable for a decision. These questions require subjects. GRE later asks how resources and value move through the relations composing and surrounding those subjects. Relational reality allows both inquiries to proceed without forcing one level to replace the other.

Ontological suspension also supports theoretical pluralism. Different disciplines may describe the same economic participant through different ontological vocabularies. Law may focus on personhood and responsibility; sociology on role and recognition; psychology on agency and cognition; computer science on system architecture; economics on choice, production, and allocation. GRE can use relational criteria to compare these descriptions where they intersect economically while allowing each domain to retain questions that exceed economic analysis. This discipline becomes decisive when artificial systems enter the picture.

1.16.5 Human–Artificial Economic Subjecthood

Artificial intelligence intensifies the subjecthood problem because economically consequential activity can increasingly be produced by systems whose legal, moral, cognitive, and institutional status differs from that of human participants. Section 1.11 showed that contemporary AI systems already participate in cognitive production, symbolic generation, workplace assistance, and organizational coordination. The question here concerns the analytical conditions under which such participation supports an economic subject position.

Four dimensions should be separated. Productive agency concerns the capacity to perform economically relevant tasks or alter outcomes. Relational continuity concerns whether counterparties repeatedly orient their conduct toward the system or toward a stable role mediated by it. Institutional recognition concerns whether rights, liabilities, accounts, contracts, or administrative consequences can be attached to the position. Normative standing concerns questions such as welfare, rights, responsibility, consciousness, or moral consideration. These dimensions can vary independently. A system may possess substantial productive agency and relational continuity while institutional and normative standing remain assigned to the human or organizational actors surrounding it.

The distinction prevents economic efficacy from carrying an ontological burden it cannot bear. Current language models can generate fluent symbolic output, while fluency itself does not establish understanding or meaning in the strong sense cautioned against by Bender and colleagues (Bender et al. 2021). Economic analysis nevertheless has reason to follow the causal role of such systems when organizations change staffing, prices, workflows, search processes, or service provision in response to them. Productive and relational effects can therefore be studied without treating those effects as evidence of human-equivalent cognition.

Institutional mediation remains central. A software agent that purchases goods, schedules labor, executes trades, recommends credit, or negotiates within a platform usually acts through accounts, permissions, assets, and liabilities ultimately connected to recognized persons or organizations. The economic subject may therefore be layered: an artificial system can occupy an operational role, a firm can bear legal responsibility, workers and users can supply inputs, and platform owners can control the infrastructure. The relevant unit depends on the question being asked. Production analysis may focus on the artificial agent's task performance; liability analysis may focus on the organization; power analysis may focus on ownership and infrastructure; welfare analysis may focus on affected humans.

This layered structure also clarifies why human and artificial economic subjects should not be treated as automatically equivalent. Human economic subjecthood is embedded in embodiment, vulnerability, social reproduction, capability development, legal rights, and forms of recognition that exceed productive output. Sen's capability perspective is relevant because it locates evaluation in substantive opportunities and the institutional conditions under which persons can act and become (Sen 1999). Artificial systems may participate powerfully in economic production without sharing the same capability, welfare, or rights framework. A relational analysis can compare economically relevant functions while preserving these normative differ-

ences.

Artificial systems can nevertheless acquire increasingly stable subject-like positions within particular economic domains. Recurrent identity, persistent memory, delegated authority, autonomous task execution, access to resources, and counterparties that adapt to system behavior can create a durable operational position. Institutional evolution may later attach new forms of recognition to such positions. The trajectory is an empirical and legal question rather than a conclusion established by technical capability alone.

The wider significance for GRE is methodological. Economic subjecthood should be analyzed through the relations that generate and stabilize participation rather than through a single universal test of entity status. Human persons, collective organizations, and artificial systems may occupy different subject positions, with different degrees of persistence, recognition, agency, and normative standing. The framework therefore preserves heterogeneous subjecthood while retaining enough structure for allocation, production, power, and responsibility to remain analyzable.

Section 1.17 takes the next step. Once subjects are understood as relationally stabilized, the central question shifts from the persistence of a subject to the capacity of relations to transform subjects, generate capabilities, alter future possibilities, and reproduce or change the conditions under which later relations form. That movement introduces the generative dimension that distinguishes GRE from a purely relational economics.

1.17 From Relational Subjecthood to Generative Relations

This section develops the transition from relational constitution to generativity. Its role in the chapter is to show why identifying the relations that stabilize an economic subject is only the first stage of analysis. The objective is to examine how persistent relations can transform subjects, generate capacities, alter the range of future possibilities, and organize forms of participation whose effects extend beyond an immediate transaction. The discussion proceeds conceptually and historically, drawing on the relational insights established in Sections 1.8 and 1.16. It first distinguishes persistence from mere recurrence, then analyzes subject transformation and capacity formation, extends the analysis to future possibility, and closes by specifying the generative structure of economic participation.

1.17.1 Persistence of Relational Structures

A relation becomes historically consequential when it persists long enough to condition later action. Persistence can arise through repeated interaction, institutional recognition, durable expectations, shared infrastructure, accumulated knowledge, legal rules, organizational routines, or material dependence. The resulting structure can outlast the particular exchange in which it first appeared. Employment relationships can generate occupational identities and skill trajectories; credit relations can alter future borrowing conditions; supply relationships can create specialized productive capacities; educational relations can reshape later opportunities; and

platform-mediated relations can reorganize access to markets, audiences, and information. The economic significance of a relation therefore depends partly on the temporal structure through which its effects continue.

Persistence should be distinguished from simple repetition. A sequence of transactions can recur without producing a durable change in the participants or in the surrounding institutional environment. A persistent relational structure has a stronger property: earlier interaction affects the conditions under which later interaction occurs. Trust, reputation, contractual routines, shared standards, specialized assets, organizational memory, and accumulated expectations are familiar mechanisms through which past relations become active conditions of the present. Granovetter's account of embedded economic action remains useful here because it directs attention to ongoing social structures rather than isolated transactions (Granovetter 1985). For GRE, the further question concerns what these ongoing structures make possible over time.

The persistence of relations can be located at several scales. A dyadic relation between worker and employer may stabilize expectations about effort, compensation, learning, and advancement. Repeated relations among firms can create interorganizational routines and shared technical standards. Relations among research institutions can stabilize channels through which knowledge, personnel, and equipment circulate. Financial relations can become embedded in clearing systems, regulatory arrangements, and expectations about liquidity. The relevant unit of analysis therefore expands as persistence becomes distributed across multiple participants.

Institutionalization is one important mode of persistence. A relation becomes institutionally durable when rules, classifications, legal forms, accounting practices, credentials, or administrative procedures reproduce it beyond the memory or intention of particular participants. Section 1.5 showed how modern economic entities are stabilized through precisely these mechanisms. The same mechanisms can stabilize relations among entities. A professional credential reproduces a relation between certified expertise and institutional access; a property title stabilizes a relation of exclusion and control; a standard contract reproduces a recognizable pattern of obligation across many interactions. Persistence can therefore be carried by symbolic and administrative infrastructures as well as by repeated personal contact.

Material complementarity can also make relations persistent. Specialized productive systems often create capacities that have value only within a wider configuration. A supplier investing in a highly specific component, a worker acquiring organization-specific knowledge, or a laboratory developing equipment around a shared protocol may become increasingly connected to a particular relational environment. The relation persists partly because the capacities generated within it are difficult to redeploy elsewhere without loss. This structure can support coordination and learning while also creating dependence and switching costs.

Persistence is normatively ambiguous. Durable relations can support trust, learning, care, coordination, and collective capacity. They can also reproduce dependency, exclusion, hierarchy, or unequal bargaining positions. The analytical importance of persistence lies in its generative potential rather than in any presumption that durability is beneficial. Once a relation modifies the conditions of later interaction, economic analysis must attend to the trajectory it produces and to the distribution of the

capacities and constraints that accumulate within it.

This temporal perspective prepares the move from relational stability to subject transformation. A subject stabilized by relations can also be changed by those relations. Persistence therefore provides the medium through which economic participation becomes formative rather than merely connective.

1.17.2 Relations and Subject Transformation

Economic subjects enter relations with histories, capacities, expectations, and institutional positions, and participation can modify each of these. The relevant transformation may concern skill, preference, identity, reputation, bargaining position, access, trust, obligation, or the practical categories through which a subject understands available action. Relations thereby participate in the production of later subjects. A relational account of economic life becomes generative when it follows this transformation across time.

Work provides a familiar example. Employment can alter a worker through training, occupational socialization, access to colleagues, exposure to organizational routines, and participation in professional communities. Human-capital theory captures one component of this process by treating education and training as investments that alter productive capacity (Becker 1964). A generative relational perspective widens the object. Skill formation occurs through relations with instructors, coworkers, institutions, technologies, standards, and bodies of knowledge. The capacity that later appears to belong to an individual can therefore have a distributed history of generation.

Organizations undergo analogous transformation. Repeated collaboration can produce routines, trust, shared vocabularies, coordination practices, and collective memory that are unavailable to participants acting separately. Powell, Koput, and Smith-Doerr's study of biotechnology is important because innovation appears within networks of interorganizational learning rather than inside fully self-sufficient firms (Powell et al. 1996). The productive subject in such settings changes through its relations: a firm's effective knowledge base and opportunity structure depend on the collaborations through which it learns and combines capabilities.

Relational transformation also occurs through institutions that classify and recognize subjects. Credentials can transform access to occupations; credit scores can change financing conditions; legal statuses can alter eligibility and responsibility; platform ratings can influence future matching and visibility. These representations can become part of the causal environment in which later subject positions are formed. Section 1.14 described the representational instability associated with symbolic abundance. The present point is temporal: once a representation affects later relations, it participates in the trajectory through which a subject is transformed.

Transformation can be cumulative and path dependent. Two subjects with initially similar resources can acquire different capacities because they enter different relational environments. Early access to strong mentors, well-connected institutions, trusted collaborators, stable employment, or high-quality infrastructure can generate later advantages that compound through subsequent relations. Conversely, exclusion from these environments can limit the opportunities through which capacities

would otherwise develop. The distribution of present resources therefore provides only a partial account of future economic position when the relations through which resources are converted into capacities differ systematically.

This dynamic also complicates the usual separation between preference and constraint. Participation can alter what subjects value, what they can imagine, which goals appear attainable, and which forms of activity become meaningful. Economic relations can therefore affect both the feasible set and the orientation of the subject within it. GRE does not require a universal theory of preference formation at this stage. It requires only the weaker historical observation that economic participation can modify the subject who later chooses, works, invests, cooperates, or withdraws.

Artificial intelligence introduces a new set of formative relations. A worker who repeatedly relies on an AI system may acquire new capabilities through augmentation, lose others through reduced practice, reorganize attention around system outputs, or become dependent on infrastructures controlled elsewhere. Organizations can similarly redesign workflows, expertise hierarchies, and decision routines around artificial systems. The long-run economic effect therefore concerns more than the productivity of a task at one moment. It also concerns the kinds of human and organizational subjects produced by sustained participation in AI-mediated relations.

Subject transformation provides the bridge to capacity formation. A generative relation becomes especially important when participation produces abilities, resources, forms of coordination, or institutional access that were unavailable to the participants before the relation developed.

1.17.3 Relations and Capacity Formation

Capacity formation gives the generative dimension a more precise economic object. A relation is generatively significant when participation helps produce a capacity that changes what a subject or collective can subsequently do. Capacities can be productive, epistemic, organizational, relational, political, or adaptive. They may reside partly in identifiable subjects and partly in the relations and infrastructures through which those subjects act.

The capability perspective offers an important point of contact because it evaluates resources through the substantive opportunities they enable rather than through possession alone (Sen 1999). GRE extends this concern temporally. The question becomes how present relations contribute to the formation of later capabilities and how unequal access to those relations shapes future opportunity. A training grant, for example, has one immediate recipient, yet its effect depends on educational institutions, mentorship, time, social support, recognition, and pathways into later work. Resource allocation and capacity generation are therefore analytically distinct stages.

Some capacities are intrinsically relational. Trust cannot be possessed by one actor in isolation in the same manner as a physical asset. Coordination capacity emerges through repeated mutual adjustment. A professional network becomes useful through reciprocal recognition and access. Collective knowledge depends on communication, shared standards, and institutional memory. Organizational resilience can depend on redundancy, distributed expertise, and the quality of relations among units. These examples show why a purely possessive vocabulary can

underdescribe capacities whose efficacy lies in structured interdependence.

Capacity can also be generated at a higher level than the participants through whose interaction it arises. A research collaboration can create a shared dataset, protocol, instrument, or body of tacit knowledge that later participants inherit. A supply network can develop standards and logistical routines that increase the productive capacity of the network as a whole. A community can establish norms and institutions that support collective action. The resulting capacity is real in its effects even when it cannot be assigned without remainder to any single member.

Relational capacity formation is often cumulative. Early participation can create the competence required to enter more advanced relations, which then generate further capability. Education, professional apprenticeship, scientific collaboration, entrepreneurship, and institutional membership frequently exhibit this structure. The trajectory can become self-reinforcing when generated capacities increase access to subsequent opportunities. The reverse is also possible: weak initial access can reduce later participation and create cumulative disadvantage.

The distribution of capacities therefore depends partly on the distribution of generative relations. Two actors receiving identical nominal resources can experience different long-run outcomes because their surrounding relations differ. Access to mentors, infrastructure, stable institutions, trusted collaborators, legal recognition, and time can determine whether a resource becomes an enduring capability. This observation is central to the later GRE treatment of allocation. Allocation affects economic futures partly by changing the relations through which capacity can be generated.

Capacity formation also clarifies why generativity should remain separate from immediate performance. A relation may reduce short-term efficiency while increasing learning, redundancy, adaptability, or future option space. Conversely, a highly optimized arrangement may produce excellent immediate output while generating little new capability among participants. GRE later develops the evaluative consequences of this distinction. At the historical level of the present chapter, the important point is that the productive significance of a relation can appear only across time.

The temporal extension of capacity leads directly to future possibility. Generated capability matters because it changes the set of relations, actions, and trajectories available later.

1.17.4 Relations and Future Possibilities

Economic relations participate in the construction of future possibility by changing which actions, affiliations, projects, and institutional positions become available. A future opportunity is rarely created by a resource alone. It emerges from a configuration that connects capacity, recognition, infrastructure, information, legal permission, social access, and material conditions. The generative effect of a relation can therefore be described partly through the possibilities it opens, preserves, redirects, or forecloses.

This perspective adds a temporal dimension to scarcity. Section 1.12 argued that contemporary scarcity increasingly concerns attention, trust, institutional access,

and generative opportunity alongside persistent material scarcity. Generative opportunity is distinctive because its value lies in what can follow from participation. Access to a research laboratory matters partly through experiments that become possible; access to a professional community matters through collaborations and recognition that may emerge; access to a stable learning environment matters through capabilities that develop over time. The economic object is therefore a pathway rather than a single consumable unit.

Future possibility is path dependent because present participation can alter later access. A first job can generate experience that changes eligibility for later work. A credit relation can create a repayment history that affects future finance. Membership in a standards body can provide information and influence over later technical rules. Participation in a platform can accumulate reputation and audience. These relations create histories that institutions later use when allocating further opportunities. Economic systems thereby convert past relations into conditions of future participation.

Possibility can also be foreclosed. Debt burdens can constrain later choices; restrictive contracts can limit mobility; dependency on proprietary infrastructure can raise switching costs; narrow credential systems can exclude forms of competence that lack recognized representation; concentrated ownership can prevent participants from converting contribution into control. The generative analysis of economic relations therefore requires attention to both expansion and contraction of future option spaces.

The same structure operates collectively. A community that loses a school, clinic, transport link, financial institution, or research facility may lose more than the immediate service. It can lose the relations through which later capabilities and projects would have formed. Conversely, shared infrastructure can generate opportunities that were not specified in the original investment. The return on infrastructure may therefore include emergent activities that could not be enumerated in advance because they depend on subsequent relations among users.

This openness distinguishes generative possibility from prediction. A generative relation can expand the range of feasible futures without determining which future will occur. Educational access can enlarge capability without fixing occupation; scientific infrastructure can enable discoveries without specifying them; social trust can support cooperation without determining its object. GRE is therefore interested in conditions that make later generation possible while retaining contingency in the trajectories that follow.

Future possibility also introduces a distributional problem. Some actors possess wide option spaces because they can draw on wealth, institutions, networks, mobility, and recognized credentials. Others confront narrow spaces even when their current consumption appears similar. A static snapshot can therefore understate inequality when it ignores the unequal capacity to generate future alternatives. The later GRE framework will treat this issue through generative opportunity, value circulation, and justice across time.

The relation between participation and possibility now becomes explicit. Participation is economically generative when it changes the conditions under which later

participation can occur. This gives the final subsection its organizing principle.

1.17.5 Generative Structures of Economic Participation

Economic participation has a generative structure when taking part in an activity changes the participant, the relation, or the surrounding institutional conditions in ways that affect later participation. This formulation extends the conventional focus on what an activity produces immediately. Employment can produce goods and income while also generating skills, reputations, dependencies, and organizational roles. Education can deliver instruction while generating curiosity, competence, social connection, and later access. Research can produce findings while creating instruments, communities, standards, and new questions. Care can meet present needs while sustaining the capacities through which persons participate in future social and economic life.

The generative structure of participation can be analyzed through several dimensions. The first is subject formation: participation can stabilize or transform the identity and position of the participant. The second is capacity formation: participation can generate abilities and resources that persist. The third is relational reproduction: participation can strengthen, weaken, or reorganize relations among actors. The fourth is institutional transformation: repeated participation can alter rules, expectations, and recognized categories. The fifth is option-space transformation: the preceding changes can expand or contract future possibilities.

These dimensions need not move together. A relation can generate productive capacity while reducing autonomy. It can expand organizational capability while concentrating control. It can strengthen trust within a group while increasing exclusion at its boundary. It can create valuable knowledge while producing dependency on infrastructure controlled by another actor. Generativity therefore describes a structure of downstream production rather than a complete normative evaluation. Later chapters address power and justice because the distribution, direction, and control of generative effects require separate analysis.

Participation can also generate relations that were not specified by the initiating objective. Collaboration undertaken to complete a project may produce a durable research network. Shared use of infrastructure may create a community of practice. A cooperative response to crisis may establish institutions that persist after the immediate problem has passed. These outcomes matter because they alter the relational environment from which later economic action begins. The economic significance of participation can therefore exceed the objective under which the activity was initially organized.

The distinction between ordinary relationality and generativity can now be stated positively. Relational analysis asks how actors, organizations, and institutions are connected and constituted through relations. Generative analysis asks how those relations produce changes that become conditions of subsequent economic life. The object includes newly generated subjects, capacities, institutions, relations, constraints, and possibilities. This temporal extension is the reason GRE requires more than a static map of economic relations.

The distinction is especially important under conditions of artificial intelligence

and symbolic abundance. An AI-mediated relation can improve immediate productivity while changing skill formation, professional recognition, dependency, organizational memory, and access to future work. A platform can facilitate exchange while reshaping market entry, reputation, data accumulation, and bargaining power. A transfer can raise present consumption while leaving generative opportunity unchanged, or it can support relations that produce durable capability. The same immediate outcome can therefore sit inside very different generative trajectories.

Generative participation also provides an alternative route for thinking about economic inclusion after the destabilization described in Section 1.15. If productive necessity weakens as a universal basis of social membership, participation can still matter through learning, care, inquiry, creation, cooperation, institution building, and other relations that generate capacities and future possibilities. This observation does not establish a normative programme or an allocation rule. It identifies an economic object that becomes more visible when productive contribution, productive necessity, and meaningful participation no longer coincide automatically.

Section 1.18 develops the analytical consequence. Once economic subjects and capacities are understood as relationally generated, an entity-centered account becomes incomplete whenever internal relations, external relations, relations among relations, downstream effects, or transformations of future generative conditions are required to explain an intervention. The next section therefore specifies the relational and generative limits of entity-centered economics before the chapter turns to the scientific conditions that make more demanding forms of analysis increasingly possible.

1.18 Relational and Generative Limits of Entity-Centered Economics

This section specifies the analytical boundary reached by entity-centered economic description after the historical developments reconstructed in Sections 1.5–1.17. Its role in the chapter is to identify the conditions under which persons, firms, households, states, and other recognized entities remain adequate units of analysis, and the conditions under which additional relational and generative description becomes necessary. The objective is therefore one of scope rather than replacement: entity-centered analysis remains powerful where entity boundaries are stable and the economically relevant effects of an intervention can be represented at the level of the recipient. The discussion proceeds from relations internal to entities, through relations surrounding them and relations among relations, to downstream intervention effects and transformations of future generative conditions. The method is conceptual and comparative, using established work on firms, embeddedness, network organization, global value chains, and entitlement to show that economically consequential structure can reside both within and between recognized entities.

1.18.1 Internal Relations within Economic Entities

An economic entity is often represented by variables attributed to the entity as a whole: income for a household, productivity for a firm, assets for a corporation, expenditure for a public body, or output for a national economy. Such variables are indispensable for comparison and administration. Their adequacy depends, however, on whether the internal relations through which the entity operates can be treated as analytically secondary for the question at hand. When internal organization shapes the conversion of resources into action, the same entity-level quantity can correspond to substantially different economic capacities.

The firm provides the clearest historical example. Coase's classic analysis of the firm begins from the observation that productive activity can be coordinated through authority inside an organization as well as through market exchange across organizational boundaries (Coase 1937). The distinction matters because a firm is more than an owner of inputs. It is a patterned arrangement of authority, employment, information, routines, and contractual relations through which inputs are combined. Two firms holding similar assets can therefore exhibit different productive capacities when their internal coordination, knowledge distribution, worker relations, or organizational routines differ.

Organizational law reinforces this point by stabilizing the entity without exhausting its internal structure. Hansmann and Kraakman show that organizational law supports standardized legal entities partly through asset partitioning, which separates organizational assets and creditor claims from those of owners or managers (Hansmann and Kraakman 2000). This legal boundary is economically consequential because it makes the organization tractable as a bearer of assets and obligations. Yet the boundary does not describe how authority, expertise, risk, reward, and decision rights are distributed inside it. A legally unified corporation can contain highly unequal or internally fragmented relational structures while appearing as one administrative recipient.

Households display a parallel difficulty. A transfer recorded at household level can alter the resources available to the household while leaving open the internal distribution of control, care obligations, time, bargaining position, and effective access. Treating the household as a single recipient can be appropriate for some fiscal or demographic questions and inadequate for questions concerning who can actually command the transferred resources. The general issue is the conversion from entity-level possession to participant-level capability. Where this conversion depends on internal relations, the entity boundary becomes an important part of the explanation rather than a neutral container.

Internal relations also shape the persistence of capacities. Knowledge can be distributed among workers and routines rather than stored in any single individual. Trust can be localized in teams. Authority can be centralized or dispersed. Care obligations can allocate time asymmetrically. Data can be accessible to some organizational positions and unavailable to others. These structures influence what an entity can do even when conventional balance-sheet or resource measures remain unchanged. A relational description therefore becomes useful when variation inside the entity explains variation in action, learning, adaptation, or access that entity-level

quantities cannot capture.

The implication is a scope condition. Entity-centered analysis is strongest when internal heterogeneity does not materially affect the causal or evaluative question. Relational analysis becomes increasingly necessary as internal organization determines the conversion of nominal resources into effective capacities. This does not dissolve the entity. It adds a second analytical resolution at which the entity is reconstructed as a stabilized configuration of relations.

1.18.2 External Relations around Economic Entities

Economic entities also derive capacities from relations that cross their recognized boundaries. Suppliers, customers, creditors, regulators, professional communities, infrastructures, knowledge networks, platforms, and public institutions can alter what an entity is able to produce, access, learn, or survive. Entity-level resources therefore acquire economic meaning partly through the external relations that make those resources usable.

Granovetter's account of embeddedness provides a foundational statement of this problem by showing that economic action is situated in ongoing structures of social relations rather than occurring among atomized actors (Granovetter 1985). The analytical consequence is especially visible where trust, reputation, repeated interaction, or social position affect exchange. A firm with modest internal resources can occupy a highly advantageous position in a dense network of reliable partners, while a better-capitalized firm can remain constrained by weak access, low trust, or peripheral position. Entity attributes and relational position therefore provide distinct information.

Interorganizational production sharpens the distinction. Powell's analysis of network forms of organization identifies arrangements in which reciprocal communication and continuing exchange coordinate activity in ways that are poorly described as either discrete market transactions or internal hierarchy (Powell 1990). Section 1.8 used this literature to reconstruct the expansion of relational production. The present concern is analytical: once production depends on continuing interorganizational relations, a firm's productive capacity cannot be inferred from its owned assets alone. It also depends on the quality, accessibility, and durability of relations through which complementary capabilities are obtained.

Global value chains extend this structure across jurisdictions and organizational scales. Gereffi, Humphrey, and Sturgeon distinguish several governance forms whose emergence depends on transaction complexity, codifiability, supplier capability, explicit coordination, and power asymmetry (Gereffi et al. 2005). The relevant unit of analysis is therefore larger than any one firm. Lead firms, suppliers, standards, logistics systems, contractual practices, and technical interfaces together produce the conditions under which production occurs. An intervention directed toward one participant can propagate through this wider configuration.

External relations are also central to effective command over resources. Sen's entitlement analysis demonstrates that access to food cannot be inferred from aggregate availability alone because persons obtain command through production possibilities, trade opportunities, state entitlements, and other socially recognized channels

(Sen 1981). The same analytical lesson generalizes cautiously: a resource can exist in the environment and remain economically inaccessible to an entity whose relational pathways to it are absent or blocked. Access is therefore a relational condition connecting resources to subjects.

This point becomes particularly important under digital and AI-mediated production. Access to compute, datasets, model interfaces, platforms, payment systems, professional networks, standards, and trusted intermediaries can determine the usefulness of nominally owned assets. External relations can also impose dependency: an organization may appear autonomous at the entity level while relying on a small number of infrastructural or platform relations for essential functions. Relational analysis reveals these conditions without requiring the abandonment of entity-level accounting.

The external limit of entity-centered analysis is reached when economically decisive capacities and constraints arise from position in a relational environment. At that point, the question changes from the quantity of resources held by an entity to the relations through which those resources become actionable, exchangeable, recognized, and reproducible.

1.18.3 Relations among Economic Relations

Economic systems frequently contain structures in which one relation changes the operation of another. These higher-order configurations cannot be represented fully by listing pairwise ties because the economically relevant object is the relation among relations. Credit conditions can alter employment relations; legal classifications can reshape market access; platform rules can mediate buyer–seller relations; professional standards can condition relations between expertise and institutional authority; and ownership structures can change how organizational decisions are connected to labor, investment, and distribution.

The distinction is easiest to see in mediated exchange. A buyer and seller can transact through a platform, yet the terms of their relation depend on a further relation between each participant and the platform. Visibility, ranking, payment, data access, dispute resolution, and reputation can all be mediated through this additional structure. The buyer–seller relation therefore changes when the platform’s rules or incentives change even if the two participants remain otherwise identical. The economically significant object is a configuration in which one relation governs, conditions, or transforms another.

Organizations contain similar higher-order structures. A manager’s authority over a worker may be conditioned by labor law, collective agreements, professional norms, internal grievance procedures, or algorithmic monitoring systems. A creditor’s claim may alter the relation between managers and investment decisions. A shareholder structure can influence how organizational surplus is connected to workers, suppliers, or research expenditure. These arrangements involve several relations whose interaction produces effects that no single dyad carries alone.

Network and value-chain research provides empirical settings in which such configurations become visible. Network forms coordinate relations among organizations through repeated exchange and reciprocal communication (Powell 1990), while

global value-chain governance connects lead-firm coordination, supplier capability, standards, and power asymmetry within structured production relations (Gereffi et al. 2005). A relation can therefore become consequential because of the other relations in which it is embedded. The same supplier tie can generate learning in one network, dependence in another, and exclusion in a third.

This higher-order perspective also clarifies why institutional change can have broad economic effects without transferring resources directly. A change in legal recognition, reporting rules, standards, or eligibility criteria can modify the relations through which many other interactions occur. Section 1.5 showed that legal personality, accounting boundaries, statistics, and administrative recognition stabilize economic entities. The same institutional devices can restructure relations among those entities by changing who may contract, who bears liability, what counts as compliant performance, or which transactions are visible to regulators and counter-parties.

Relations among relations create a methodological demand for explicit boundary and scale choices. An analyst can always enlarge the relational map, and no practical inquiry can include every connection. The relevant question is whether omitting a higher-order relation changes the explanation of the outcome under study. GRE therefore treats higher-order analysis as problem-dependent. It becomes warranted where the behavior of a focal relation varies systematically with the configuration of surrounding relations.

The resulting picture remains compatible with entity-level economics. Entities continue to matter as legal, administrative, and organizational condensations. The additional analytical layer concerns the architectures through which their relations are coupled. This layer becomes central whenever intervention in one relation changes the operation of others.

1.18.4 Downstream Effects of Economic Intervention

Entity-centered evaluation commonly begins with a defined intervention and a defined recipient: a transfer to a household, a subsidy to a firm, a loan to an organization, a tax on an income stream, or an investment in a public body. This structure is often appropriate for immediate incidence. Its generative limit appears when the intervention changes relations whose later effects extend beyond the initial recipient and evaluation horizon.

A resource transfer can generate several distinct downstream pathways. It may alter bargaining position, enable entry into a network, support education or training, change creditworthiness, reduce dependence on a particular employer, permit migration, create time for care or inquiry, or allow an organization to invest in standards and relationships that affect later collaboration. Two transfers with the same nominal value can therefore produce different trajectories because they enter different relational configurations.

The distinction between receipt and command is again useful. Sen's entitlement framework shows that resource outcomes depend on socially organized channels through which persons gain effective command (Sen 1981). In a generative setting, the question extends further: effective command at one moment can alter the chan-

nels available later. Access to credit can create a repayment record; access to education can change occupational networks; access to infrastructure can change market participation; access to a platform can generate data and reputation that affect future visibility. An intervention can therefore modify the relations through which later interventions or opportunities are converted into capability.

Downstream effects can also propagate across entities. Support to one firm may sustain suppliers, preserve specialized knowledge, alter local employment, or strengthen a technological standard. A procurement decision can reorganize a supply network. A platform rule can change incentives for thousands of participants. A financial intervention can alter expectations and counterparties beyond its formal recipient. The immediate recipient remains administratively important, while the causal object expands to include the relational pathways through which the intervention propagates.

Evaluation horizons shape whether these effects appear. A short-horizon assessment may correctly report immediate consumption, output, employment, or solvency while omitting changes in learning, trust, dependency, network position, institutional access, or option availability that emerge later. Longer horizons create their own uncertainty and attribution problems. GRE therefore does not presume that downstream effects can always be measured precisely. It treats temporal extension as a question of analytical adequacy: when plausible downstream mechanisms are central to the intervention's significance, the evaluation should state which are included, which are excluded, and what evidential confidence attaches to them.

The same discipline applies to negative downstream effects. An intervention can create dependency, crowd out alternative relations, increase concentration, lock participants into proprietary infrastructure, or shift risk to actors outside the original evaluation boundary. Entity-centered accounting may record a local gain while relational analysis detects a redistribution of constraint elsewhere. Such cases are especially important when the initial recipient and the later bearer of cost are different entities.

Downstream analysis therefore changes the unit of evaluation from a recipient-state pair to a trajectory through a relational system. The intervention remains identifiable, and the recipient remains important, yet the relevant economic question includes what relations the intervention forms, strengthens, weakens, or redirects over time.

1.18.5 Transformation of Future Generative Conditions

The deepest limit of entity-centered analysis appears when economic activity changes the conditions under which future economic activity can be generated. At this point the object is no longer only a downstream outcome. It is the transformation of the relational, institutional, material, or epistemic conditions that structure later possibilities.

Capacity formation provides one route. A relation can generate skills, trust, organizational routines, infrastructure, or shared knowledge that later activities presuppose. Section 1.17 described these processes as generative because participation changes what subjects and institutions can subsequently do. The same logic applies to

constraints. A relation can produce dependency, exclusion, standardized interfaces, proprietary lock-in, debt obligations, or institutional classifications that narrow later options. Generativity therefore includes the production of enabling and constraining conditions.

Institutional persistence amplifies these effects. Once a relational arrangement is stabilized through law, accounting, standards, administrative categories, or organizational routines, it can shape later transactions that were absent from the original decision. A financing rule can influence which organizations survive long enough to innovate. A credentialing system can determine who enters a profession. A platform's data architecture can shape later visibility and bargaining power. A property regime can influence which forms of cooperation or exclusion become ordinary. These structures operate as inherited conditions for actors who did not choose their initial form.

The transformation of future conditions is especially important for value and allocation. A present allocation can generate capacities that increase later access to resources, or it can support relations whose benefits circulate beyond the initial recipient. Conversely, a present extraction can weaken the relations required for future generation even when current output remains high. The later Parts of this book develop these questions through relational allocation, heterogeneous value, generative return, spacetime, power, and justice. The historical argument of the present chapter is narrower: contemporary economic organization increasingly makes these future-conditioning effects difficult to treat as peripheral.

Artificial intelligence intensifies the issue because current arrangements can shape the distribution of future productive intelligence. Decisions about compute, data, model access, research infrastructure, intellectual property, standards, and institutional participation can influence which actors are able to learn from and contribute to later technological development. Immediate productivity effects are therefore only one dimension of the economic significance of AI-related investment and allocation. The configuration of relations around artificial productive capacity can alter the future distribution of capability itself.

A generative analysis also requires restraint. Future conditions are uncertain, relational pathways branch, and interventions can produce unintended effects. Strong claims about long-horizon generation therefore require stronger evidence than descriptions of present allocation. GRE's later methodology distinguishes association, enablement, contribution, co-generation, and stronger forms of attribution so that temporal extension does not become speculative license. The present section establishes only the analytical reason for asking the question.

The resulting scope condition can be stated positively. Entity-centered economics remains highly effective when the relevant outcomes are well represented by attributes of recognized entities within a specified horizon. Relational analysis becomes necessary when internal organization, external position, or higher-order relational structure materially affects those outcomes. Generative analysis becomes necessary when present relations transform the subjects, capacities, constraints, institutions, or possibilities through which later economic life unfolds. These analytical layers are complementary resolutions of economic reality rather than mutually exclusive schools.

This distinction completes the historical movement from the modern entity-centered order reconstructed in Sections 1.3–1.6 to the relational and generative problems made increasingly visible by globalization, digital mediation, artificial intelligence, symbolic abundance, and changing subjecthood. Sections 1.19 and 1.20 now turn from historical necessity to historical possibility: the scientific formalisms, computational resources, and analytical vocabularies that make more demanding relational and generative inquiry increasingly feasible.

1.19 Scientific Conditions for Relational Analysis

This section identifies the scientific conditions that have made relational economic analysis increasingly tractable. Its role in the chapter is historical and methodological: the preceding sections established why contemporary economic problems increasingly exceed an exclusively entity-centered description, while the present section asks which analytical developments make more explicit study of relations feasible. The objective is to specify a repertoire rather than to prescribe a single formalism. The discussion proceeds from dynamical systems and complex-systems approaches to network and higher-order representations, field-based descriptions, and multiscale or background-sensitive formalisms. Each subsection distinguishes the analytical possibility supplied by a scientific language from the stronger claim that economic reality literally possesses the ontology of the formal system used to describe it.

1.19.1 Dynamical Systems

Dynamical-systems theory enlarges economic analysis by making trajectory, stability, feedback, and qualitative change explicit objects of study. A static description records the state of an economic entity or relation at a selected moment. A dynamical description asks how that state changes, which states are reachable from it, which configurations persist after disturbance, and how local changes accumulate along a trajectory. The analytical language is mature across the natural and applied sciences, where nonlinear systems are studied through state spaces, flows, fixed points, periodic behavior, bifurcations, and sensitivity to initial conditions (Strogatz 2024). Its relevance to relational economics lies in the possibility of treating relations and relational configurations as time-indexed variables whose evolution matters to the economic outcome.

The gain is especially clear when a relation has memory. Employment relations accumulate experience and expectation; credit relations acquire repayment histories; supply relations generate specialized routines; platform relations accumulate reputation and data; institutional relations carry prior classifications into later access. A cross-sectional description can identify that a relation exists, while a dynamical description can ask how its present form depends on the path by which it was reached. The trajectory becomes economically relevant because two configurations that appear similar at one moment may respond differently to later disturbance when their histories have generated different capacities, expectations, or constraints.

Dynamical systems also provide a precise distinction between stability and value.

Stability concerns how a configuration responds to perturbation. It does not establish that the configuration is efficient, just, desirable, or generative. This distinction is important for GRE because persistent economic structures can include durable cooperation, durable exclusion, resilient infrastructure, or resilient dependency. A dynamical description can characterize persistence without silently converting persistence into normative endorsement.

Nonlinearity further expands the range of analyzable relational phenomena. When the effect of one relation depends on the state of other relations, aggregate behavior may fail to equal a simple sum of isolated effects. Feedback can amplify or damp disturbances; thresholds can make similar interventions consequential in one configuration and negligible in another; interaction among several processes can produce regime changes absent from any component considered separately. Such possibilities matter in economies organized through interdependent firms, financial claims, standards, infrastructures, and institutional expectations.

The methodological contribution remains bounded. A dynamical model requires a defensible specification of states, variables, boundaries, and transformation rules. Social and economic systems also include reflexive subjects whose interpretation of a model can enter the subsequent dynamics. The availability of dynamical-systems theory therefore establishes a scientific possibility: economic relations can be studied as evolving configurations rather than as static background conditions. It does not by itself establish which dynamical representation is appropriate to a particular economic problem. That question belongs to model construction, evidence, and later methodological chapters.

1.19.2 Complex Systems

Complex-systems research provides a second condition for relational analysis by directing attention to systems whose macroscopic behavior depends on interaction among heterogeneous, adaptive, and partially autonomous components. The relevant scientific shift is analytical rather than terminological. A system can display regularities at one scale that are difficult to infer from isolated components because interaction, feedback, adaptation, and boundary formation contribute to the collective pattern. Holland's work on complex adaptive systems, for example, treats signals, boundaries, and evolving organization as central to the behavior of systems ranging from biological organization to markets and governments (Holland 2012).

For economic analysis, this perspective weakens the presumption that explanation must begin and end with attributes of independently specified units. Firms can adapt to competitors and regulators; households respond to institutions and changing prices; standards alter which technologies are compatible; market participants learn from one another; organizations modify routines in response to observed outcomes. The state of one participant is therefore partly conditioned by a surrounding configuration that is itself changing in response to participants. Complexity-oriented analysis provides a vocabulary for studying this reciprocal dependence without requiring that every collective regularity be reduced immediately to a single representative agent or equilibrium condition.

The approach also makes heterogeneity analytically productive. Differences

among actors in resources, expectations, information, capabilities, and network position can affect system-level behavior because interaction transmits and transforms those differences. Heterogeneity can produce clustering, specialization, segmentation, cascades, or uneven adaptation. In a relational economy, the distribution of positions can therefore matter as much as an average attribute. A system composed of actors with the same mean income, skill, or productivity can exhibit different dynamics when the relations among them differ.

Complex-systems approaches are particularly useful for distinguishing local rule from global pattern. A participant may act according to a relatively simple decision rule while the aggregate configuration exhibits persistence, volatility, concentration, or coordination that no participant designed. This possibility resonates with the historical argument developed earlier in the chapter. Modern economic institutions often generate structures whose consequences exceed the horizon of the actors participating in them. The scientific significance of complexity research is that such distributed generation can be represented and investigated without assigning the collective pattern to a central designer.

Adaptation introduces another important dimension. Economic actors respond to outcomes, and their responses alter the environment to which later actors adapt. The resulting feedback can change the effectiveness of a rule over time. A regulatory threshold, ranking metric, trading strategy, or recommendation system can have one effect before widespread adaptation and another after participants reorganize around it. Complexity-oriented analysis therefore encourages researchers to treat observation, strategy, and institutional response as potentially endogenous components of the system.

The term *complexity* can easily become too broad to discipline analysis. GRE therefore uses complex-systems language only when the local problem exhibits the relevant structure: interaction, heterogeneity, feedback, adaptation, multilevel organization, or emergent collective regularity. Complexity is a scientific resource for relational inquiry rather than an explanation by itself. Its historical importance lies in the development of formal and computational methods capable of treating interdependence as a primary analytical object.

1.19.3 Network and Higher-Order Models

Network science provides the most direct formal language for representing economic relations among identifiable entities. Nodes represent analytically specified units and edges represent specified relations among them. Once such a representation is justified, analysis can address connectivity, centrality, clustering, community structure, path length, assortativity, diffusion, robustness, and the interaction between network topology and dynamical processes. Newman's review of complex networks documents the development of these concepts across social, technological, and biological systems and emphasizes the study of both network structure and dynamics on networks (Newman 2003).

The relevance to economics is immediate because many economic phenomena already have relational form. Credit links connect borrowers and lenders; supply chains connect buyers and suppliers; ownership links connect shareholders and

firms; labor mobility connects organizations; communication links connect teams; trade links connect jurisdictions; platform relations connect several participant groups through an intermediary. A network representation can therefore preserve relational information that disappears when each entity is summarized only by its own attributes.

Network models also make positional effects explicit. Two firms with similar observable resources can occupy very different positions in a supply network. Two researchers with similar individual capabilities can have different access to knowledge because their collaborative neighborhoods differ. Two households with similar income can face different opportunity structures because transport, care, credit, or social-support connections differ. In each case, an entity-level variable remains relevant, while network position adds information about the pathways through which resources, risks, information, and opportunities can move.

Pairwise networks have a clear limitation: many economic relations involve groups, institutions, or coordinated sets of participants whose effects cannot be represented adequately as collections of independent dyads. A project consortium, household, standards committee, cooperative, research team, or multilateral agreement can impose conditions on a group as a group. Higher-order network research has developed explicit representations for such cases through hypergraphs, simplicial complexes, and related structures. Battiston and colleagues survey this literature and show that interactions among three or more units can generate dynamical behavior that is not captured by pairwise links alone (Battiston et al. 2020).

Higher-order models are especially relevant to the GRE concern with relations among relations. A contract can depend on a regulatory institution; a supply relation can depend on a standard jointly maintained by several organizations; a worker's access to employment can depend on credentialing relations linking educational institutions, professional bodies, and employers. Such structures can be flattened into pairwise edges, but flattening can erase the condition that the relevant effect exists only when several relations are jointly present. Higher-order representations make that joint dependence visible.

These models nevertheless remain representations. Nodes must be chosen, edges must be defined, interaction order must be specified, and the temporal stability of the representation must be justified. A legal entity, household, platform account, institution, or community can function as a node for one question and as an internally structured relational system for another. Network analysis therefore does not eliminate the boundary problem; it renders the boundary decision explicit and makes its consequences inspectable. This is a scientific advance because relational structure can be formalized, compared, and tested while preserving awareness that the graph is an analytical construction.

1.19.4 Field-Based Representations

Field-based representations become useful when economically relevant conditions are distributed across a domain and cannot be localized adequately in a finite set of isolated links. In mathematics and mathematical physics, field descriptions assign quantities to locations in a continuous or discretized domain and study how

local states vary, interact, and propagate. Partial differential equations provide a developed formal language for representing such spatially or continuously distributed dynamics, including nonlinear propagation, diffusion, transport, and wave-like processes (Evans 2010). GRE draws from this scientific repertoire at the level of representation: some economic conditions may be better described as distributed environments than as attributes attached to individual entities.

Several economic objects have this character. Accessibility can vary continuously across geographic space; labor-market opportunity can depend on local density and transport; information exposure can vary across communication environments; regulatory intensity can differ across jurisdictions; trust or reputational climate can be unevenly distributed; infrastructure can create gradients of connection and cost. In such cases, the relevant condition surrounds many actors at once and changes the consequences of local relations. A field representation can encode this distributed context without assigning the whole effect to one neighboring entity.

Field language also helps distinguish a relation from the conditions under which the relation operates. The same contractual form can have different consequences in regions with different institutional capacity. The same amount of credit can generate different outcomes where market access differs. The same platform rule can affect participants differently when local alternatives, network density, or legal protections vary. Treating the surrounding condition as a field makes it possible to model systematic variation in relational effect across position.

The approach can be extended beyond geographic space when the relevant domain is abstract. A field can be defined over a state space, capability space, or other structured domain if distance and neighborhood have defensible meanings. Such constructions can represent how similar configurations experience similar pressures or how local changes propagate through a structured possibility space. Their usefulness depends on whether the chosen geometry captures an empirically meaningful relation rather than merely providing a visually attractive mathematical surface.

Field-based representation therefore carries a stronger burden of justification than ordinary metaphor. A model must specify what the field variable represents, what domain indexes it, what counts as locality, how interactions propagate, and how the representation connects to evidence. GRE's ontological suspension is important here. The success of a field model would show that a distributed representation is analytically productive for a given problem; it would not establish that an economy is literally a physical field. The scientific condition consists in the availability of mature mathematics for distributed, locally coupled systems and in the possibility of adapting that mathematics under explicit domain-specific assumptions.

1.19.5 Multiscale and Background-Sensitive Formalisms

Relational economic systems operate across several scales at once. Individuals participate in households, organizations, networks, institutions, markets, jurisdictions, and transnational systems; processes at each level can constrain or enable processes at others. Multiscale analysis supplies concepts for studying how organization, variation, and response differ across scale rather than assuming that one analytical resolution is sufficient. Bar-Yam's multiscale treatment of complex systems formalizes the

idea that effective response depends on matching the scale of variation in a system and has been applied to biological and social organization (Bar-Yam 2004).

The economic importance of scale is visible in ordinary examples. A workplace policy can affect an individual immediately while altering organizational culture slowly. A local supply disruption can propagate into sectoral production when many firms share dependencies. A financial position can be safe at the level of one institution while contributing to correlated systemic exposure. A household decision can be rational under prevailing prices while repeated decisions across many households alter the market conditions on which those choices depended. Relational analysis therefore requires the ability to move among levels without presuming that a pattern observed at one scale can be transferred unchanged to another.

Multiscale formalisms also clarify why aggregation can both reveal and conceal. Coarse-graining can expose collective regularities invisible at the level of individual events, while the same operation can erase heterogeneity and local dependence that matter to causal interpretation. Disaggregation can recover local structure while making system-level constraints harder to see. The choice of scale is therefore part of the model rather than a neutral technical detail. A relational account should report which distinctions are preserved at the selected resolution and which are suppressed.

A related scientific development concerns formalisms in which the background against which dynamics unfold is itself allowed to vary. General relativity provides a paradigmatic case: spacetime geometry is dynamical rather than a fixed stage, and background-independent approaches in gravitational physics develop descriptions in which the geometry relevant to motion is part of the system being modeled (Rovelli 2004). GRE does not infer a social ontology from this physical result. The methodological significance is narrower: modern science contains coherent formalisms in which the structure conditioning local dynamics need not be treated as permanently external and fixed.

This possibility matters for relational economics because economic backgrounds are often historically produced. Market boundaries change; infrastructures are built and abandoned; legal categories are revised; standards alter compatibility; platforms reorganize access; institutions modify the relations they regulate. A model that holds the surrounding structure fixed can be adequate over a limited interval, while a longer-horizon analysis may need the background itself to become variable. Section 1.20 develops this issue in generative terms by asking how states, relations, constraints, and transformation rules can coevolve.

Background sensitivity also disciplines comparison. A relation observed in one institutional setting may not preserve its effects when transplanted into another because the surrounding rules, infrastructures, histories, and scales differ. Formal models can represent this contextual dependence through parameters, boundary conditions, changing topology, or evolving geometry. The appropriate device depends on the problem. The common principle is that the relation cannot always be interpreted independently of the structure in which it is embedded.

The scientific opening described across this section is therefore plural. Dynamical systems make trajectories and stability analyzable; complex-systems approaches make interaction, heterogeneity, feedback, and adaptation central; networks and

higher-order models represent explicit relational structure; field approaches represent distributed conditions; multiscale and background-sensitive formalisms permit analysis across levels and under changing contextual structure. None of these formalisms establishes GRE by itself. Together they change the historical feasibility of a relational economics by providing mathematical and conceptual languages in which relations can become explicit objects of analysis. Section 1.20 turns to the stronger problem of generativity, where the analytical task includes the endogenous transformation of the structures and rules through which later economic change occurs.

1.20 Scientific Conditions for Generative Analysis

This section identifies the scientific conditions that make generative economic analysis increasingly possible. Its role in the chapter is to move beyond the explicit representation of relations developed in Section 1.19 toward a stronger analytical problem: economic relations can change the structures, constraints, and transformation rules under which later economic activity occurs. The objective is to specify the scientific languages capable of representing this endogeneity without treating every historical change as an instance of one universal mechanism. The discussion proceeds through endogenous dynamics, structure-changing processes, path dependence, coevolution, and the evolution of transformation rules. The method is comparative and conceptual. It draws on evolutionary economics, institutional economics, path-dependent dynamics, and complex-adaptive-systems research to identify analytical possibilities that later GRE chapters may formalize more precisely.

1.20.1 Endogenous Dynamics

Generative analysis begins where the conditions of change enter the dynamics being studied. A model can describe an economic system dynamically while leaving its transformation law fixed: prices, quantities, employment, or network positions may change through time according to a stable rule. Such a model is already richer than a static comparison because it represents trajectories. Generative analysis asks a further question. Economic activity can alter the conditions that shape later activity, so the variables governing subsequent change may themselves become endogenous to the history of the system.

Evolutionary economics supplies a developed precedent for this orientation. Nelson and Winter organize economic analysis around processes in which firm behavior, routines, search, innovation, and competitive selection jointly determine industrial change through time rather than around instantaneous optimization toward a predetermined equilibrium (Nelson and Winter 1982). The analytical importance of this move for GRE lies in its treatment of economic behavior and industrial structure as mutually consequential. Firms act within inherited routines and market conditions, yet search, innovation, imitation, investment, and differential survival can alter the distribution of routines and capabilities that later firms confront.

Endogeneity in this stronger sense can appear through several channels. Learning can change an actor's future response to a similar situation. Investment can create infrastructure that changes the feasible action set. Repeated exchange can build trust

or dependency. Regulation can produce new categories to which future regulation refers. Platform participation can generate data that improve the platform's later matching or ranking capability. A subsidy can expand a technology sufficiently to lower its later cost, and the lower cost can alter subsequent adoption. In each case, present activity modifies a condition relevant to the next round of activity.

Institutional economics adds another level of endogeneity. North treats institutions as humanly devised constraints that structure political, economic, and social interaction while also developing an account of institutional change in which organizations, incentives, and accumulated choices affect the evolution of those constraints (North 1990). This creates an analytical loop important for GRE: institutions condition economic action, economic action generates incentives and organizational responses, and those responses can contribute to institutional persistence or transformation. The conditioning structure therefore belongs to the history of the system rather than remaining permanently external to it.

Endogenous dynamics should still be specified locally. Some constraints can remain effectively fixed over the time horizon of a study; some institutional changes arrive through exogenous shocks; some actor responses are well represented by stable rules. GRE therefore does not require every model to endogenize every background condition. The scientific condition established here is the availability of frameworks in which a conditioning variable can become part of the dynamics when the research question requires it. This possibility is the first step from relational description toward generative analysis.

1.20.2 Structure-Changing Processes

Generative economic processes can alter more than the values attached to a fixed set of variables. They can change the structure through which variables are related. A supply relation can be created or terminated; a firm can merge with another; a market can acquire a platform intermediary; a legal category can make a new organizational form available; an infrastructure investment can connect previously separated regions; a technological standard can render previously incompatible components interoperable. The analytical object is therefore sometimes a changing architecture rather than movement inside an unchanged architecture.

This distinction matters because parameter adjustment and structural change carry different implications. A fall in a transaction cost changes the strength of an existing relation. The creation of a new intermediary changes which relations exist and how other relations are routed. A wage increase changes the terms of an employment relation. The legal recognition of a new form of worker, firm, property, or digital asset can change the categories through which economic relations are constituted. Generative analysis requires formalisms capable of representing these alterations in topology, membership, boundary, rule, or institutional status.

Evolutionary approaches again provide an important scientific precedent. Nelson and Winter's framework treats technological and organizational change as processes in which variation, search, selection, and retained routines can alter the composition and behavior of industries over time (Nelson and Winter 1982). Complex-adaptive-systems research broadens the representational possibility by treating signals, bound-

aries, and hierarchical organization as structures that can themselves develop and reorganize (Holland 2012). These traditions differ in object and formal detail, yet both make structural transformation an analytical problem rather than a residual category for changes that a fixed-state model cannot absorb.

Network and higher-order models discussed in Section 1.19 can also be made generative when the relational topology evolves. Nodes may enter or leave; edges may form, strengthen, weaken, or dissolve; higher-order groups may assemble or fragment; network position can affect the probability of subsequent tie formation. The same principle applies beyond networks. Field boundaries can shift, institutional classifications can be rewritten, and the scales at which coordination occurs can change. The scientific requirement is a representation in which the structure at one time can influence, and be influenced by, the processes unfolding within it.

Structure-changing analysis also disciplines causal interpretation. When an intervention alters the architecture through which later effects propagate, the relevant causal question extends beyond the immediate response of a fixed system. The intervention may create new channels of diffusion, new dependencies, new exclusion boundaries, or new organizational capacities. Two interventions with identical short-term outputs can therefore generate different long-term trajectories when one leaves the surrounding structure unchanged and the other rewrites it. This distinction becomes central later in GRE's treatment of relational allocation and generative value.

1.20.3 Path Dependence and Historical Trajectories

Generative analysis requires historical trajectories because the effect of a present relation can depend on the sequence through which the current configuration was produced. Path dependence provides a scientific language for this dependence on prior development. Arthur's analysis of competing technologies shows how increasing returns can make early events consequential for later adoption and can produce lock-in under specified conditions (Arthur 1989). The contribution is methodological as much as substantive: a historically contingent event can matter because it changes the returns, expectations, compatibility relations, or learning environment confronting later choices.

A path-dependent account differs from a simple statement that history matters. The relevant claim identifies a mechanism through which earlier events alter later transition possibilities. Increasing returns, sunk investments, institutional complementarity, network effects, learning, or accumulated expectations can make the consequences of an action depend on when it occurs and on what has occurred before. A trajectory therefore contains information that a comparison of endpoints can lose.

Institutional development displays the same structure in a different register. North's account of institutional change emphasizes that existing institutions shape the incentives and organizations through which later change proceeds, giving institutional development a cumulative and historically conditioned character (North 1990). The implication for GRE is that economic relations can inherit constraints generated by earlier relations even after the original circumstances have disappeared. Ownership forms, labor classifications, accounting conventions, technical standards, transport networks, and financial infrastructures can continue structuring action

because later investments and expectations have formed around them.

Historical trajectories also permit branching. A configuration can support several feasible continuations, and relatively small differences in timing, sequencing, or local interaction can direct development toward different regions of the possibility space. This does not require the claim that every historical outcome is radically contingent. Some paths are strongly constrained; others are open across a wider range. Generative analysis therefore benefits from distinguishing structural persistence, path dependence, branching possibility, and exogenous disturbance rather than placing all historical change under a single label.

For GRE, the importance of path dependence extends to value and subject formation. Skills acquired under one technological regime can shape access to the next. A reputation accumulated in one institutional setting can affect later credit or employment. A community's prior exclusion can constrain current participation even after formal barriers are removed. A platform's early relational centrality can attract further participants and data. The generative consequence of a relation is therefore partly temporal: it can modify the conditions under which later relations become possible, valuable, or difficult to escape.

Path dependence also has a methodological limit. Lock-in claims can be too easily inferred retrospectively from persistence alone. Arthur's mechanism depends on specific increasing-return processes rather than on the mere fact that one outcome endured. GRE should therefore identify the mechanism through which prior states affect later transition probabilities, costs, or available options. Historical narrative and formal path dependence can complement one another only when the connection between sequence and later constraint is made explicit.

1.20.4 Coevolution of States, Relations, and Constraints

A generative economic system can contain several mutually changing layers. Actor states change, relations among actors change, and the constraints governing those relations change. Treating these layers separately can be useful for local analysis, while longer-horizon explanation often requires their coevolution. The scientific challenge is to represent feedback across layers without assuming that one layer permanently determines the others.

Consider an employment system. Workers acquire skills and experience; firms reorganize tasks and technologies; labor-market relations alter bargaining positions and mobility; legal institutions revise classifications and protections; training systems respond to changing demand. Each layer conditions the others. A change in technology can alter the value of a skill, which changes hiring patterns, which changes incentives for training, which changes the future distribution of skills, which can alter the technologies firms find profitable to adopt. A single state variable can describe one slice of this process, while a coevolutionary account follows the reciprocal modification of several kinds of structure.

The same logic applies to finance, platforms, and knowledge systems. Credit histories influence lending relations, lending relations shape investment, investment changes productive capability, and new capabilities alter future creditworthiness. Platform rules shape interaction, participant behavior generates data, data influence

algorithmic ranking, and revised ranking changes participant behavior. Research institutions allocate attention and resources, resulting inquiry changes the stock of knowledge, and new knowledge changes which questions appear feasible or valuable. In each case, generative relations operate through recursive coupling among state, relation, and constraint.

Complex-adaptive-systems research supplies conceptual resources for such recursive organization. Holland's treatment of evolving signals and boundaries emphasizes that adaptation can occur through changing interactions and organizational structures rather than through movement of an isolated variable alone (Holland 2012). Evolutionary economics similarly treats behavior, search, capability, and competitive structure as historically linked processes (Nelson and Winter 1982). GRE draws from these approaches a methodological possibility: the object of explanation can be a coupled trajectory in which the units, relations, and constraints relevant to later behavior are partly generated by earlier behavior.

Coevolution also limits simple attribution. When several layers change together, a downstream outcome can be jointly generated by processes that are difficult to isolate as independent causes. A regulatory change may alter firm behavior only because technological capabilities have changed; a technological opportunity may matter only because institutions recognize a new organizational form; a new organizational form may diffuse only because financing relations support it. Generative analysis therefore requires language for contribution, enablement, mediation, and co-generation in addition to strong single-cause attribution. Later methodological chapters develop those distinctions explicitly.

The coevolutionary perspective remains compatible with analytical decomposition. Researchers can hold some layers fixed to study a local mechanism, compare periods in which one layer changes faster than another, or construct reduced models for particular questions. The crucial scientific advance is the ability to relax those simplifications when they become explanatorily consequential. States, relations, and constraints can be modeled as jointly historical objects rather than as permanently separated levels.

1.20.5 Evolution of Transformation Rules

The strongest form of generative analysis treats the rule governing change as historically variable. A conventional dynamical representation specifies how a current state is transformed into a later state under a given transition rule. Many economic applications can legitimately hold that rule fixed over the period of interest. A generative system can require another level of description when learning, institutional change, technological innovation, or reflexive adaptation modifies the rule through which subsequent states are produced.

Economic routines provide a concrete illustration. Nelson and Winter treat organizational routines as persistent elements of firm behavior while also giving search and innovation a role in changing the routines and capabilities that shape later behavior (Nelson and Winter 1982). The analytical implication is subtle. A firm's current action may be generated by an inherited routine, while the outcome of that action can contribute to search, learning, imitation, or selection that changes the routine governing

future action. The transformation process therefore has a history of its own.

Institutional rules exhibit a similar recursion. Property definitions, accounting standards, eligibility criteria, contractual forms, and regulatory classifications can shape economic action. Repeated use of those rules can generate disputes, workarounds, constituencies, complementary investments, or new information that later motivates rule revision. Once revised, the institution changes the mapping from economic circumstance to permitted or rewarded action. Institutional change is therefore generative in a stronger sense when it alters the rule through which later economic relations are constituted or evaluated.

Adaptive digital systems make this issue especially visible. A ranking or recommendation rule can be updated from observed behavior; participants then adapt to the updated rule; their adaptation generates new observations that affect subsequent updates. The operative transformation rule is consequently embedded in a feedback process. Economic analysis that treats the algorithm as a fixed parameter can be adequate for a short interval, while analysis of longer-run behavior may need to represent the update procedure, participant adaptation, and institutional oversight as part of the system.

The evolution of transformation rules marks the boundary between a merely dynamic account and the more demanding conception of generativity developed in this book. Generativity does not require that every transformation rule change continuously. It requires analytical openness to the possibility that present relations can modify the mechanisms, constraints, classifications, or rules through which future relations are produced. This is the scientific condition that later permits GRE to study economic spacetime as historically generative rather than as a fixed arena containing changing entities.

The section's five conditions can now be stated together. Endogenous dynamics make conditioning variables part of the process; structure-changing models allow relational architectures to evolve; path dependence gives prior trajectories causal relevance; coevolution links changing states, relations, and constraints; evolving transformation rules make the mechanisms of change themselves historical objects. These scientific resources do not yield a complete generative economics, and none supplies a universal formalism. They establish a historical possibility: contemporary economic analysis possesses increasingly developed languages for asking how economic activity changes the conditions of its own continuation. Section 1.21 turns from scientific form to computational capacity and asks what makes these richer representations practically analyzable at scale.

1.21 Computational Conditions of Contemporary Analysis

This section identifies the computational conditions that make the richer scientific representations introduced in Sections 1.19 and 1.20 practically analyzable. Its role is distinct from the previous two sections. Scientific formalisms establish that relational and generative structures can be represented; computational infrastructures determine whether those representations can be explored at useful scale, temporal resolution, and heterogeneity. The objective is to reconstruct a historical shift in fea-

sible economic inquiry rather than to equate computational power with explanatory adequacy. The discussion proceeds from the expansion of computational capacity to large-scale simulation, dynamic network computation, agent-based and multiscale modeling, and emerging paradigms that make increasingly complex models differentiable, composable, and computationally tractable. Throughout, computational feasibility is treated as an enabling condition for inquiry rather than as evidence that a model is empirically correct.

1.21.1 Expansion of Computational Capacity

The first computational condition is the sustained expansion of the scale at which numerical procedures can be executed. Gordon Moore's 1965 observation concerning rapidly increasing component density in integrated circuits became a historical marker for a broader technological trajectory in which increasingly capable processors, memory systems, storage, networking, and software infrastructures reduced the cost of many forms of computation (Moore 1965). The significance for economic analysis lies less in any single hardware metric than in the resulting expansion of the model classes that can be investigated routinely.

Earlier analytical methods often benefited from strong reductions because computational constraints made large heterogeneous systems difficult to represent directly. Representative agents, aggregate state variables, stationary environments, sparse interaction structures, and short simulation horizons can all be theoretically justified in appropriate domains. They also reduce computational burden. As computational capacity expands, however, some reductions become choices rather than unavoidable technical necessities. Researchers can preserve more heterogeneity, maintain explicit interaction structures, examine longer trajectories, repeat analyses across parameter configurations, and compare several model specifications rather than committing immediately to a single reduced representation.

The increase in feasible computation changes the relation between model complexity and empirical discipline. A richer model can carry more of the structure that the research question considers consequential: multiple actor types, institutional rules, evolving networks, spatial structure, delayed effects, and several temporal scales. The same computational expansion also makes it easier to construct models whose internal complexity exceeds their evidential support. The historical opening is therefore double. More relational structure can be represented, while the space of underdetermined models also becomes larger.

Computation further changes the practice of uncertainty analysis. A model need not be examined at one parameter vector or under one deterministic scenario. Ensembles can explore sensitivity to initial conditions, stochastic shocks, behavioral rules, boundary assumptions, and alternative institutional specifications. This makes uncertainty more explicit when the ensemble is designed around substantively meaningful alternatives. It can also create an appearance of comprehensiveness when the sampled alternatives omit the main source of uncertainty. Computational abundance expands the set of questions that can be asked; it does not decide which uncertainties deserve representation.

For GRE, the main historical consequence is that relational detail has become less

dependent on analytical compression than in earlier computational environments. A model can increasingly retain the identity of several classes of relations, preserve temporal ordering, and follow feedback across scales. This capacity does not remove the need for abstraction. It changes the reason for abstraction: simplification should be defended by the analytical role it serves rather than by computational necessity alone.

1.21.2 Large-Scale Simulation

Simulation provides a second computational condition because it allows researchers to investigate systems whose aggregate consequences are difficult to derive analytically from local rules. Epstein and Axtell's *Growing Artificial Societies* provided an influential demonstration in the social sciences by constructing artificial populations in which macro-level patterns emerge from repeated interaction among heterogeneous agents (Epstein and Axtell 1996). The methodological importance of this approach lies in the ability to make an explicit set of micro-level rules generate a trajectory that can then be inspected, perturbed, and compared with alternatives.

Large-scale simulation is especially useful when a research question concerns interaction effects. A rule governing one actor may have little meaning in isolation because its consequences depend on the simultaneous behavior of many others. Credit defaults can propagate through balance-sheet relations; supply interruptions can move through production networks; migration decisions can alter local opportunity structures; institutional rules can induce adaptation that changes the effect of the rule itself. Simulation permits these interactions to be represented as processes rather than collapsed immediately into an equilibrium condition or comparative-static result.

The growth of computational capacity also makes repeated simulation more important than a single illustrative run. A stochastic model can be executed across many seeds; a parameterized model can be explored across ranges; several institutional designs can be compared under the same disturbance; rare but consequential trajectories can be searched for rather than assumed absent. Such ensembles are particularly relevant to generative analysis because the object of interest may be a distribution of possible trajectories, the frequency of transition between regimes, or the conditions under which a local perturbation produces a system-level effect.

Simulation also supports counterfactual reasoning under explicit assumptions. A model can compare a historical baseline with altered network structure, different access rules, alternative timing, or modified institutional constraints. The comparison is valuable because it forces the counterfactual mechanism to be stated operationally. Its evidential standing remains conditional. A simulated counterfactual shows what follows within the model from the altered assumptions; it does not establish that the real economy would have followed the same path. Calibration, validation, historical plausibility, and comparison with observed trajectories remain separate tasks.

This distinction is essential for GRE. Simulation can make generative hypotheses visible by showing how a proposed relational mechanism could produce a downstream configuration. It can also reveal that an intuitively plausible mechanism fails to generate the expected effect once interactions and feedback are made explicit.

Both outcomes are useful. Neither converts synthetic output into empirical confirmation. Computational simulation is therefore best understood as a controlled environment for mechanism exploration, sensitivity analysis, and the comparison of possible trajectories.

1.21.3 Dynamic Network Computation

Network analysis becomes more powerful when computation can follow relations that change through time. A static network records a topology for a selected interval. Many economic relations, however, form, dissolve, recur, switch direction, or change weight. Temporal-network research formalizes this problem by representing contacts or ties with explicit timing and duration, making the order of interactions part of the analytical object (Holme and Saramäki 2012). The computational condition is important because temporal ordering substantially enlarges the information required to describe a network and the number of possible paths through it.

Dynamic network computation changes several familiar economic questions. In finance, the relevant exposure structure can evolve as positions are opened and closed. In production, suppliers can be replaced, duplicated, or vertically integrated. In labor markets, workers move among organizations and carry skills and information across firm boundaries. On digital platforms, recommendation, ranking, and participant behavior can change the effective network on a much shorter timescale. The economic consequence of a relation can therefore depend on when it exists, what preceded it, and which other relations are simultaneously active.

Temporal computation also makes path-sensitive diffusion analyzable. A sequence of edges can transmit information, liquidity, disruption, reputation, or opportunity only when their timing permits the sequence to occur. Two networks with the same aggregate number of edges can support different propagation because the edges appear in different orders. This matters for GRE because the generative effect of a relation may depend on whether it arrives before or after a complementary capability, institutional recognition, or competing pathway has formed.

The same computational logic extends to networks whose topology coevolves with behavior. Participants can create or sever relations in response to outcomes, and those relational changes alter later incentives. A supplier may diversify after disruption; a lender may reduce exposure after loss; a platform participant may migrate after changes in ranking; a research group may form new collaborations after acquiring a technique. Computation permits the network itself to become an endogenous state variable rather than a fixed background for action.

Dynamic network models nevertheless inherit the representational limits described earlier. The existence of a time-stamped edge still depends on a definition of the relation, a boundary around the system, and data that capture the relation with sufficient resolution. Computational tractability cannot recover relations that were never observed, distinguish causal influence from coincident timing by itself, or decide whether a higher-order configuration should be represented as several edges or as a distinct collective relation. The historical advance is the capacity to compute with changing topology; the epistemic burden of defining and validating the topology remains.

1.21.4 Agent-Based and Multiscale Modeling

Agent-based computation provides a fourth condition by permitting heterogeneous actors, local rules, institutional constraints, and interaction structures to coexist within one executable model. Agent-based computational economics explicitly developed this possibility as an alternative way to study decentralized economic systems whose aggregate behavior is generated through interaction, learning, strategic adaptation, conventions, organizations, and institutional arrangements (Tesfatsion 2006). The approach complements the complex-systems perspective of Section 1.19 by giving the interacting components an operational computational representation.

The value of agent-based modeling for GRE lies in the ability to preserve heterogeneity that may be generatively consequential. Workers can differ in skills and network position; firms can differ in routines and financing; households can differ in obligations and access; institutions can impose different rules across categories; artificial agents can possess capabilities unavailable to human participants. These differences can be represented together with local interaction rules, allowing the analyst to ask whether macro-level concentration, segregation, coordination, instability, or capacity formation emerges from the specified configuration.

Agent-based representation does not require an ontological commitment to isolated atomic individuals. An “agent” can be an analytical object whose behavior is partly constituted by relational position, organizational membership, inherited resources, institutional classification, or internal state. Organizations, households, collective bodies, and algorithmic systems can also be represented as agents at appropriate scales. The computational framework is therefore compatible with a relational account so long as the model does not treat the agent’s attributes as independent of the relations the theory claims are constitutive.

Multiscale modeling extends the same logic when economically relevant processes operate at different resolutions. Local transactions can occur rapidly, organizational routines can change more slowly, institutions can evolve over longer intervals, and infrastructure can impose constraints across several scales. A computational model can couple these layers explicitly, allowing fast processes to generate inputs into slower processes and slower structures to condition the space within which faster interactions occur. Such coupling is particularly important for the fast–slow historical perspective developed elsewhere in GR, while GRE need not assume that every economic system admits a clean separation of timescales.

The difficulty lies in validation. A model that includes heterogeneous agents, networks, institutions, and several scales can fit many qualitative patterns for many different reasons. More computational detail can increase realism in one dimension while reducing identifiability overall. Agent-based and multiscale models therefore require careful documentation of which mechanisms are load-bearing, which parameters are calibrated, which structures are empirical, and which are exploratory. Their historical significance is that such questions can now be investigated computationally with far fewer reductions than before; their scientific standing still depends on evidence and comparative model evaluation.

1.21.5 Emerging Computational Paradigms

Contemporary computation is also changing the form of modeling itself. One important development is automatic differentiation, which evaluates derivatives of computer programs efficiently and has become a central component of modern machine-learning systems and differentiable programming (Baydin et al. 2018). The broader significance is that increasingly complex computational pipelines can be treated as composable objects whose parameters may be estimated, optimized, or linked to inference procedures without requiring every derivative or intermediate transformation to be derived manually.

For relational and generative analysis, this opens several possibilities. A simulation can be connected to parameter estimation; a network process can be embedded within a larger model; a surrogate model can approximate a computationally expensive component; and an end-to-end pipeline can propagate information across several stages. These possibilities reduce the technical distance between formal specification, simulation, estimation, and sensitivity analysis. They also make it easier to construct hybrid models that combine institutional rules, network dynamics, statistical components, and learned functions.

Parallel and distributed execution further alters the feasible scale of inquiry. Rather than treating a simulation as a single sequence of calculations, researchers can divide parameter sweeps, Monte Carlo replications, network partitions, or independent scenarios across many processors or machines. Accelerator-based computation makes some matrix- and tensor-intensive models practical at scales that would previously have been prohibitive. Cloud infrastructures can make such resources available without local ownership of equivalent hardware. These developments broaden access to some forms of computation while also introducing new dependencies on software ecosystems, specialized hardware, infrastructure providers, and technical expertise.

Machine-learning methods introduce another change by allowing components of a model to be estimated from data when an explicit functional form is unavailable or deliberately left flexible. This can support prediction, emulation, dimensionality reduction, or the approximation of expensive simulations. It can also weaken interpretability when learned components become more complex than the mechanism the model was intended to clarify. GRE therefore treats learned components as one computational resource among others. Their use should follow the epistemic role of the model: prediction, mechanism reconstruction, counterfactual exploration, or decision support impose different requirements on transparency and validation.

The emergence of these paradigms does not imply convergence toward a single computational methodology. Some economic questions remain best served by transparent low-dimensional models; others benefit from simulation, networks, agent-based systems, or learned approximations. The historical change lies in the expansion of feasible combinations. Formal components can increasingly be composed, differentiated, simulated, and compared within one computational environment. This makes generative relational questions more tractable while also making model governance, reproducibility, provenance, and infrastructure dependence more important.

The five conditions developed in this section define a computational opening

rather than a computational solution. Expanding capacity permits richer representations; large-scale simulation explores interaction and counterfactual trajectories; dynamic network computation follows changing relations; agent-based and multi-scale models preserve heterogeneity across levels; and emerging paradigms make complex computational systems increasingly composable. None resolves the evidential problem of whether the modeled relations correspond to the world. Section 1.22 therefore turns from computational feasibility to evidential conditions: the data infrastructures, longitudinal records, continuous observation, and distributed evidence systems through which relational and generative claims may be confronted with empirical traces.

1.22 Evidential Conditions of Contemporary Governance

This section identifies the evidential conditions that make more adaptive forms of economic governance and institutional decision-making increasingly feasible. Its role in the chapter is narrower than a general theory of governance. The objective is to reconstruct how changes in data production, longitudinal linkage, distributed observation, evidence updating, and feedback-oriented decision processes alter the practical possibility of following relational and generative economic processes through time. The discussion proceeds from continuous data collection to longitudinal infrastructures, distributed sources of observation, iterative evidence revision, and adaptive decision processes. The method is historical and institutional: the section treats contemporary evidence infrastructures as enabling conditions for GRE while preserving the distinction between evidence, interpretation, judgment, and legitimate action.

1.22.1 Continuous Data Collection

A first evidential condition is the movement from episodic measurement toward more continuous forms of observation. Traditional censuses, surveys, audits, and periodic administrative reports remain indispensable because they are designed around explicit statistical purposes and can provide carefully structured observations. Contemporary public sectors, however, increasingly operate alongside administrative databases, digital service records, transaction logs, sensor streams, platform traces, and machine-generated records that are produced as part of ongoing activity. The World Bank's 2021 account of public-intent data explicitly includes administrative, census, survey, citizen-generated, and machine-generated sources and emphasizes their roles in policy design, execution, monitoring, and evaluation (World Bank 2021). The OECD similarly describes a data-driven public sector as one in which data become an integral resource for policy making, service delivery, organizational management, and innovation (Ooijen et al. 2019).

The methodological change is temporal as much as quantitative. A periodic observation compares a small number of states separated by an interval. A more continuous evidential infrastructure can preserve intermediate changes, timing, sequence, and recurrence. This matters for relational analysis because a relation can form, weaken, reverse, or disappear between conventional observation points. It matters even more for generative analysis because the sequence through which capacities,

constraints, and institutional responses develop may be part of the explanation. A record of endpoints can show that a transition occurred; a denser temporal record can sometimes reveal the pathway through which it occurred.

Continuous observation also changes the practical relation between evidence and intervention. When evidence arrives only after a long reporting cycle, institutions often evaluate effects retrospectively. When administrative or operational indicators are updated more frequently, deviations can become visible during implementation. Such visibility does not by itself determine whether a deviation matters, whether the underlying measure is valid, or whether action should follow. It changes the feasible timing of inquiry. Institutions can potentially distinguish a transient fluctuation from a persistent change sooner, compare the development of several trajectories, and identify where additional investigation is needed before the end of an intervention cycle.

The increase in observational frequency introduces corresponding risks. Data generated continuously are often generated for operational rather than analytical purposes. Definitions can change as software, eligibility rules, administrative procedures, or institutional priorities change. Missingness may become systematic because participation in digital systems is unequally distributed. High-frequency measurement can also encourage attention to what changes rapidly while slower structures remain difficult to observe. A rich stream of immediate indicators may therefore produce an evidential bias toward fast variables precisely when the economic process of interest depends on institutional, demographic, ecological, or cultural variables that move more slowly.

Continuous data collection should therefore be treated as an expansion of temporal access rather than as complete observation. It increases the density of traces available to analysis while leaving unresolved questions of validity, provenance, representativeness, privacy, and interpretation. For GRE, its historical significance lies in making trajectories easier to reconstruct. The evidential object can increasingly become a process unfolding through time rather than a pair of states compared before and after an event.

1.22.2 Longitudinal Evidence Infrastructures

The second condition is the development of infrastructures capable of relating observations across time. Longitudinal evidence differs from repeated measurement when records can be connected to persons, organizations, places, programs, institutions, or relational configurations in ways that preserve temporal continuity. The analytical gain is substantial. Repeated cross-sections reveal population-level change; longitudinal systems can additionally show trajectories, persistence, transitions, and heterogeneous sequences within the changing population.

Contemporary data-governance initiatives increasingly recognize integration and linkage as institutional questions rather than purely technical tasks. The World Bank's *World Development Report 2021* develops the idea of an integrated national data system in which data production, exchange, and use are connected across public institutions and, under appropriate arrangements, with academic, private-sector, civil-society, and individual participants (World Bank 2021). The OECD's account of

a data-driven public sector likewise emphasizes data governance across the public sector rather than isolated use within single agencies (Ooijen et al. 2019). These developments create the possibility of observing economic processes whose relevant evidence is distributed across organizational boundaries.

Longitudinal linkage is especially important for generative questions because effects can appear after substantial delay. An educational intervention may affect labor-market position years later; infrastructure can alter patterns of firm formation only after complementary investments accumulate; institutional recognition can affect credit access, organizational survival, or intergenerational opportunity over long horizons. A short evaluation window may correctly measure immediate outputs while missing later relational or capacity effects. Longitudinal evidence creates the possibility of following these effects without assuming that the value of an intervention is exhausted by its first measurable consequence.

The same infrastructure also supports the analysis of path dependence. If later outcomes depend on the sequence through which earlier states were reached, an evidential system must preserve more than current status. It must retain enough historical structure to distinguish trajectories that end at similar observed states but arrive there through different institutional histories. This requirement links evidence design to the historical arguments developed in Section 1.20: path dependence is analytically meaningful only when the history required to identify the path has been preserved.

Longitudinal evidence is therefore inseparable from institutional memory. Data definitions, identifiers, classifications, and access rules must persist long enough for records to remain comparable, or their changes must themselves be documented. A category that changes meaning over time can create false continuity when the same label is treated as stable. Conversely, a change in classification may be evidence of institutional transformation rather than a nuisance to be harmonized away. The evidential infrastructure must preserve both observations and the history of the categories through which those observations were produced.

This point is central to GRE's emphasis on future re-epistemization. Evidence retained for one analytical purpose may later be revisited under different concepts or methods. Longitudinal infrastructure therefore has a generative epistemic function of its own: it preserves the possibility that future inquiry will reconstruct present processes differently. The value of preservation is not limited to improving current prediction. It also maintains future access to historical relational structure.

1.22.3 Distributed Observation

A third evidential condition is the distribution of observation across many institutions and forms of participation. Economic reality is rarely observed from a single point. Tax administrations, statistical offices, regulatory agencies, firms, financial institutions, researchers, civil-society organizations, households, digital platforms, sensors, and citizens can all produce traces relevant to the same economic process. The World Bank describes the contemporary data-governance ecosystem as extending beyond government to the private sector, civil society, academia, and other actors, and it emphasizes the institutional problem of enabling data to be combined and exchanged

while safeguarding rights, security, and integrity (World Bank 2021).

Distributed observation matters because different observers have access to different manifestations of the same relational system. Administrative records may reveal recognized membership and formal transfers. Surveys may reveal experiences and expectations that administrative systems do not record. Transaction data may reveal realized exchanges with high temporal resolution. Qualitative fieldwork may reveal meanings and informal relations that standardized records cannot represent. Citizen-generated evidence can make visible local conditions that centralized institutions observe poorly. None of these sources provides a neutral view from nowhere; their combination can nevertheless reveal discrepancies that would remain invisible within a single evidential regime.

The analytical opportunity is therefore triangulation across positions. When several sources converge despite different production processes, confidence in some claims may increase. When they diverge, the divergence can become substantive evidence. A statistical indicator may report stable service delivery while interview evidence documents changing barriers to access. Administrative records may show formal program participation while transaction data reveal that effective use is concentrated among a subset of recognized participants. The discrepancy may indicate measurement error, category mismatch, selective observation, or a genuine difference between formal recognition and realized relation.

Distributed observation also introduces a relational problem among evidence producers themselves. Data are generated under different mandates, incentives, legal constraints, business models, technical standards, and temporal rhythms. Combining them requires interfaces: shared identifiers, interoperable formats, provenance records, agreements over access, and institutions capable of resolving conflict among sources. Evidence infrastructure is therefore itself a relational system. Its architecture influences which observations can be connected, which remain isolated, which actors gain interpretive authority, and which absences become structurally persistent.

This makes evidential plurality both a strength and a governance challenge. A system with many observers can reduce dependence on one institutional viewpoint, yet concentration in key infrastructures can create new forms of asymmetry. Platforms with privileged access to behavioral data may see relations unavailable to public institutions or researchers. Public agencies may possess administrative authority that makes their classifications consequential even when alternative descriptions exist. Communities may hold contextual knowledge that has little institutional standing. The distribution of evidence production is consequently linked to the distribution of epistemic power, a theme developed more fully in later parts of the book.

For GRE, distributed observation expands the possibility of reconstructing relational reality from multiple manifestations. It also reinforces ontological and epistemic restraint. Agreement among sources can strengthen an inference; disagreement can reveal the dependence of representation on institutional position. The existence of many observations does not eliminate the inverse problem of moving from manifestations to the relational structures that generated them.

1.22.4 Evidence Updating

The fourth condition is the institutional capacity to revise evidential judgments as new information becomes available. Evidence-informed policy making is defined by the OECD broadly enough to include statistics, data, research evidence, and evaluation in decisions to plan, implement, and, where relevant, alter public policies and programs (OECD 2020a). This formulation is important because it treats evidence as part of an iterative process rather than as a single justificatory input delivered before action begins.

Updating has several analytically distinct forms. New observations can change estimates within an existing model. They can reveal that a previously measured relation has weakened or reversed. They can expose a missing variable, a classification error, or a boundary condition that the original analysis omitted. In stronger cases, evidence can undermine the representation itself and require a change of model, category, or theory. GRE's generative perspective makes the last possibility especially important because the system under study may change the relations that made an earlier model adequate.

An updating infrastructure must therefore preserve revision rather than merely replace earlier states of knowledge. Versioned datasets, documented methodological changes, model histories, evaluation reports, and explicit records of revised assumptions allow later researchers to reconstruct why a decision or interpretation changed. Without such provenance, an institution can update its current estimate while losing the history of its own learning. The result is a paradoxical form of informational abundance in which current dashboards improve while the trajectory of interpretation becomes harder to recover.

Evidence updating also requires a distinction between evidential revision and automatic policy response. An estimate can change without changing the relevant decision because the difference is too small, the intervention is costly to reverse, normative commitments constrain the available actions, or several evidential sources point in different directions. Conversely, a modest evidential change can become decision-relevant when a system is close to an institutional threshold or when delay carries large costs. Evidence informs judgment; it does not mechanically substitute for judgment.

This distinction becomes more important when evidence arrives at different frequencies. Operational indicators may update daily, survey estimates quarterly, fiscal data annually, and structural outcomes only over several years. Treating every update as equally decision-relevant would allow high-frequency indicators to dominate simply because they arrive more often. A generative-relational approach instead asks which timescale corresponds to the process being governed and whether the evidence updates the fast state, the slower relational architecture, or the assumptions linking the two.

The scientific significance of evidence updating is therefore deeper than responsiveness. It institutionalizes revisability. A governance system capable of learning from new evidence can revise representations, evaluations, and interventions without assuming that the original model was final. This does not guarantee learning; institutional incentives can suppress inconvenient findings, methods can remain

locked into outdated categories, and rapid updating can amplify noisy indicators. It creates a historical possibility for governance to remain epistemically open while acting under incomplete knowledge.

1.22.5 Adaptive Decision Processes

The fifth condition is the incorporation of feedback into decision processes. The OECD's policy framework on public governance describes monitoring information as a resource that can feed decision-making, track implementation, and support adjustment, and it emphasizes feedback loops and evaluation across the policy cycle (OECD 2020c). Its cross-country work on policy evaluation likewise treats evaluation as a source of accountability, learning, and improved decision-making rather than as a purely retrospective exercise (OECD 2020b). These institutional practices make it increasingly feasible to treat some interventions as revisable processes rather than fixed decisions followed by a single final assessment.

Adaptive decision processes are especially relevant where causal structure is uncertain and the consequences of intervention unfold over time. An institution can begin with a provisional model, monitor selected consequences, compare observed and expected trajectories, revise the model, and alter the intervention when predefined or deliberatively assessed conditions are met. The practical logic resembles the research-practice relation developed later in this book: action generates evidence, evidence revises understanding, and revised understanding changes subsequent action.

Adaptation requires more than frequent data. The evidence must arrive in a form that can be interpreted relative to the decision. Institutions need authority to revise implementation, procedures for recording why revisions occur, and safeguards against the continuous redefinition of objectives in response to short-term indicators. An adaptive process also needs temporal discipline. Some interventions require enough time for effects to emerge; premature adjustment can prevent the very relation being evaluated from stabilizing. Other interventions create rapidly compounding harms for which delayed response is costly. The appropriate update interval is therefore a property of the intervention and the relevant dynamics rather than a universal preference for faster feedback.

Reversibility further conditions adaptation. When an action can be adjusted at low cost, institutions can learn through incremental changes more readily. When interventions alter legal status, infrastructure, ecological conditions, or long-horizon expectations, later correction may be partial or impossible. In such cases, evidence updating still matters, but adaptive governance must operate through staged commitments, safeguards, contingency planning, or more demanding *ex ante* justification. The availability of real-time evidence does not make irreversible action experimental in the same sense as a reversible pilot.

Adaptive decision processes also remain normatively and politically situated. Evidence can indicate consequences and uncertainty; it does not decide which consequences deserve priority, whose losses require protection, or which distributions are acceptable. These questions lead directly to Section 1.23, which examines the political and normative conditions under which relational governance becomes institution-

ally meaningful. The evidential architecture developed here supplies information for decision; it does not supply the legitimacy or normative ordering of the decision itself.

The historical opening is therefore specific. Contemporary institutions can increasingly observe economic processes at higher frequency, connect evidence across time and organizations, revise interpretations as evidence accumulates, and alter some decisions during implementation. These capacities make relational and generative governance more technically conceivable because trajectories and feedback can enter institutional practice. They also expose new problems of surveillance, data concentration, representational power, model dependence, and temporal mismatch. GRE treats this evidential expansion as a condition of possibility for adaptive economic analysis and action, while retaining the distinction between observing a generative system and possessing the knowledge or authority required to control it.

1.23 Political and Normative Conditions of Relational Governance

This section identifies the political and normative conditions that have made relational questions increasingly admissible within economic governance. Its role in the chapter is historical and disciplinary: Sections 1.19–1.22 reconstructed scientific, computational, and evidential capacities, whereas the present section concerns the standards by which institutions decide whose relations, capabilities, interests, and temporal claims enter the field of evaluation. The objective is to show how political participation, human-rights recognition, capability-oriented development, commons governance, and environmental or intergenerational normativity widened the range of economically relevant objects beyond administratively isolated recipients and contemporaneous outputs. The discussion proceeds through those five developments in sequence. Its method is reconstructive rather than prescriptive. The section treats these traditions as historical conditions that make relational governance thinkable and institutionally legible; systematic governance theory remains outside the scope of this chapter and is reserved for later GRE interfaces with Generative Relational Governance.

1.23.1 Political Participation

Political participation changes the position from which governance problems are formulated. An administrative model can treat persons primarily as recipients, taxpayers, employees, consumers, regulated entities, or statistical units. Participatory arrangements introduce an additional status: persons and organized groups can enter the process through which public problems, priorities, and institutional responses are interpreted. This change matters for relational analysis because affected parties often possess information about dependencies, informal practices, exclusion mechanisms, and downstream effects that remain weakly represented in administrative categories.

The institutionalization of participation as a governance concern is visible in

twentieth-century rights instruments. Article 25 of the International Covenant on Civil and Political Rights recognizes for citizens the right and opportunity to take part in the conduct of public affairs, directly or through freely chosen representatives, alongside electoral and public-service rights (United Nations General Assembly 1966). The significance for the present argument lies in the formal recognition of participation as part of the structure of public affairs. Governance is thereby evaluated partly in terms of relations between institutions and those subject to, represented within, or participating in public decisions.

Participation also alters the evidential architecture described in Section 1.22. Distributed observation acquires a political dimension when experiences and local knowledge enter institutional deliberation through hearings, consultation, representative bodies, associations, unions, community organizations, or other channels. Administrative evidence can reveal formally recorded transfers and statuses; participants can identify how those arrangements operate in practice. A benefit formally available to all eligible persons may be difficult to obtain because of language, time, location, documentation, digital access, stigma, or dependence on intermediaries. These conditions are relational because the effective outcome depends on the configuration linking claimant, institution, infrastructure, and other actors.

The analytical value of participation therefore extends beyond preference aggregation. Participants can contest the definition of the object itself. A housing problem may be framed as insufficient units, insecure tenure, spatial exclusion, financing constraints, discriminatory access, or inadequate transport relations. Each formulation identifies a different relational structure and therefore a different possible intervention. Participatory processes can expose these competing formulations before an administrative category becomes fixed as the sole representation of the problem.

Participation also has clear limits. Access is uneven, organized interests possess different resources, participation can be costly, and institutional forums can reproduce existing asymmetries. A meeting or consultation does not establish that every affected relation has become visible. Nor does participation remove the need for expertise, legal authority, or accountable decision. For GRE, the historical importance of participatory norms is more specific: they weaken the assumption that the economic object of governance can be fully specified from the administrative center before those situated within its relational field are heard.

This condition matters especially for generative questions. The long-run effects of an intervention may include changes in trust, capability, dependency, organizational learning, and future access. Such effects often appear first in the experiences of participants rather than in the primary output metric. Political participation creates institutional channels through which some of these effects can become objects of public interpretation and later evidence.

1.23.2 Human Rights and Institutional Recognition

Human-rights frameworks provide a second condition by institutionalizing forms of recognition that are not exhausted by economic role. The Universal Declaration of Human Rights states that everyone has a right to recognition everywhere as a person before the law and articulates equality before the law among its foundational guar-

antees (United Nations General Assembly 1948). The present chapter does not derive a complete normative theory from these provisions. Their historical significance for GRE lies in the expansion of institutional standing: a person enters governance as a bearer of recognized claims whose relevance cannot be reduced to ownership, productivity, income, contractual position, or contribution to measured output.

This development complicates purely recipient-centered economic description. A transfer system can be efficient in reaching a target population while still affecting privacy, family relations, freedom of association, legal recognition, or access to remedy. A labor arrangement can increase output while altering dependence, voice, or exposure to arbitrary exclusion. A data infrastructure can improve administrative coordination while creating new asymmetries in visibility and classification. Once recognized rights become part of the institutional environment, these dimensions become relevant constraints on the design and evaluation of economic governance.

Institutional recognition is itself relational. A right requires more than the internal possession of an attribute by an isolated individual. Its practical effect depends on relations among the person, legal rules, administrative bodies, courts or review mechanisms, documentation systems, and other actors capable of honoring or violating the recognized claim. Formal recognition can therefore be represented as a stabilized institutional relation whose effectiveness varies with access, enforcement, and the organization of remedies. This does not reduce rights to economic relations; it clarifies why rights matter to economic institutions even when their normative foundation is external to economics.

The distinction is especially important for heterogeneous subjects. Section 1.16 separated economic subjecthood from final metaphysical or moral personhood. Human-rights traditions reinforce the need to preserve that separation. Economic systems may assign roles to persons, organizations, or artificial agents for purposes of contract, production, or allocation, while normative and legal standing follows different criteria. GRE therefore requires care whenever economic functionality is used as a proxy for broader status. Productive capacity does not settle legal recognition, and administrative recognition does not settle moral standing.

Human-rights frameworks also expand the temporal stability of claims. Economic positions can change rapidly: employment can end, assets can be lost, and market participation can fluctuate. Rights-based recognition is designed to persist across many such changes. This creates an institutional layer whose continuity can constrain short-horizon optimization. From a generative perspective, such continuity can affect future options by preserving access to institutions, remedies, participation, and protections during periods of economic transition.

The resulting contribution to relational governance is a boundary condition rather than a policy formula. Rights language identifies some relations and statuses as institutionally salient even when they are poorly captured by current output measures. GRE can analyze the economic consequences of those relations while leaving their ultimate normative justification to political philosophy, law, and human-rights theory.

1.23.3 Capability and Human Development

The capability and human-development traditions provide a third condition by shifting evaluation from resources alone toward substantive opportunities. Sen's capability-oriented account treats development in terms of the expansion of substantive freedoms and the effective opportunities persons possess to pursue valued ways of living (Sen 1999). This change is directly relevant to relational economics because identical resource bundles can generate different practical possibilities when persons occupy different social, institutional, bodily, geographic, or infrastructural positions.

Capability analysis changes the evaluative unit without dissolving the importance of resources. Income, assets, services, and public provision remain means through which capabilities can be supported. Their significance depends on conversion conditions. A transport subsidy has little mobility effect where transport infrastructure is absent. Digital access has different consequences when literacy, language, hardware, electricity, or institutional recognition differ. Formal educational access has different value when care obligations, discrimination, health, or local labor-market structures alter the feasible use of education. These examples make visible the relational environment through which resources become effective opportunities.

This shift also creates a bridge between entity-centered and relational evaluation. The person remains a central normative referent, while the explanation of effective freedom requires attention to the relations surrounding that person. The analytical question becomes wider than resource possession. It includes the social and institutional conditions through which possession is converted into the ability to act. GRE can therefore inherit an important discipline from capability analysis: resources, outcomes, and opportunities should remain distinct analytical objects.

The capability perspective also introduces plurality into economic evaluation. If development concerns several kinds of functioning and freedom, a single aggregate output measure provides only a partial view. Health, education, mobility, participation, security, and other dimensions can interact without being reducible to one immediate metric. For GRE, the importance of this plurality is methodological as well as normative. Heterogeneous values may require different evidential forms and may circulate through different relational pathways. Later chapters extend this problem into value conversion and partial commensurability; the present section identifies the historical tradition that made plural opportunity structures a legitimate economic concern.

A generative extension follows when present capabilities affect the production of future capabilities. Education can generate learning capacity; political participation can generate organizational experience; secure institutional recognition can expand future access; health can affect later employment and care relations. Capability at one moment can therefore become an enabling condition for capability at another. This temporal extension should be handled cautiously because long-horizon attribution is difficult. It nevertheless reveals why capability-oriented evaluation is congenial to a generative framework: the object of interest can include the changing possibility set through which later economic action becomes feasible.

The relationship between GRE and the capability tradition remains one of dialogue

rather than identity. Capability analysis supplies a normative and evaluative vocabulary centered on substantive opportunity. GRE adds questions about relational generation, circulation, conversion, and historical transformation. The two can overlap substantially while preserving different theoretical commitments.

1.23.4 Commons and Collective Governance

Commons scholarship supplies a fourth condition by demonstrating that shared-resource problems can be organized through institutional forms more diverse than a binary opposition between centralized control and individualized private ownership. Ostrom's comparative work on common-pool resources analyzes self-organized and self-governed arrangements, including both long-enduring institutions and cases of fragility or failure (Ostrom 1990). The historical importance of this research lies in making collective rule formation, monitoring, boundary definition, conflict resolution, and institutional adaptation legitimate objects of economic analysis.

The commons perspective is relational at several levels. Resource use depends on relations among appropriators, relations between users and the resource system, and relations between local arrangements and wider authorities. The relevant economic object is therefore difficult to represent as a resource plus a set of independent claimants. Rules concerning access, monitoring, sanctions, contribution, and conflict resolution alter how the group interacts and how the resource is reproduced over time. Institutional performance depends on the configuration of these relations.

This perspective also widens the concept of governance capacity. Coordination can emerge from participants who construct and revise rules rather than receiving a fully specified institutional design from an external authority. Such self-organization does not imply spontaneity without structure. Ostrom's analysis emphasizes institutions, information, credible commitments, monitoring, and the conditions under which participants can organize collective action (Ostrom 1990). The result is an empirical challenge to analytical frameworks that treat collective governance as conceptually secondary to state administration or market exchange.

For GRE, commons research is especially relevant to the distinction between allocation and generativity. A shared resource can be allocated among current users while the institutional relations required for future reproduction are simultaneously strengthened or weakened. Rules that maximize present extraction may reduce future resource availability or institutional trust. Rules that preserve the resource can impose present costs while maintaining later options. Evaluation therefore requires attention to the relation between current distribution and the reproduction of the relational system through which future use remains possible.

The commons tradition also provides a useful warning against idealizing relational organization. Collective governance can fail, exclude outsiders, reproduce internal hierarchy, or become fragile under changing conditions. Ostrom's own comparative programme includes failures as well as successes (Ostrom 1990). Relational density therefore has no automatic normative priority. GRE treats the structure, distribution, revisability, and generative effects of relations as empirical and normative questions rather than assuming that more relation is inherently better.

This restraint clarifies the contribution of commons scholarship to the present

chapter. Its historical significance is the expansion of the institutional imagination available to economics. Shared resources can be governed through architectures in which users, rules, monitoring, and collective learning form an evolving system. That possibility supplies an important precedent for GRE's later examination of relational allocation, collective capacity, and institutional generativity.

1.23.5 Environmental and Intergenerational Normativity

Environmental and intergenerational norms provide a fifth condition by extending economic evaluation across temporal and ecological boundaries. The World Commission on Environment and Development's 1987 report, *Our Common Future*, articulated sustainable development through the relation between present needs and the ability of future generations to meet their own needs, while explicitly linking environmental limits to technology and social organization (World Commission on Environment and Development 1987). This formulation helped stabilize a normative vocabulary in which future consequences and ecological conditions became integral to development rather than external aftereffects.

The temporal extension changes the structure of the economic subject. Future persons cannot participate in current markets, vote in current institutions, or contract with present actors, yet current investment, extraction, infrastructure, emissions, debt, and technological choices can alter their opportunity sets. Intergenerational evaluation therefore requires institutions to represent interests that are temporally absent from ordinary exchange. This is a relational problem because current actions create relations of inheritance, dependence, exposure, and constraint across time even when the later participants are not presently available as counterparties.

Environmental problems introduce a related spatial and systemic extension. Ecological processes cross property boundaries, jurisdictions, and accounting periods. Resource depletion, pollution, biodiversity loss, and climate effects can connect actors who never transact directly. The relevant causal structure can therefore extend beyond the legal boundaries of the entity generating an immediate economic output. Environmental normativity makes these externalized relations harder to exclude from evaluation simply because they fall outside the original contract or reporting unit.

The combination of environmental and intergenerational concerns also changes the meaning of economic efficiency across horizons. An arrangement can produce high current output while degrading ecological or institutional conditions required for later production. Conversely, present investment can maintain capacities whose returns become visible only over long periods. GRE's generative vocabulary is well suited to this temporal structure because it distinguishes present outcomes from the conditions under which future outcomes can be generated. The normative judgment about how present and future claims should be weighted remains a separate problem.

This separation is important. Generativity does not itself determine goodness. A system can generate durable forms of domination as readily as durable capabilities; an environmentally persistent arrangement can remain unjust; preservation can conflict with present claims of people facing urgent deprivation. Environmental and

intergenerational evaluation therefore requires explicit normative reasoning about distribution, rights, risk, recognition, and acceptable tradeoffs. GRE can contribute an account of how current relations transform future possibility while leaving the ranking of competing claims open to normative theory.

The five developments reconstructed in this section collectively widen the space within which relational governance can be conceived. Participation makes affected relations relevant to problem formation; human-rights recognition gives institutional standing to persons beyond their market roles; capability analysis directs attention toward effective opportunities and conversion conditions; commons research establishes collective institutional arrangements as substantive economic objects; and environmental or intergenerational normativity extends evaluation across ecological systems and historical time. None of these traditions yields GRE by itself. Together they create a political and normative environment in which economic governance can increasingly be asked to account for relations, heterogeneous forms of standing, future possibilities, and the conditions through which economic life is reproduced. Section 1.24 turns next to the development of *nomos* and institutional form: the legal and organizational structures through which such expanded objects of recognition can acquire durable institutional expression.

1.24 Development of Nomos and Institutional Form

This section identifies the institutional developments through which relational and collective economic claims acquire durable forms of recognition. Its role in the chapter is to connect the political and normative opening reconstructed in Section 1.23 with the legal and organizational forms that make such claims actionable over time. The objective is to show how *nomos*, used here in a deliberately broad sense for the operative normative and institutional order that assigns statuses, claims, duties, boundaries, and procedures, changes the range of economic relations that can be represented and governed. The discussion proceeds from legal recognition of economic subjects to collective and relational interests, then examines fiduciary, cooperative, and commons-based institutions before turning to adaptive and revisable governance. The method is institutional and comparative. It treats legal and organizational forms as historically developed devices for stabilizing relations, allocating authority, preserving claims, and structuring revision, while leaving the systematic theory of governance to later GRG work.

1.24.1 Legal Recognition of Economic Subjects

Legal recognition supplies one of the principal mechanisms through which an economic subject becomes durable across transactions. Section 1.5 showed that modern economic entities become administratively legible through legal personality, property, contract, accounting, and statistical classification. The present subsection develops the institutional implication of that argument. Recognition gives continuity to a position that can own assets, incur liabilities, enter contracts, appear in accounts, receive transfers, and bear responsibilities across time. Economic analysis can then treat that position as sufficiently stable for allocation and coordination even when

the persons, assets, or internal relations composing it change.

Organizational law illustrates the structure clearly. Hansmann and Kraakman emphasize the role of organizational law in separating organizational assets and creditor claims from the personal asset positions of owners and managers (Hansmann and Kraakman 2000). This legal partitioning stabilizes a boundary through which an organization can persist as an economic unit across changes in membership and management. The System of National Accounts similarly treats institutional units as basic economic and statistical units capable of owning assets, incurring liabilities, making economic decisions, and entering transactions in their own right (European Commission et al. 2009). The legal and statistical vocabularies differ, yet both rely on institutional criteria that transform a changing configuration of persons and resources into a recognizable economic position.

Recognition therefore participates in the production of the economic object it later regulates. Once an entity is recognized as a corporation, public body, household, nonprofit institution, or other institutional unit, contracts, accounts, taxes, liabilities, transfers, and statistics can attach to that recognized position. The resulting stability supports coordination across actors who need not reconstruct the entity's entire internal relational history before every transaction. Recognition reduces uncertainty by supplying a durable institutional address for claims and obligations.

The same mechanism also creates selective visibility. Legal and administrative forms foreground the relations that are necessary to maintain the recognized position: ownership, authority, liability, membership, contractual capacity, reporting boundaries, and related incidents. Other relations can remain external to the formal identity of the unit even when they matter economically. Care relations within households, dependencies in supply chains, informal obligations, platform-mediated access, or ecological dependencies can shape the capabilities of an entity while receiving weaker recognition in the formal representation.

The development of nomos therefore changes economic ontology at the practical level. The point concerns institutional reality rather than ultimate metaphysics. A legal form can make a configuration sufficiently stable and causally efficacious for economic coordination, while the broader social and relational constitution of that configuration remains open to analysis. This institutional stabilization is one historical condition for entity-centered economics, and it also creates the starting point from which relational analysis can ask which constitutive relations remain outside the recognized boundary.

1.24.2 Recognition of Collective Interests

Collective interests extend institutional recognition beyond a simple aggregation of individually held claims. Their significance for GRE lies in the possibility that an institution can assign standing, assets, decision rights, or continuity to a group whose economic activity depends on coordinated relations among members. The relevant development concerns the creation of forms through which a collective can persist and act even as individual participants enter, leave, or change their contributions.

The corporation is one familiar example, though collective recognition extends beyond shareholder-owned firms. Associations, cooperatives, mutuals, unions, pension

arrangements, foundations, public bodies, and commons institutions organize interests whose practical content depends on continued relations among multiple participants. Organizational forms can provide a durable account of membership, authority, assets, obligations, and procedures for collective decision. The recognized collective thereby acquires a temporal continuity that individual contracts alone may reproduce only at substantial cost.

Cooperative institutions provide a particularly explicit example of collective economic recognition. The International Cooperative Alliance defines a cooperative as an autonomous association of persons who voluntarily unite to meet common economic, social, and cultural needs through a jointly owned and democratically controlled enterprise (International Cooperative Alliance 1995). The institutional form links economic participation to membership and democratic control rather than treating invested capital as the sole basis of authority. The form therefore stabilizes a collective interest while preserving identifiable relations between members, enterprise, capital, and decision procedures.

Collective recognition also changes the possible recipient of economic allocation. Resources can be directed toward a jointly owned reserve, a common infrastructure, a pension fund, a mutual insurance pool, or an institution whose purpose concerns a class of members rather than an isolated owner. The economic consequence is important: the same nominal transfer can have different effects depending on whether it becomes private property, common capital, an indivisible reserve, or an institutional asset governed under membership rules. Collective forms therefore make visible the organization of the recipient, not merely its legal name.

The analytical implication for GRE concerns persistence and aggregation. A collective interest gains economic relevance when the institutional form can preserve some relation or capability across turnover in individual participants. The collective then becomes a legitimate object of analysis because decisions taken at the collective level can affect access, bargaining position, risk sharing, investment, and future participation. Collective recognition does not establish that every group claim is normatively valid. It establishes that economic institutions have developed devices through which certain shared claims can be stabilized, governed, and carried through time.

This development widens the space of economic subjecthood introduced in Section 1.16. Subject positions can be individual, organizational, or collective, and their stability can depend on different institutional mechanisms. GRE can therefore examine how a form of collective recognition alters the relations through which resources, risks, authority, and returns circulate.

1.24.3 Recognition of Relational Interests

Relational interests concern claims whose practical meaning depends on a structured relation among parties rather than on possession by a single isolated holder. Their institutional significance becomes visible when economic law or organizational design protects a beneficiary relation, membership relation, stewardship relation, access relation, dependency, or continuing obligation whose value cannot be described fully as an immediately transferable asset. The objective of this subsection is to identify the institutional possibility of recognizing such interests while preserving the distinction

between a relation and an entity created to administer it.

Trust law offers a useful illustration. Hansmann and Mattei analyze the trust as an organizational form that structures relations among parties to the trust and between those parties and third parties, with asset partitioning playing a central role in the form's economic function (Hansmann and Mattei 1998). The legal arrangement separates dimensions of control, benefit, and creditor access that coincide more closely under simple ownership. A beneficiary's economic interest therefore depends on an institutionalized relation to assets managed through another position. The interest is intelligible through the relation among trustee, beneficiary, trust assets, and third parties.

Membership provides a second illustration. Cooperative rights derive their practical meaning from a person's relation to the cooperative and to the procedures through which the cooperative is controlled. The ICA principles connect membership to participation in policy and decision making, member economic participation, autonomy, and cooperation among cooperatives (International Cooperative Alliance 1995). These rights have an economic content that differs from a detached claim to an asset because their exercise depends on continuing membership and institutional procedures.

Commons institutions provide a third form. Ostrom's comparative work documents arrangements in which groups establish rules concerning access, use, monitoring, sanctioning, and collective choice around common-pool resources (Ostrom 1990). The relevant interest is structured through relations among users, resource conditions, monitoring practices, and institutional rules. Access to the resource can therefore depend on membership and rule-governed relations rather than on exclusive ownership by a single actor.

These examples suggest a cautious extension of economic representation. Some interests can be represented more adequately by specifying the relation through which the claim is sustained. A transfer to a beneficiary, a member, or a resource user changes more than the recipient's balance sheet when it also alters membership, control, dependence, monitoring obligations, or future access. Relational analysis becomes useful where these continuing relations materially shape the effect of the transfer.

Institutional recognition can nevertheless reify what it stabilizes. A fluid relation may be converted into a fixed legal category, a standardized entitlement, or an administratively bounded membership class. Such stabilization can be necessary for enforceability and coordination, while also narrowing the variation that the institution can perceive. GRE therefore treats recognition as a generative intervention into the relation itself. The act of making an interest legible can alter the boundaries, participants, duties, and future trajectories of that interest.

The resulting problem is central to later relational allocation theory. Economic institutions require stable addresses for claims, yet generative relations can change through time. A relationally sensitive nomos therefore needs forms that can recognize continuing interests while leaving room for revision when the relations supporting those interests change.

1.24.4 Fiduciary, Cooperative, and Commons-Based Institutions

Fiduciary, cooperative, and commons-based institutions demonstrate several ways in which economic organization can distribute ownership, control, benefit, and responsibility across differentiated positions. Their importance for the present chapter lies in institutional variety. Modern economic ordering already contains forms that depart from the simple image of one owner controlling one asset for one private purpose. These forms provide concrete precedents for GRE's later inquiry into allocation toward relations, shared infrastructures, collective capacities, and future beneficiaries.

Fiduciary structures separate authority from beneficial interest under a rule-governed relation. Trust law can partition assets and structure claims among trustees, beneficiaries, and third parties in ways that ordinary bilateral contracting may reproduce only imperfectly (Hansmann and Mattei 1998). The institutional form permits resources to be administered under purposes and duties that persist beyond a particular transaction. Pension funds, charitable trusts, and other fiduciary arrangements differ substantially in law and function, yet they share the broader feature that control over assets and entitlement to benefits can occupy distinct institutional positions.

Cooperatives organize a different separation. Their institutional identity places membership, joint ownership, and democratic control at the center of the enterprise. The ICA principles include voluntary and open membership, democratic member control, member economic participation, autonomy, cooperation among cooperatives, and concern for community (International Cooperative Alliance 1995). These principles provide an explicit example of an economic organization whose architecture links capital to a wider structure of membership and collective decision. The form does not guarantee equal outcomes or successful governance. It demonstrates that economic control and return can be institutionally organized through relations other than proportional investor ownership.

Commons-based institutions add another configuration. Ostrom's cases show that users of common-pool resources can, under some conditions, develop durable rules for access, monitoring, conflict resolution, sanctioning, and collective choice (Ostrom 1990). Their significance lies in the institutionalization of reciprocal and continuing relations around a shared resource. The resource remains economically consequential through a pattern of use and governance that cannot be reduced to a sequence of isolated transfers.

Taken together, these forms show that institutional architecture can distribute economic functions across multiple roles. Asset control, beneficial claim, membership, monitoring, participation, stewardship, and residual return can be attached to different positions and linked through rules. GRE can therefore treat institutional form as part of the causal structure of allocation. A resource entering a trust, cooperative, or commons arrangement enters a different generative configuration from the same resource entering an individually controlled account.

The comparison also supplies a methodological caution. Fiduciary, cooperative, and commons institutions arise from distinct legal histories and solve different problems. Their inclusion in one subsection identifies a family resemblance in the distri-

bution of economic roles, rather than an equivalence among the forms. Later GRE chapters can ask which structural features matter for generativity, participation, return, or justice without treating any existing institution as a ready-made model for a generative relational economy.

The historical lesson is therefore modest and important. Institutional pluralism already exists within modern economic orders. The development of GRE does not require inventing relational organization from an empty institutional field. It requires learning how existing forms distribute relations, which capacities they preserve, which asymmetries they create, and how their rules affect the circulation and return of value.

1.24.5 Adaptive and Revisable Governance

Adaptive and revisable governance provides the final institutional condition considered in this section. Relational and generative systems change while institutions act upon them, so a governance form that can monitor consequences, evaluate implementation, revise rules, and reopen decisions possesses a different temporal structure from a one-time allocation. The objective here is to identify institutional revisability as a historically developed governance capacity, while maintaining the distinction between revisability and a complete theory of adaptive governance.

Institutional revision begins with explicit procedures for observation and feedback. The OECD's framework on sound public governance treats monitoring as a source of information during implementation and connects monitoring and evaluation to adjustment of policy and resource allocation (OECD 2020c). Its comparative work on policy evaluation similarly presents evaluation as a mechanism for accountability, learning, and improved decision making (OECD 2020b). These practices create formal channels through which observed consequences can re-enter the decision process.

Revisability matters because the object of governance can change during implementation. A program can alter incentives, participation, organizational routines, information flows, or the composition of the population it serves. A rule designed for one relational configuration can therefore become mismatched to the configuration partly generated by the rule itself. Monitoring then has a generative significance: the institution observes a system in which its own intervention has become one of the causal relations.

Adaptive capacity also depends on the architecture of amendment. Institutions vary in who can initiate revision, what evidence can count, whose testimony has standing, which decisions are reviewable, how quickly rules can change, and which commitments remain protected from short-horizon adjustment. These procedural differences shape the relation between stability and learning. Rapid revision can make an institution responsive while reducing predictability; rigid rules can preserve expectations while allowing mismatch to accumulate. The relevant design problem concerns the organization of temporal responsiveness across different kinds of rule and different timescales of change.

Existing institutions already embody partial solutions. Cooperative principles can be interpreted and updated through living guidance while preserving a continuing

institutional identity (International Cooperative Alliance 2017). Commons arrangements can contain collective-choice and monitoring procedures through which rules respond to local conditions (Ostrom 1990). Public-governance frameworks increasingly formalize monitoring, evaluation, and revision across policy cycles (OECD 2020c; OECD 2020b). These examples differ in authority and domain, yet each illustrates an institutional attempt to combine continuity with revisability.

For GRE, revisability has a further implication. If institutions participate in generating the relations they govern, evaluation cannot be confined to whether an intervention reached its initial target. It must also ask what relations, capacities, dependencies, and future options emerged through implementation. The answer can justify revision even when the initial output metric was achieved. This creates an institutional bridge between the evidential conditions of Section 1.22 and the generative analysis developed in Sections 1.17–1.20.

The development of nomos therefore expands the practical possibility of relational economics in two directions. Institutional forms can recognize subjects, collectives, and continuing relational interests; revision procedures can preserve the capacity to learn when those recognized forms change through time. The same developments also create new risks of reification, unequal recognition, procedural capture, and excessive administrative complexity. Sections 1.25 and 1.26 turn to these limits directly by examining the remaining epistemic and governance constraints on relational analysis.

1.25 Contemporary Limits of Relational Knowledge

This section specifies the epistemic limits that remain after the scientific, computational, evidential, political, and institutional openings developed in Sections 1.19–1.24. Its role in the chapter is to prevent the historical possibility of richer relational analysis from being mistaken for complete access to relational reality. The objective is to distinguish five persistent limits: partial observation, dependence on representation and model choice, causal underdetermination, boundary and scale dependence, and the temporal incompleteness of generative evaluation. The discussion is epistemological and methodological. It treats these limits as conditions under which relational knowledge must be produced, qualified, compared, and revised, while reserving the fuller epistemology of GRE for Chapter 5.

1.25.1 Partial Observation

Relational economic analysis begins from evidence that is necessarily selective. Some relations are formally recorded because institutions require contracts, accounts, registrations, transactions, tax records, administrative classifications, or platform logs. Other relations remain informal, tacit, private, strategic, episodic, or distributed across several organizations. Even where a relation leaves traces, the traces may register only a subset of its dimensions: a payment can record transfer without recording dependence; an employment contract can record legal status without recording informal authority; a platform interaction can record clicks without recovering the meaning assigned to them by participants. The expansion of data

infrastructures therefore increases observational capacity while leaving observation partial.

The limitation is partly institutional. Observation follows the categories and purposes of the apparatus that produces the record. Administrative systems collect information relevant to eligibility, compliance, taxation, service delivery, or organizational management; firms collect information relevant to production, customers, risk, or control; platforms record events that can be captured through their technical architecture. These records can be highly informative, yet their coverage follows the operational needs of the institution. Relations that matter economically can remain weakly observed when no reporting system has reason, authority, or technical capacity to record them.

The limitation is also distributed. Hayek's account of dispersed knowledge emphasized that economically relevant knowledge exists in incomplete and frequently contradictory fragments across many participants rather than as a single integrated dataset available to one observer (Hayek 1945). Contemporary digital infrastructures can aggregate quantities that were previously unavailable, but aggregation does not eliminate dispersed local knowledge. Tacit routines, contextual expectations, unrecorded obligations, local meanings, and strategic interpretations can remain distributed even in data-rich environments.

Observation can also be asymmetric. Powerful organizations may observe workers, consumers, suppliers, or users more extensively than those participants can observe the organizations that classify and act upon them. Regulators may receive disclosures unavailable to ordinary market participants; platforms may infer behavior from traces that users cannot inspect; organizations can combine datasets across domains while individuals encounter only the immediate interface. Relational knowledge therefore has a distribution as well as a quantity. The question concerns who can observe which relations, at what resolution, and under what rights of access.

Partial observation does not imply that relational claims are arbitrary. It requires explicit evidential scope. A study can state which relations are directly observed, which are inferred, which are represented through proxies, which are inaccessible, and which may be missing systematically. Section 1.22 showed that contemporary evidence infrastructures make such distinctions increasingly tractable. The epistemic discipline is to treat improved observation as an expansion of access rather than as completion of the object.

1.25.2 Representation and Model Dependence

Observed traces become relational knowledge only after they are represented. A representation identifies the entities or positions to be distinguished, the relations to be recorded, the attributes attached to them, the temporal resolution of observation, and the boundary within which the description is constructed. This dependence is present in statistical tables, legal categories, network graphs, field models, agent-based simulations, and qualitative relational maps alike. Relational analysis therefore inherits the epistemic problem already identified for economic categories in Section 1.1: analytical objects become visible through historically and methodologically specific forms of description.

Statistical and administrative classifications provide an obvious case. Desrosières showed historically how statistical reasoning depends on conventions of classification, coding, equivalence, and aggregation that make heterogeneous cases comparable (Desrosières 1998). Hacking likewise emphasized that categories and styles of reasoning have histories and can participate in the formation of the objects they classify (Hacking 2002). For GRE, the implication is that a relational dataset never contains relations in an entirely pre-analytical form. It contains observations organized through decisions about what counts as an actor, event, tie, transaction, household, firm, sector, location, or period.

Formal relational models make these decisions especially explicit. A network representation requires nodes and edges; a higher-order representation requires specification of group interactions; a field representation requires a domain, variables, locality, and boundary conditions. Newman and Battiston and colleagues show how different relational formalisms preserve different structures and support different dynamical questions (Newman 2003; Battiston et al. 2020). A pairwise graph can be appropriate for one problem and lose essential information for another in which an interaction exists only through a group or institutional configuration.

Model dependence therefore concerns the conditional character of inference. A conclusion can be robust within one representation and change when the same empirical material is organized differently. The appropriate response is comparison rather than the search for a representation free of assumptions. Where several defensible models exist, GRE can ask which distinctions each preserves, which it suppresses, and which conclusions survive translation across representations. Later methodology chapters develop this principle through triangulation, sensitivity analysis, and model plurality.

Representation also feeds back into practice. Once a model becomes an administrative category, performance indicator, credit classification, eligibility rule, or platform score, it can change behavior and institutional treatment. Model dependence then becomes generative as well as epistemic: the representation partly enters the relational system it was designed to describe. This possibility strengthens the requirement that GRE state the standing of its models precisely. A model can be descriptively useful without being ontologically exhaustive, and its institutional use can create consequences not contained in its original evidential purpose.

1.25.3 Causal Underdetermination

Relational association does not by itself identify the mechanism that generated it. Two organizations may display similar network positions for different historical reasons; a strong association between membership and outcome may reflect selection, common institutional exposure, reciprocal influence, or an unobserved relation; a policy intervention may coincide with change while several other processes evolve simultaneously. The problem becomes more demanding in generative systems because relations, subjects, institutions, and constraints can coevolve, allowing causes and conditions to change while the observed process unfolds.

The econometric identification literature provides a disciplined language for this limitation. Manski argues that empirical analysis should distinguish what the avail-

able data identify from what follows only after additional assumptions are introduced, and develops methods for learning under incomplete data and partially identified treatment responses (Manski 2008). The relevance to GRE is methodological rather than doctrinal. Rich relational data can narrow the space of plausible explanations while leaving several mechanisms compatible with the observations. Stronger causal claims require stronger design, temporal evidence, institutional knowledge, or assumptions whose credibility must be stated rather than hidden inside the model.

Generative analysis adds historical underdetermination. Path-dependent systems can reach similar visible states through different sequences, and similar interventions can produce different consequences when they enter different institutional trajectories. North's account of institutional change and Arthur's treatment of increasing returns both make historical sequence analytically consequential (North 1990; Arthur 1989). The present state therefore need not uniquely reveal the path that generated it, while a path reconstructed from records may still admit competing mechanism accounts.

Causal underdetermination is especially important for relational intervention. An observed improvement after strengthening one relation can coexist with changes in resources, expectations, network position, institutional recognition, or external conditions. A claim that the relation generated the improvement should therefore identify the mechanism, timing, relevant alternatives, and evidence that discriminates among them. Where the evidence supports only contribution, enablement, or association, the language of the claim should remain correspondingly limited.

This limitation does not reduce GRE to causal skepticism. It specifies an evidential gradient. Some cases support strong identification; others support partial identification, mechanism plausibility, process evidence, or bounded counterfactual claims. The research task is to align claim strength with evidential strength. A generative framework is most useful when it makes additional mechanism questions visible, while remaining willing to leave several explanations open when the evidence does not discriminate among them.

1.25.4 Boundary and Scale Dependence

Relational descriptions depend on where the system boundary is drawn and at what scale the system is observed. An organization's internal network can appear dense when suppliers, regulators, households, and infrastructures are excluded, while the same organization can occupy a peripheral or dependent position in a wider production network. A household can appear as one allocation unit in national accounts and as a differentiated set of care, income, authority, and dependency relations at a finer resolution. Neither description is automatically defective. Each answers a different question by preserving a different set of distinctions.

The statistical literature provides a classic warning against transferring relationships across levels of aggregation. Robinson demonstrated that correlations calculated for grouped populations can differ materially from corresponding relationships among individuals and that aggregate associations therefore do not license direct inference about individual behavior (Robinson 1950). The lesson extends beyond ecological correlation. Relations observed at one scale can be generated by compositions,

constraints, or interactions that disappear when the analysis moves to another scale.

Complex-systems research supplies a complementary perspective. Bar-Yam's multiscale framework treats structure as scale-dependent and emphasizes that the organization visible at one resolution can differ from that visible at another (Bar-Yam 2004). Network analysis likewise depends on the population of nodes and ties admitted to the graph (Newman 2003). Centrality, clustering, brokerage, modularity, and dependency can all change when the boundary expands, when a layer is added, or when a higher-order interaction is represented explicitly.

Boundary selection also has normative and political consequences. A firm-level productivity analysis can exclude unpaid household labor on which workers depend; a project-level cost calculation can exclude environmental effects outside the administrative jurisdiction; a platform-level analysis can omit alternative infrastructures that determine user exit; a national analysis can externalize transnational labor or ecological relations. The boundary therefore determines which effects become internal to evaluation and which remain external to it.

GRE should consequently treat boundary and scale as reportable analytical choices. Robust relational claims should indicate whether their conclusions persist across plausible alternative boundaries and resolutions. Where conclusions change, the variation is itself informative: it identifies the scale or relational neighborhood at which the mechanism appears. This discipline makes multiscale disagreement a source of knowledge rather than a defect to be concealed by choosing one preferred level.

1.25.5 Temporal Limits of Generative Evaluation

Generative evaluation faces a temporal limit because the object of evaluation includes consequences that have not yet fully emerged. A transfer can alter capacity only after training, trust, infrastructure, or institutional learning accumulates. A new dependency can remain invisible while short-run output rises. A legal or organizational reform can change incentives immediately while its effects on membership, bargaining position, or future options unfold over years. The evaluation window therefore participates in the result.

Historical economics already contains several mechanisms that make timing consequential. North emphasizes that institutions shape subsequent incentives and paths of change, while Arthur shows how early events can be amplified through increasing returns and produce later lock-in (North 1990; Arthur 1989). These mechanisms imply that a present intervention may acquire significance through a trajectory whose later stages are unavailable at the moment of decision. The evaluator observes a prefix of the process rather than the completed process.

The problem is sharper for generative value because later effects can include changes in the capacity to generate further relations and options. An educational opportunity may alter later learning relations; infrastructure can generate complementary investment; institutional access can create future membership and information; extraction can weaken capacities whose loss becomes visible only after repeated use. Short observation windows can therefore favor effects that manifest quickly and discount effects whose causal pathway is delayed or distributed across

scales.

Long-horizon evaluation also encounters changing contexts. The institutional environment, technology, value structure, and subject positions relevant to an intervention can change while the intervention's effects unfold. A model calibrated to the initial context can therefore lose adequacy before the trajectory ends. This is one reason why adaptive evaluation and institutional learning, discussed in Sections 1.22 and 1.24, are historically important: they permit judgments to be revised as later evidence arrives rather than treating the first evaluation as final.

Temporal incompleteness places a limit on certainty rather than on evaluation itself. GRE can distinguish immediate outcomes from delayed effects, preserve the assumptions under which projections are made, identify indicators of intermediate mechanism, and reopen judgments when new stages of the trajectory become observable. Intergenerational and environmental evaluation makes this requirement especially visible because the subjects affected by present action can include persons who cannot participate in present evidence production or decision procedures (World Commission on Environment and Development 1987).

The five limits developed in this section define an epistemic boundary around the contemporary opening reconstructed in the chapter. More data leave observation partial; richer formalisms remain representation-dependent; relational patterns can support several causal accounts; boundaries and scales shape what becomes visible; and generative evaluation remains temporally incomplete while the future is still being produced. These limits do not cancel the case for GRE. They determine the form that a responsible relational economics must take: explicit about observation, conditional about models, disciplined about causal language, transparent about scale and boundary, and revisable across time. Section 1.26 extends the argument from knowledge to governance by examining the practical constraints that arise when institutions attempt to act within systems they can only partially know.

1.26 Contemporary Limits of Relational Governance

This section identifies the practical limits that remain when relational knowledge is translated into governance. Its role in the chapter is to complement the epistemic limits developed in Section 1.25: even when relevant relations can be observed with reasonable confidence, governing them requires administrative categories, legally actionable statuses, computationally tractable representations, normative judgments, and institutional authority over observation and classification. The objective is to specify five recurrent constraints on relational governance: administrative reification, legal dependence on stable categories, computational complexity, normative plurality, and asymmetries in the power to observe and classify. The discussion is institutional and diagnostic. It does not develop a general governance theory, which belongs to the later GRG programme; it clarifies why richer relational knowledge does not automatically yield frictionless or normatively determinate intervention.

1.26.1 Administrative Reification

Relational governance requires administrative action, and administrative action requires objects that can be named, compared, recorded, assigned responsibility, and linked to procedures. This requirement creates a recurring tension. Relations are often continuous, overlapping, historically variable, and distributed across more than one institution, while administrative systems operate through discrete records, statuses, eligibility categories, reporting units, and decision rules. Governance therefore converts a relational field into administratively actionable objects. The conversion is necessary for implementation, but it can also stabilize the representation more strongly than the underlying relation warrants.

Modern statistical and administrative systems provide a clear illustration. Statistical practice depends on conventions that establish equivalence, coding, classification, and aggregation among heterogeneous cases (Desrosières 1998). Standardized quantitative procedures also became institutionally attractive because they support impersonal and transportable judgment (Porter 1995). These achievements make large-scale administration possible. They also establish a practical grammar in which only some aspects of relational life become legible enough to govern.

The problem can be described as administrative reification: a representation introduced to make a relation governable can acquire the practical standing of the relation itself. A household category may become the unit through which eligibility is determined; an employment status may determine access to benefits; a reporting entity may define which transactions are consolidated; a statistical sector may determine how economic activity is aggregated. The administrative object then influences the distribution of resources and responsibilities, so its boundary has causal consequences even when the underlying relational configuration remains more complex.

Hacking's historical ontology is useful for specifying the feedback involved. Classifications can become part of the conditions under which people and institutions understand, organize, and respond to the classified object (Hacking 2002). The point for GRE is not that every category mechanically creates what it names. It is that classificatory schemes can enter the relations they describe through eligibility rules, compliance procedures, accounting boundaries, organizational incentives, and social recognition. Administrative representation can therefore become one component of the generative environment.

Relational governance cannot remove this problem simply by adding more categories. Finer classification increases descriptive resolution, but it also increases administrative complexity and may create additional boundaries whose practical effects require justification. Nor does replacing discrete categories with continuous scores eliminate reification. A score can become an equally powerful administrative object when access, risk, priority, or surveillance is tied to it. The central governance question concerns how a representation is used, how contestable its boundary remains, and how easily the institution can revise it when the relations it governs change.

The practical implication is a discipline of representational modesty. Administrative categories should be treated as instruments with stated scope, provenance, and revision procedures. Where a category carries high-stakes consequences, gov-

ernance should preserve some capacity to inspect cases that fit the administrative representation poorly. This principle does not dissolve the need for categories. It recognizes that governance acts through abstractions whose convenience can otherwise conceal their generative effects.

1.26.2 Legal Dependence on Stable Categories

Law intensifies the demand for stability because rights, duties, liabilities, ownership claims, contracts, jurisdiction, and procedures require sufficiently determinate subjects and objects. Economic relations may change rapidly, yet legal institutions must decide who bears an obligation, who controls an asset, which entity enters a contract, which claimant has standing, and which boundary determines responsibility. Relational governance therefore encounters a second limit: the more fluid the underlying relations become, the more difficult it is to preserve both relational sensitivity and legal determinacy.

Organizational law illustrates the productive side of stable legal categories. Hansmann and Kraakman show how organizational forms create standardized legal entities and partition assets and creditor claims, making complex collective activity governable through relatively stable legal boundaries (Hansmann and Kraakman 2000). Trust law can likewise separate control, benefit, and claims against assets in ways that organize relations among trustees, beneficiaries, settlors, and third parties (Hansmann and Mattei 1998). These forms do more than describe economic relations. They constitute durable positions through which those relations can be coordinated over time.

The same stability can become constraining when relational reality changes more quickly than legal form. A firm may reorganize production across contractors, platforms, subsidiaries, and algorithmic systems while liability remains attached to formally recognized entities. A worker may occupy economically dependent positions that fall imperfectly within inherited employment categories. A digital platform may mediate transactions across jurisdictions whose legal classifications do not align. The governance problem is therefore temporal as well as conceptual: legal categories need enough persistence to support expectation and accountability, while economic relations continue to evolve.

Accounting and statistical systems reveal analogous tensions. The IFRS Conceptual Framework treats the reporting entity as a boundary constructed for financial reporting and allows that boundary to include a single entity, part of an entity, or multiple entities (IFRS Foundation 2018). The System of National Accounts similarly relies on institutional units capable of owning assets, incurring liabilities, entering contracts, and making economic decisions (European Commission et al. 2009). Such frameworks show that institutional practice already works with analytically constructed boundaries. Relational governance can therefore revise boundaries, but it must do so through procedures that preserve comparability, responsibility, and legal intelligibility.

The limit is consequently structural rather than merely conservative. Governance cannot continuously redefine every subject and relation at the speed of the underlying economy without destabilizing the expectations that institutions themselves sus-

tain. Legal and administrative forms have characteristic revision times. Their relative slowness can support coordination by making obligations predictable. The same slowness can generate mismatch when new relations become economically consequential before the corresponding legal categories evolve.

GRE should therefore treat legal stability and relational sensitivity as jointly necessary institutional properties whose balance varies by domain. Some contexts require durable categories because responsibility must remain traceable across time. Others require flexible recognition because fixed categories systematically omit economically dependent or newly mediated relations. The analytical task is to identify which features require continuity, which can be revised, and what institutional process governs the transition. Systematic theories of legal change belong outside GRE, but the economic consequences of such temporal and classificatory mismatch remain central to relational analysis.

1.26.3 Computational Complexity

Relational governance also faces a computational limit. As the number of relevant actors, ties, higher-order interactions, temporal states, institutional constraints, and possible interventions increases, the space of relationships that could matter expands rapidly. Contemporary computation makes much richer analysis possible, as Section 1.21 established, yet increased computational capacity does not remove the need to simplify. Governance decisions must operate with finite models, finite observation windows, and finite time for deliberation.

Network analysis already demonstrates how relational structure creates analytical burdens absent from isolated-unit models. Newman surveys methods for representing large systems through networks, while higher-order network research shows that pairwise edges can fail to preserve interactions that exist only at the level of groups or simplicial structures (Newman 2003; Battiston et al. 2020). Adding temporal variation increases the problem further because the relevant structure is then a sequence of changing relational configurations rather than a single graph. The computational object can become very large before uncertainty in measurement or causality is even considered.

Complex-systems analysis adds another difficulty: local interventions can propagate through feedback, adaptation, and reorganization. Bar-Yam's multiscale treatment emphasizes that different structures become visible at different scales and that organization at one scale cannot be inferred straightforwardly from another (Bar-Yam 2004). A governance model that represents only one scale may therefore miss interactions that dominate elsewhere. A model that attempts to include every scale can become too complex to interpret, calibrate, or use within a decision process.

Computational complexity is consequently also a problem of model selection. Governance must choose which relations to resolve, which to aggregate, which time horizon to simulate, which behavioral responses to include, and which uncertainty to represent. Agent-based models can explore heterogeneous interaction, while dynamical and network models can reveal trajectories, feedback, and topological dependence. Each representation reduces the system differently. Richer models may expose mechanisms hidden by simpler ones, yet they can also become harder to validate and easier

to overfit to assumptions rather than evidence.

The practical limit becomes especially visible when a model is used for intervention. Decision makers often need an answer before exhaustive computation, data collection, or model comparison is possible. In such settings the relevant standard is not maximal descriptive detail but sufficient robustness for the decision at hand. A relational governance process should therefore identify which conclusions survive plausible changes in boundary, scale, representation, and parameter assumptions. Later chapters develop this through sensitivity analysis, model plurality, and adaptive revision.

Computational capacity thus changes the frontier of feasible governance without eliminating incompleteness. The historical significance of contemporary computation lies in making previously inaccessible relational structures analyzable. The corresponding discipline lies in refusing to treat computational resolution as equivalent to causal certainty or institutional wisdom. The more complex the model, the greater the need to state what it omits, which assumptions drive its results, and which decisions would change if another representation were adopted.

1.26.4 Normative Plurality

Relational governance is further limited by normative plurality. A relational description can show that an intervention changes access, dependence, capability, trust, participation, risk, ecological conditions, or future possibilities. The description does not by itself determine how these consequences should be ranked. Relations can support several legitimate values simultaneously, and changes that improve one dimension can burden another. Governance therefore requires normative judgment in addition to relational knowledge.

The capability approach provides one important illustration. Sen shifts attention from resources alone toward the substantive opportunities people possess to achieve valued ways of living (Sen 1999). This move makes conversion conditions and heterogeneous circumstances analytically visible, but it does not reduce all valued functionalities to a single universally fixed metric. Human-rights traditions introduce another vocabulary centered on recognition, equality, participation, and protected claims (United Nations General Assembly 1948; United Nations General Assembly 1966). Environmental and intergenerational frameworks extend evaluation to conditions affecting future persons and ecological systems (World Commission on Environment and Development 1987). Commons scholarship adds concerns with collective rule making, participation, and durable institutional arrangements (Ostrom 1990). These frameworks overlap, yet they are not interchangeable.

Normative plurality becomes particularly demanding when consequences are distributed across time and scale. An intervention can improve immediate output while reducing long-horizon ecological resilience. A classification can improve administrative consistency while excluding a minority whose circumstances fit poorly. A platform policy can increase average efficiency while concentrating dependence on an intermediary. A commons institution can protect collective resources while imposing participation burdens unevenly. Relational analysis reveals these connections more clearly, but greater descriptive richness can increase the number of values that enter

evaluation.

This is a limit on optimization rather than a rejection of it. Optimization is appropriate where the objective, constraints, and metric are sufficiently specified and normatively legitimate for the local problem. The difficulty arises when governance projects heterogeneous values into a single scale without examining what the conversion suppresses. GRE therefore needs to distinguish between locally commensurable objectives and domains in which value translation remains partial, contested, or institutionally imposed. The later chapters on heterogeneous value and justice develop this problem directly.

Normative plurality also limits claims about generativity. A system capable of generating more relations, options, or future capacities is not therefore morally superior. Generativity describes a capacity of a system; goodness requires additional normative premises. A relation can be highly generative while reproducing domination, environmental harm, or exclusion. Conversely, some constraints can be ethically justified precisely because they limit what may be generated. Relational governance must therefore keep generative evaluation and normative evaluation analytically connected without collapsing one into the other.

The governance implication is procedural as well as philosophical. Where values cannot be reduced credibly to one metric, institutions need ways to expose the relevant dimensions, represent affected positions, preserve disagreement, and revise decisions as consequences become clearer. Participation, rights, public reasoning, and institutional contestability become important partly because normative plurality cannot be resolved by data alone. GRE can analyze the economic effects of these arrangements while leaving the choice among comprehensive moral theories open.

1.26.5 Power over Observation and Classification

The final limit concerns power over the production of governable reality. Relational governance depends on observation, categorization, measurement, and model construction. The actors controlling these processes can influence which relations become visible, which subjects are recognized, which outcomes are measured, and which interventions appear administratively available. Observation and classification therefore have a political economy of their own.

The history of statistical reasoning makes this visible. Desrosières shows that statistical categories depend on conventions that make heterogeneous cases comparable, while Porter demonstrates how quantitative standards acquire authority through institutional practices of objectivity (Desrosières 1998; Porter 1995). Hacking adds that classifications can participate in shaping the objects and persons to which they are applied (Hacking 2002). Together, these traditions imply that the infrastructure of observation is not a neutral window placed outside economic relations. It is itself composed of institutions, technologies, standards, professional roles, and access rights.

Contemporary data infrastructures intensify this issue because observational capacity is unevenly distributed. Section 1.25 noted that platforms, firms, regulators, public agencies, researchers, and individuals can possess very different access to the same relational system. The World Bank's treatment of data-governance ecosystems explicitly emphasizes that data use depends on institutional arrangements, interop-

erability, safeguards, and relations among multiple actors (World Bank 2021). OECD work on data-driven public sectors similarly treats data governance as a system-wide institutional problem rather than a purely technical matter (Ooijen et al. 2019). The ability to combine, interpret, and operationalize data is therefore itself an economic capability.

Power can enter at several points. An institution may determine which categories are officially recognized; a platform may define the events its interface records; a rating system may choose the variables that enter a score; an employer may observe worker performance at greater resolution than workers can observe organizational decision rules; a public authority may possess administrative records unavailable to the persons classified by them. These asymmetries affect both knowledge and action because the observing institution can convert its representation into eligibility, pricing, ranking, surveillance, resource allocation, or enforcement.

Classification power should nevertheless be distinguished from arbitrary control. Institutions often require standardized categories to coordinate at scale, preserve legal equality, maintain comparability, and administer scarce resources. The relevant GRE question concerns the relational conditions under which classificatory authority becomes accountable. Important dimensions include transparency about category construction, rights to correction or appeal, separation between measurement and automatic sanction where appropriate, access to provenance, and the possibility of revising categories whose consequences diverge from their intended scope.

The same point applies to models. A relational model used only for research has one kind of influence; the same model embedded in a decision system can acquire allocative power. The transition from description to intervention changes the normative and institutional burden carried by the model. Governance therefore needs to track who constructs the representation, who can inspect it, who bears its errors, and who possesses authority to revise it. These questions connect epistemology directly to economic power without implying that every act of measurement is domination.

Taken together, the five limits of this section qualify the practical promise of relational governance. Administrative systems require simplification; law requires stability; computation requires tractable models; normative plurality requires judgment; and observational infrastructures distribute classificatory power unevenly. None of these limits makes relational governance impossible. They determine the conditions under which it can remain accountable, revisable, and proportionate to what is actually known. The historical opening identified in the preceding sections is therefore accompanied by a corresponding discipline: greater relational visibility expands the space of possible intervention while also making the institutional consequences of representation harder to ignore.

1.27 The Double Character of Contemporary Technical Capacity

This section synthesizes the technical and institutional developments examined in Sections 1.19–1.26 by identifying their double character. Contemporary infrastructures expand the capacity to observe relations, predict outcomes, simulate alterna-

tives, and assist decision-making. The same infrastructures can also intensify surveillance, behavioral steering, infrastructural dependence, and concentration of analytical power. The objective is to state this duality without assigning technology an intrinsic political direction. The discussion proceeds through five coupled developments: observation and surveillance, prediction and behavioral intervention, computational assistance and concentration, artificial intelligence as both analytical capacity and economic power, and the relation between technical capability and epistemic humility. The method is historical and institutional: technical affordances are treated as capacities whose economic consequences depend on ownership, access, authority, institutional embedding, and the relations through which technical outputs become actionable.

1.27.1 Observation and Surveillance

Contemporary data infrastructures make a wider range of economic relations observable at finer temporal and organizational resolution. Administrative records can be linked longitudinally; platforms record interaction histories; firms collect transaction and workflow data; sensors and digital systems produce machine-generated traces; and public institutions increasingly organize data across previously separated administrative domains. These developments provide important conditions for relational analysis because relations that once appeared only through episodic surveys or retrospective reconstruction can increasingly be observed as evolving patterns (Einav and Levin 2014; World Bank 2021; Ooijen et al. 2019).

Observation acquires a different institutional character when visibility is asymmetric and linked to consequential action. A platform may observe participants' transactions and interactions at a level of detail unavailable to the participants themselves. An employer may observe task timing, communication patterns, or performance indicators while workers possess limited visibility into the models used to evaluate them. A public authority may combine administrative sources that no individual subject can assemble. In these cases, expanded observation becomes part of a relation between observer and observed rather than a purely epistemic improvement.

The distinction between observation and surveillance therefore turns on relational conditions as much as on the technical act of data collection. Relevant conditions include who observes whom, whether visibility is reciprocal, which purposes authorize collection, who controls the resulting records, whether subjects can inspect or correct representations, and which decisions can be made from the collected data. The same technical infrastructure can support accountability in one configuration and disciplinary control in another. OECD and World Bank work on data-driven public systems accordingly places institutional safeguards, governance arrangements, and relations among data holders and users alongside the expansion of analytical capacity (Ooijen et al. 2019; World Bank 2021).

Algorithmic management provides a concrete economic illustration. Kellogg, Valentine, and Christin document how computational systems can participate in directing, evaluating, and disciplining work, showing that the relevant technology is embedded in organizational relations of authority rather than operating as a detached measurement device (Kellogg et al. 2020). The observational capacity is eco-

nomically consequential because it can be converted into scheduling, performance evaluation, task assignment, or sanction.

For GRE, the important historical change is therefore an increase in the resolution at which economic relations can be rendered observable together with an increase in the asymmetry with which such observation may be distributed. Relational visibility creates new possibilities for research and governance while also making control over observation itself an economic resource. The analytical task is to trace the institutional relation that converts visibility into action.

1.27.2 Prediction and Behavioral Intervention

The second development concerns the coupling of prediction with intervention. Improvements in data availability and computational modeling allow institutions to estimate future outcomes at a scale and frequency that would previously have been impractical. Prediction can support screening, forecasting, risk assessment, resource prioritization, and other decisions in which action depends on expectations about future states. Manski's treatment of prediction and decision emphasizes the importance of distinguishing what data identify from what follows only after additional assumptions, a distinction that remains central when predictive systems enter practical decision processes (Manski 2008).

Prediction becomes economically generative when the predicted classification changes the conditions under which the subject subsequently acts. A risk score may alter access to credit or insurance. A ranking may alter visibility in a marketplace. A forecast may change inventory, investment, or staffing. A recommendation system may reorganize attention. A workplace model may alter assignments, targets, or evaluation. The technical output therefore participates in the relational system that produces the next round of data.

This feedback means that predictive performance and causal understanding occupy different analytical positions. A model may rank future outcomes accurately while leaving the mechanism producing those outcomes only partially specified. Once a prediction is used to intervene, the intervention can alter incentives, opportunities, expectations, and observed behavior. The subsequent data then reflect both the underlying process and the institution's prior decisions. Relational governance therefore requires attention to the pathway through which prediction becomes action rather than treating predictive accuracy as a complete account of policy or organizational adequacy.

Algorithmic control in organizations illustrates this coupling. Systems that predict performance or classify workers can also structure the tasks, metrics, and incentives through which later performance is produced (Kellogg et al. 2020). The distinction is especially important in settings where subjects adapt strategically to measures or where institutional decisions affect the opportunities that later appear in the data. Prediction can then become one component of a recursively generated environment.

The double character of predictive capacity lies in this recursion. Better prediction can reduce local uncertainty and improve the timing or targeting of decisions. The same capacity can intensify behavioral intervention when predictions are embedded in institutions with authority over access, ranking, work, prices, or eligibility. GRE

therefore treats prediction as an epistemic capacity whose economic consequences depend on the relation between model, decision-maker, affected subject, and subsequent feedback.

1.27.3 Computational Assistance and Computational Concentration

Contemporary computation also combines diffusion of analytical assistance with concentration of upstream productive capacity. At the level of use, software, cloud services, machine-learning tools, and generative systems can reduce the time and expertise required for search, coding, drafting, simulation, and data analysis. Experimental and workplace evidence on generative AI already shows that assistance can raise productivity in bounded tasks and that gains can be heterogeneous across users (Noy and Zhang 2023; Brynjolfsson et al. 2025). These developments can broaden access to forms of analytical work that once required larger teams or more specialized training.

The upstream conditions of frontier computation display a different structure. Advanced model development depends on large-scale compute, specialized hardware, data infrastructure, engineering expertise, and substantial capital expenditure. The 2026 AI Index reports that industry produced more than ninety percent of notable AI models in 2025 and documents rapid growth together with substantial concentration in global AI compute infrastructure (Stanford Institute for Human-Centered Artificial Intelligence 2026). The diffusion of interfaces to many users can therefore coexist with concentration of the infrastructural capacity on which those interfaces depend.

These two movements should be analyzed together. A small organization may gain access to powerful analytical assistance through an external service while becoming more dependent on a small number of infrastructure providers. A researcher may gain computational capacities that would once have required institutional-scale resources while losing visibility into training data, model updates, or service continuity. A firm may reduce internal analytical costs while increasing switching costs or exposure to upstream pricing and access decisions. Local capability expansion can therefore occur together with systemic dependency.

Computational concentration is not an automatic consequence of computational progress. Open models, shared infrastructure, public research institutions, interoperability, and falling hardware costs can diffuse capability. The present historical configuration nonetheless makes the relation between downstream accessibility and upstream concentration analytically important. GRE needs to distinguish the availability of a tool to users from ownership or control of the productive infrastructure that makes the tool possible.

The distinction also matters for value capture. Assistance can generate productivity gains across a broad set of users while infrastructure owners capture rents through access fees, complementary services, proprietary interfaces, data advantages, or control over standards. The distribution of gains therefore depends on the institutional pathways through which computational capacity is accessed and monetized. Technical diffusion and economic concentration can proceed simultaneously because they operate at different layers of the production structure.

1.27.4 Artificial Intelligence as Analytical Capacity and Economic Power

Artificial intelligence sharpens the double character of contemporary technical capacity because it can function simultaneously as an analytical instrument and as an economically controlled productive asset. As an analytical instrument, AI can expand the capacity to search, summarize, classify, generate hypotheses, write code, compare scenarios, and interact with complex information environments. The productivity evidence reviewed in Section 1.11 shows that these systems can complement workers and redistribute access to task-relevant expertise in specific settings (Brynjolfsson et al. 2025; Noy and Zhang 2023).

Analytical capability becomes economic power through institutional conversion. A model that can classify customers, workers, suppliers, applicants, or content has limited allocative consequence until an organization can connect its classifications to prices, access, visibility, employment, credit, procurement, or enforcement. A system capable of generating high-quality recommendations has limited bargaining power unless an actor controls the infrastructure, interface, or organizational position through which those recommendations shape decisions. The distinction between capability and power therefore remains essential.

Several conversion interfaces are especially important. Ownership links technical capacity to claims over revenue and assets. Platform position links analytical capacity to control over access and intermediation. Organizational authority links prediction to task assignment and evaluation. Infrastructure ownership links computational capacity to dependency and pricing. Data access can reinforce model quality and observational asymmetry. These relations can combine, allowing a technically capable system to become part of a durable structure of economic influence.

The concentration documented in the AI Index is therefore relevant to economic power without establishing a deterministic relation between frontier capability and social dominance (Stanford Institute for Human-Centered Artificial Intelligence 2026). Concentrated model development and compute create one possible basis for asymmetry; law, market structure, interoperability, public infrastructure, competition, organizational design, and user exit determine how that asymmetry is converted into power. The same technical system can occupy different economic positions under different institutional arrangements.

AI also complicates the location of agency inside economic systems. A firm may delegate analysis to a model while retaining formal decision authority. Workers may rely on AI assistance while remaining responsible for output. Platforms may embed model judgments in ranking systems that influence participants without presenting the model as an independent economic subject. These arrangements distribute analytical agency across humans, organizations, and artificial systems. GRE therefore benefits from treating AI first as a relationally embedded capacity and then asking which subject positions, claims, dependencies, and responsibilities are generated around it.

This approach also preserves the distinction developed in Section 1.16 between economic subjecthood and final metaphysical personhood. An artificial system can become economically consequential through its position in productive and decision

relations even where legal or moral personhood remains absent. The economic problem concerns how capability is institutionalized, who controls its deployment, who bears its errors, who captures its returns, and which forms of dependency follow.

1.27.5 Technical Capability and Epistemic Humility

The expansion of technical capacity increases the need for epistemic discipline. Richer data, larger models, faster simulation, and more capable artificial systems permit analysis of economic structures that were previously inaccessible. They also increase the number of representations, model choices, generated hypotheses, and intervention pathways that researchers and institutions must evaluate. Technical capability can reduce some forms of uncertainty while creating new dependencies on complex systems whose limitations are difficult to observe directly.

The distinction between symbolic performance and epistemic adequacy remains central. Bender and colleagues caution that fluent language generation should not be treated as equivalent to understanding, and the argument applies more broadly to the interpretation of machine-generated analytical output (Bender et al. 2021). A system can produce coherent classifications, explanations, forecasts, or policy drafts whose evidential basis, causal adequacy, or domain validity still requires independent examination. The lower marginal cost of producing analysis can therefore increase the scarcity of verification and judgment.

Contemporary AI evaluation provides a second illustration. The 2026 AI Index documents rapid capability gains together with declining transparency for some frontier systems and growing difficulty in maintaining benchmarks that remain informative as model performance changes (Stanford Institute for Human-Centered Artificial Intelligence 2026). Improved technical performance does not remove the need to ask what a benchmark measures, how training and evaluation data relate, which failure modes remain hidden by aggregate scores, and how results transfer into institutional settings.

Relational and generative systems add another source of uncertainty because observation and intervention can change the system being analyzed. A model embedded in allocation, ranking, or governance can alter the behavior that later supplies its evidence. A classification can reshape the subject position it describes. A successful intervention can transform the relation that made the original model valid. The practical value of a model therefore depends partly on whether institutions can detect when the conditions supporting it have changed.

Epistemic humility in GRE is consequently a positive methodological discipline. It includes calibrated claim strength, explicit model assumptions, provenance, validation against alternative representations, preservation of uncertainty, rights of contestation where models affect subjects, and willingness to revise conclusions as relations evolve. Computational and AI assistance can strengthen each of these practices when used to compare models, search evidence, test sensitivity, and document provenance. Their contribution is greatest when technical capacity expands the range of questions that can be examined while institutional arrangements preserve the capacity to recognize error.

The double character of contemporary technical capacity can now be stated in the

terms required by this chapter. The same historical developments that make relational and generative economics increasingly possible also increase the capacity to observe, classify, predict, intervene, and concentrate control over the infrastructures through which economic reality is represented. Technical progress therefore widens both the epistemic reach of economic inquiry and the institutional stakes of economic representation. This combination supplies an important part of the historical opening for GRE developed in the next section: the contemporary economy is simultaneously more relationally observable and more dependent on the governance of the very instruments that make such observation possible.

1.28 Historical Opening for Generative Relational Economics

This section draws together the historical argument developed across Sections 1.1–1.27. Its role is to explain why Generative Relational Economics becomes intelligible as a distinct research programme under the present historical configuration. The objective is narrower than a claim of historical inevitability. GRE is treated as one possible theoretical response to a conjunction of transformations in productive forces, inherited relations of production, scarcity, allocation, subjecthood, scientific representation, computation, evidence, and institutional form. The discussion therefore proceeds through six linked developments: transformation of productive forces, tension within inherited relations of production, transformation of scarcity and allocation, reorganization of productive dependence, destabilization of entity-centered subjecthood, and the emergence of economic problems whose object is increasingly relational and generative. The method is historical-material and reconstructive. It asks which material and institutional changes make certain economic questions more difficult to contain within inherited abstractions, while preserving the possibility that other theoretical responses to the same historical condition remain available.

1.28.1 Transformation of Productive Forces

The contemporary opening for GRE begins with a transformation in productive forces. Marx's historical-material formulation linked changes in productive forces to tensions within existing relations of production, while leaving open the concrete form through which such tensions are resolved (Marx 1859). Chapter 1 has used this formulation as a historical orientation rather than as a deterministic law. The relevant contemporary change is the expansion of productive capacity through digital networks, data infrastructures, automated cognition, machine learning, large-scale computation, and globally distributed organizational systems.

Digitalization lowers important costs of search, copying, communication, tracking, and verification, thereby changing which transactions and forms of coordination are economically feasible (Goldfarb and Tucker 2019). Artificial intelligence extends this transformation into cognitive and symbolic tasks. Task-based analyses of automation emphasize that technologies can displace some forms of labor while creating complementary or reinstated tasks elsewhere, so technological change should be studied through changing task structures rather than through a simple substitution narrative

(Acemoglu and Restrepo 2019). Experimental and workplace evidence on generative AI similarly indicates substantial productivity effects in bounded cognitive tasks, with gains distributed unevenly across workers and contexts (Noy and Zhang 2023; Brynjolfsson et al. 2025).

The historical significance of these developments lies in the kind of productive capacity they make increasingly available. Earlier mechanization primarily expanded the scale and speed of physical production. Contemporary systems also expand the production, recombination, classification, translation, and circulation of symbols. They therefore intervene in domains previously associated with expertise, communication, research, administration, and professional judgment. The result is a productive system in which material, informational, symbolic, organizational, and computational capacities are more tightly coupled than in earlier industrial configurations.

The transformation is also uneven. Access to advanced analytical interfaces can diffuse rapidly while upstream infrastructures remain highly concentrated. Frontier model development depends on capital-intensive compute, specialized hardware, data infrastructure, and organizational capability, and recent AI-industry evidence documents continuing concentration in frontier model production and compute capacity (Stanford Institute for Human-Centered Artificial Intelligence 2026). Productive expansion therefore coexists with new dependencies on infrastructure providers, model developers, platforms, and institutions able to finance and govern high-cost computational systems.

This combination matters for GRE because it changes the analytical object of production. Productive capacity is increasingly difficult to locate within a single worker, firm, machine, or asset. It emerges from relations among human expertise, software, models, data, platforms, organizational routines, legal permissions, energy and compute infrastructures, and transnational supply networks. The transformation of productive forces therefore creates a historical setting in which relational configuration becomes more directly productive and in which the capacity of relations to generate later capabilities becomes economically salient.

1.28.2 Tension within Inherited Relations of Production

The transformation of productive forces becomes theoretically consequential when it is placed beside inherited relations of production. Modern economic institutions developed around property, contract, firms, employment, administratively recognizable recipients, and relatively stable categories of productive contribution. These forms remain indispensable. They organize responsibility, allocate rights, structure investment, and make complex economies legally and administratively actionable. Their historical effectiveness is one reason entity-centered economic analysis became so powerful.

Contemporary productive organization places increasing pressure on the assumption that productive contribution can be attributed cleanly to bounded entities. Firms remain important, yet production may depend on platform ecosystems, contractual networks, external research communities, data contributors, open-source infrastructures, cloud providers, supply chains, and distributed knowledge relations. Coase's

analysis of firms already treated organizational boundaries as an institutional response to the comparative costs of using markets and internal direction rather than as a natural partition of economic reality (Coase 1937). Granovetter's account of embeddedness later emphasized that economic action is situated within ongoing social relations rather than exhausted by atomized actors or anonymous markets (Granovetter 1985). Contemporary digital production intensifies the practical relevance of both observations.

A second tension concerns employment. Wage labor historically connected productive contribution, income, organizational membership, and social recognition. Automation and AI can loosen these connections by altering which tasks require human labor and by changing the relative productive advantage of workers across tasks (Acemoglu and Restrepo 2019; Korinek and Stiglitz 2019). The resulting problem is not reducible to the quantity of employment. It concerns the institutional architecture through which access to income, insurance, training, status, and participation has been attached to employment. A productive transformation can therefore become an allocative and recognitional transformation even when total output rises.

A third tension concerns ownership and control. Productivity gains generated by collective infrastructures can be appropriated through legally concentrated claims. Evidence on the rise of highly productive "superstar" firms provides an important structural precedent: technological and organizational changes can increase aggregate productivity while simultaneously increasing concentration and altering labor shares (Autor et al. 2020). The specific mechanisms of AI-era concentration remain empirical questions, but the broader analytical point is already established. Productive abundance does not determine the distribution of control over productive assets or the claims on the surplus they generate.

These tensions do not imply that inherited relations of production have become obsolete. They indicate that the mapping from productive contribution to recognized ownership, income, authority, and participation becomes increasingly mediated by institutional relations. GRE enters this space by asking how economic relations convert productive capacities into claims, dependencies, exclusions, and future possibilities. The historical opening therefore lies in the growing explanatory importance of the conversion architecture connecting productive forces to institutional positions.

1.28.3 Transformation of Scarcity and Allocation

The same historical transformation alters the structure of scarcity. Classical and neo-classical economics developed powerful languages for problems in which scarce resources must be allocated among competing uses. That problem remains fundamental. Material resources, energy, land, housing, time, specialized labor, compute, and institutional capacity remain scarce. Contemporary technical change therefore does not abolish scarcity. It changes its distribution across domains and, in some domains, the relation between abundance at one layer and scarcity at another.

Symbolic production provides the clearest example. The marginal cost of producing text, images, code, summaries, and other representations can fall dramatically when generative systems are available. Yet the abundance of symbols increases the burden placed on attention, verification, interpretation, provenance, and trust. Si-

mon's formulation that an abundance of information consumes attention becomes more important as the production of candidate messages, claims, and artifacts becomes cheaper (Simon 1971). Arrow's analysis of trust as an economically consequential institution likewise indicates why systems of exchange cannot be understood through information volume alone (Arrow 1972). Where provenance and authenticity become uncertain, the capacity to verify and trust may become more valuable precisely because symbolic supply has expanded.

Allocation problems therefore move beyond the transfer of a nominally scarce resource to a recognized recipient. Access to attention, institutional gateways, trusted channels, computational infrastructure, and relations capable of generating later opportunities may become decisive. Sen's capability approach is relevant here because it distinguishes possession of resources from the substantive opportunities a person can actually realize under specific conversion conditions (Sen 1999). GRE extends this concern temporally: present allocations can change the relations and capacities through which later opportunities are generated.

This temporal extension changes the meaning of scarcity. A resource can be abundant at the point of production while the relations required to convert it into learning, recognition, authority, or participation remain scarce. Knowledge artifacts can be plentiful while trustworthy epistemic communities remain difficult to build. Computational tools can be widely accessible while frontier compute remains concentrated. Information can be abundant while institutional attention remains rationed. A system can therefore experience simultaneous abundance and scarcity at different relational layers.

The historical opening for GRE lies partly in this layered structure. Allocation remains central, but the analytically relevant object increasingly includes the relation through which a resource becomes effective and the downstream conditions the allocation helps reproduce. The question shifts from the amount transferred to the generative pathway through which a transfer becomes capacity, dependence, participation, exclusion, or future option. This shift provides the historical bridge from an allocation-centered economics toward a generative relational one.

1.28.4 Reorganization of Productive Dependence

The development of artificial generativity introduces a further historical possibility. Political economy has long examined societies in which persons depend on goods and services generated through the activity of others and in which division of labor, property, employment, exchange, and monetary claims organize access to those outputs. This description is not offered as an exhaustive origin theory of markets or money. It identifies a persistent structural problem: subjects are not materially self-sufficient, and access to socially generated capacities and products must therefore be organized through institutions. Marx's later political writings are relevant here because they connect distributive arrangements to the organization of the conditions of production rather than treating consumption shares as an isolated problem (Marx 1994).

Artificial intelligence, especially when considered together with robotics and automated infrastructures, makes it possible to ask how this structure changes when an expanding range of useful outputs can be generated without another person's direct

labor. The present evidence concerns bounded tasks rather than a fully autonomous productive system, and task-based analyses caution against treating automation as a single process of total substitution (Acemoglu and Restrepo 2019; Korinek and Stiglitz 2019). The historical significance lies instead in the possibility opened by the trajectory: some forms of instrumental dependence on another human's productive activity may weaken as artificial systems become capable of producing, coordinating, or mediating a wider range of goods and services.

Such a development would not by itself dissolve political-economic dependence. Dependence can migrate. A subject who no longer requires another person's labor may still require access to models, compute, energy, data, robotic systems, land, networks, standards, legal permissions, or organizational infrastructures controlled by others. Productive abundance at the level of output can therefore coexist with scarcity at the level of access. The replacement of a human generator by an artificial generator does not constitute a transformation of the underlying relation when the capacity to use, modify, maintain, or redirect the generative system remains strongly conditioned by asymmetric claims or permissions.

This distinction reveals three analytically different layers. One concerns access to what has already been generated. A second concerns access to the capacity through which new outputs can be generated. A third concerns participation in shaping the conditions under which generation becomes possible and develops over time. Those conditions include material infrastructures, legal rules, knowledge, education, technical standards, organizational forms, relational networks, trust, institutional recognition, and increasingly the computational systems through which productive possibilities are opened or closed. The layers interact, but distribution at one layer does not determine distribution at the others.

The historical implication is that a reduction in necessary human labor need not produce a post-political-economic society. It can instead move the political-economic problem toward the organization of access to generative capacity and toward the formation, opening, closure, and reproduction of generative conditions. A society could become less dependent on human labor while remaining highly dependent on infrastructures whose use and evolution are unequally controlled. Conversely, broad access to productive outputs would remain incomplete if subjects had little capacity to participate in shaping the systems on which that access depends.

GRE treats this reorganization as an opening rather than as a technological destiny. It does not infer a particular institutional order from advances in artificial intelligence, nor does it assume that labor, exchange, money, or property disappear when artificial productive capability increases. The narrower historical claim is that AI makes a deeper political-economic question increasingly visible: who can enter, mobilize, influence, maintain, and alter the processes through which future economic possibilities are generated? Later parts of the book develop the implications for capital, power, exploitation, access, and governance. At this point, the question matters because it helps explain why the economic object of inquiry can no longer be exhausted by the distribution of already generated outputs.

1.28.5 Destabilization of Entity-Centered Subjecthood

The transformation of productive forces and scarcity also destabilizes inherited forms of economic subjecthood. Modern economic institutions became organized around subjects that could be recognized as workers, owners, consumers, households, firms, creditors, debtors, and public authorities. These roles were never exhaustive descriptions of the persons or organizations occupying them. They were nevertheless sufficiently stable for contract, taxation, accounting, employment, welfare provision, and statistical administration.

AI complicates this architecture because systems without settled legal or moral personhood can perform economically significant tasks, generate outputs, mediate decisions, and participate in organizational workflows. Bender and colleagues' critique of large language models remains important precisely because fluent symbolic performance does not settle questions of understanding, agency, or personhood (Bender et al. 2021). GRE therefore has reason to distinguish economic participation from final ontological status. A system may have economic effects without thereby receiving the standing of a human person, legal subject, or moral patient.

Human subjecthood is destabilized from the opposite direction. If productive tasks once central to employment can be performed more efficiently by artificial systems, the relation among work, contribution, income, recognition, and membership becomes less secure. Becker's human-capital framework captured an industrial and post-industrial world in which investments in education and skill were strongly connected to productive capacity and earnings (Becker 1964). Contemporary AI can alter the return structure of particular skills, and the social consequences depend on how institutions reorganize work, ownership, training, access, and participation.

The relevant distinction is between productive contribution, productive necessity, and subject worth. A person can contribute without being comparatively necessary for a task; a person can lose comparative productive advantage without losing the capacity for experience, relation, learning, care, judgment, or participation. Sen's emphasis on substantive freedom helps expose the insufficiency of reducing social standing to productive superiority (Sen 1999). The economic problem is therefore institutional as much as technological: inherited systems may continue to allocate income and recognition through roles whose historical relation to productive necessity is changing.

A relational account of subjecthood becomes useful under these conditions because it asks how economic positions are generated and stabilized through relations of recognition, responsibility, membership, dependence, and access. Such an account does not deny persons, firms, or organizations as entities. It asks which relations make those entities economically actionable and how changes in those relations alter the subject positions available in the economy. The destabilization of entity-centered subjecthood thus becomes one route by which contemporary conditions open space for a relational economic framework.

1.28.6 Emergence of Generative Relational Economic Problems

The preceding transformations converge in a class of problems whose object is neither a static entity nor a fixed relation. The central question becomes how economic relations generate later subjects, capacities, institutions, constraints, dependencies, and possibilities. This is the specifically generative dimension of the historical opening.

Several developments make such questions increasingly visible. Networked production means that the productivity of an actor can depend on the configuration of relations in which it is embedded. Institutional change means that rules can alter incentives while also changing the categories and relations through which later action occurs (North 1990). Evolutionary economics provides a language for routines, search, innovation, and changing populations of organizational practices (Nelson and Winter 1982). Path-dependent models show how early contingencies and increasing returns can shape later feasible trajectories (Arthur 1989). Contemporary network and multiscale methods provide additional tools for representing changing relational structures across levels (Newman 2003; Bar-Yam 2004).

The economic significance of generativity appears when present arrangements change the conditions of later generation. Education can alter future capabilities. Employment can generate professional networks and institutional access. Ownership can generate collateral, bargaining power, and future investment capacity. Platform participation can generate reputation and dependence simultaneously. Research infrastructures can generate new knowledge while concentrating access to data or compute. Allocation can therefore be evaluated partly through the futures it makes easier, harder, or impossible.

These problems exceed a static relational description because the relation itself may change through use. A cooperative relation can build trust or exhaust it. A labor relation can generate skill, dependence, identity, or exit capacity. A credit relation can expand productive capacity while changing future claims and bargaining positions. An institutional category can stabilize participation while reifying the boundary it recognizes. Generative analysis therefore asks how relations reproduce, transform, or dissolve the conditions under which later relations become possible.

GRE emerges historically at the intersection of necessity and possibility. It becomes more necessary because productive forces, scarcity, subjecthood, and power are increasingly organized through complex relations whose downstream effects matter economically. It becomes more possible because scientific, computational, evidential, and institutional capacities make such relations more observable and analyzable than before. Neither movement is complete. Relations remain partially observed, causal structures remain underdetermined, formal models remain selective, and governance remains constrained by law, normativity, computation, and power.

The historical opening should therefore be read as an invitation rather than as a conclusion. Contemporary conditions make a generative relational economics increasingly thinkable and potentially useful. They do not establish its superiority in advance. GRE must still demonstrate that relational and generative descriptions improve explanation, comparison, evaluation, or practical understanding in concrete economic problems. Section 1.29 therefore closes the chapter by stating the possibili-

ties and limits under which this historically situated research programme proceeds.

1.29 Historical Possibilities and Limits of Generative Relational Economics

This section closes Chapter 1 by stating the historical possibility conditions, analytical scope, and persistent limits under which Generative Relational Economics proceeds. Its role is to convert the chapter's historical reconstruction into a disciplined statement of theoretical standing. The objective is to specify what contemporary conditions make GRE increasingly feasible, which forms of analysis those conditions enable, which institutional developments permit relational and generative questions to become actionable, and which epistemic and normative limits remain constitutive. The discussion proceeds through relational observability, generative analytic capacity, institutional possibility, epistemic and normative limits, and the historical situatedness of GRE itself. The method is synthetic and reflexive. It draws on the chapter's historical, institutional, methodological, and normative analyses while treating GRE as one historically generated response among possible responses to the contemporary economic condition.

1.29.1 Relational Observability

A relational economics becomes practically possible only to the extent that economically relevant relations can be observed, reconstructed, or responsibly inferred. Contemporary economies generate a density of traces that earlier periods could rarely sustain at comparable scale. Administrative records, transaction logs, platform interactions, longitudinal surveys, organizational databases, supply-chain records, machine-generated telemetry, and public statistical infrastructures can preserve sequences of interaction rather than only isolated outcomes. Data infrastructures therefore enlarge the empirical space in which economic relations can be studied over time.

This expansion of observability matters because relations are rarely exhausted by the attributes of the entities they connect. A firm may possess a given stock of capital while occupying a highly dependent position in a supply network. A worker may have a given qualification while lacking access to the institutions through which that qualification becomes economically effective. A platform may mediate exchange while simultaneously structuring search, ranking, reputation, and access. Network analysis, longitudinal linkage, and higher-order relational models make some of these structures increasingly representable (Newman 2003; Battiston et al. 2020). Public-data frameworks likewise increasingly recognize administrative, survey, citizen-generated, and machine-generated data as complementary sources within wider evidence ecosystems (World Bank 2021; Ooijen et al. 2019).

Relational observability remains mediated. Data are produced through classifications, measurement conventions, legal permissions, technical architectures, and institutional incentives. Desrosières and Porter show in different ways that statistical objects depend on conventions of classification and comparability rather than emerg-

ing as transparent copies of the world (Desrosières 1998; Porter 1995). A database may therefore make a relation visible by first deciding what counts as an actor, event, category, transaction, or boundary. The resulting representation can be highly useful while remaining historically and institutionally constructed.

The distinction between a relational trace and the relation itself is therefore essential. Repeated communication may indicate collaboration, dependence, supervision, conflict, or routine administration depending on context. A contractual tie may formalize only part of the effective relation. Absence of recorded interaction may reflect genuine absence, informal coordination, inaccessible records, strategic concealment, or a measurement system designed around different categories. Section 1.25 therefore treated partial observation and representation dependence as persistent epistemic conditions rather than temporary technical inconveniences.

GRE consequently treats contemporary observability as a historical opening rather than as epistemic completion. More relations can be followed across time, organizations, and scales than was previously feasible, and this makes relational economic questions empirically more tractable. The same evidential expansion also increases the need to document provenance, classification, missingness, boundary choice, and the institutional production of data. Relational observability becomes scientifically valuable when the conditions of observation remain part of the analysis.

1.29.2 Generative Analytic Capacity

Relational observation supplies a map of connections and configurations; generative analysis requires an additional capacity to investigate how those relations produce later states, structures, constraints, and possibilities. Contemporary scientific and computational developments make this second task more feasible by allowing trajectories, endogenous change, feedback, heterogeneous interaction, and historical dependence to become explicit analytical objects.

Several methodological developments contribute to this capacity. Dynamical systems provide languages for trajectories and stability; complex-systems approaches examine emergence under heterogeneous local interaction; network and higher-order models represent changing relational topology; agent-based models permit heterogeneous actors and institutions to coevolve; and multiscale approaches make it possible to distinguish structures visible at different levels of aggregation (Strogatz 2024; Holland 2012; Newman 2003; Battiston et al. 2020; Tesfatsion 2006; Bar-Yam 2004). Evolutionary economics adds a specifically economic vocabulary for routines, search, innovation, and changing organizational populations, while path-dependent models clarify how early contingencies and increasing returns can alter later feasible trajectories (Nelson and Winter 1982; Arthur 1989).

These developments allow GRE to ask a stronger class of questions. A labor relation can be studied through the skills, networks, bargaining positions, and exit capacities it generates over time. An allocation can be studied through the institutional access or dependence it creates downstream. A platform relation can be studied through reputation, lock-in, and future option structure. A legal category can be studied through the subject positions and claims it makes actionable. Generative analysis therefore concerns the transformation of conditions under which later economic action occurs,

rather than only the immediate output associated with a present state.

Contemporary computation expands the feasible scale of this inquiry. Large simulations, temporal-network computation, differentiable pipelines, and hybrid empirical-modeling systems allow analysts to explore model families that would once have been prohibitively costly (Holme and Saramäki 2012; Baydin et al. 2018). Yet computational capacity changes the size of the model space more quickly than it resolves the problem of model adequacy. A simulation can represent a mechanism precisely while representing the wrong mechanism. A dynamic network can reproduce observed patterns while omitting the institutional meaning of the links. A path-dependent model can generate lock-in without establishing that the empirical case follows the proposed feedback process.

Generative analytic capacity must therefore remain tied to evidential discipline. The relevant gain is the ability to formulate, compare, and interrogate richer mechanisms, not the ability to eliminate historical contingency or uncertainty. Manski's distinction between what data identify and what additional assumptions contribute remains especially important when models become more elaborate (Manski 2008). GRE uses expanded analytic capacity to sharpen questions about generation, propagation, and transformation while preserving explicit uncertainty about mechanism, boundary, scale, and historical transfer.

1.29.3 Institutional Possibility

A theoretical framework becomes economically consequential only when institutions can recognize at least some of the objects it describes. Contemporary institutional development provides partial precedents for treating collective relations, shared assets, fiduciary responsibilities, capabilities, common resources, environmental constraints, and adaptive revision as legitimate objects of economic organization. These precedents do not constitute a ready-made GRE institutional order. They show that modern *nomos* already contains forms through which economic action can be organized around structures more complex than isolated ownership by a single bounded recipient.

Trust law, for example, can partition assets and organize claims among trustees, beneficiaries, and third parties through a durable legal structure (Hansmann and Mattei 1998). Cooperative institutions can connect economic participation to jointly owned and democratically controlled organizations (International Cooperative Alliance 1995). Commons scholarship demonstrates that groups can construct rule systems for governing shared resources under conditions that vary across successful and unsuccessful cases (Ostrom 1990). Human-development approaches shift evaluative attention from resource possession alone toward substantive capability and opportunity (Sen 1999). Environmental and intergenerational frameworks extend the relevant horizon beyond present output toward the conditions inherited by future participants (World Commission on Environment and Development 1987).

These institutional forms matter for GRE because they demonstrate several kinds of possibility. Economic claims can be attached to collective arrangements. Decision rights can be distributed. Stewardship can be distinguished from unrestricted alienation. Beneficial interests can differ from formal control. Rules can be revised when

ecological or social conditions change. Resources can be evaluated partly through the capabilities they enable. None of these arrangements is intrinsically generative or just; each can succeed, fail, exclude, or produce unintended dependencies. Their significance lies in demonstrating that the institutional grammar of economic life is historically variable.

Adaptive and evidence-informed governance provides a further possibility. Contemporary public-governance frameworks increasingly emphasize monitoring, evaluation, evidence updating, and institutional learning across policy cycles (OECD 2020a; OECD 2020b; OECD 2020c). Such arrangements create institutional space for decisions to be revised as downstream effects become visible. For GRE, this is particularly relevant because generative evaluation is temporally incomplete. If an intervention changes capabilities, relations, constraints, or option spaces over time, institutional arrangements require procedures for observing those effects and reopening prior judgments.

Institutional possibility therefore concerns the capacity to make relational and generative objects actionable without assuming that every relation should be formalized. Legal recognition can stabilize a relation while also reifying it; measurement can make a collective visible while narrowing its meaning; adaptive governance can increase responsiveness while also expanding surveillance or administrative reach. GRE consequently treats institutional development as an enabling condition whose design remains normatively and politically contestable.

1.29.4 Epistemic and Normative Limits

The historical possibility of GRE is inseparable from limits that cannot be removed by greater technical sophistication alone. Relations remain partially observable, causal mechanisms remain underdetermined, boundaries and scales remain analytically consequential, and generative trajectories remain temporally incomplete. Section 1.25 established these limits at the level of knowledge, while Section 1.26 showed how they reappear in governance through administrative abstraction, legal categories, computational tractability, normative plurality, and unequal power over observation and classification.

The first limit concerns inference. Rich relational data do not automatically distinguish causation from selection, common causes, strategic adaptation, or unobserved structure. Longitudinal evidence can improve temporal reconstruction without guaranteeing identification. Larger models can represent more mechanisms while creating additional degrees of freedom. The appropriate response is calibrated claim strength: association, mechanism evidence, historical reconstruction, counterfactual support, and formal plausibility should be distinguished rather than compressed into a single language of proof. Manski's analysis of identification provides a useful discipline for this restraint (Manski 2008).

The second limit concerns scale and boundary. An observed relation can change meaning when the analytical boundary expands. A cooperative relation within one organization may depend on extraction elsewhere in a supply chain. A beneficial local intervention may transfer risk across jurisdictions or generations. Robinson's demonstration that aggregate and individual correlations may differ remains a clas-

sic warning against treating patterns at one scale as direct evidence at another (Robinson 1950). GRE therefore treats multiscale analysis as a source of additional questions rather than a guarantee of a privileged final scale.

The third limit is normative. Generativity does not establish goodness. A relation can generate capability while also generating dependence; an institution can stabilize trust while excluding outsiders; an innovation can expand aggregate option space while closing options for particular groups. Sen's capability framework is useful precisely because evaluative questions depend on the substantive freedoms available to differently situated persons rather than on resource totals alone (Sen 1999). Rights, participation, recognition, historical responsibility, and intergenerational consequences introduce normative dimensions that cannot be inferred mechanically from system dynamics.

The fourth limit concerns power. Observation, classification, prediction, and intervention are themselves relational capacities. Actors able to define categories, control data infrastructures, set model assumptions, allocate compute, or determine which evidence enters institutional processes can shape the reality that governance subsequently treats as given. Contemporary technical capacity therefore has the double character developed in Section 1.27: it can widen analytical access while concentrating the infrastructures through which economic reality is represented and acted upon.

These limits define GRE's practical discipline. The framework requires sufficiently strong claims to be useful and sufficiently explicit limits to remain revisable. Relational and generative analysis should be introduced where it changes explanation or evaluation, and its additional complexity should be justified by the economic problem under study. Epistemic humility is therefore an operational condition of the framework rather than a concluding disclaimer.

1.29.5 Generative Relational Economics as a Historically Situated Theory

GRE is itself part of the historical process it seeks to analyze. Its concepts arise within a period marked by global production networks, digital intermediation, artificial intelligence, symbolic abundance, transformed scarcity, intensified data infrastructures, expanding computational capacity, new forms of concentration, and growing concern with capabilities, commons, environment, and intergenerational effects. The framework therefore has no privileged position outside history from which to describe an economy that remains historically dynamic.

This situatedness clarifies the status of the chapter's central claim. Relations have always mattered in economic life. Earlier economic theories were not simply deficient because they often foregrounded entities, prices, resources, or allocation. Those abstractions were historically powerful because property, firms, households, employment, contracts, accounts, and administrative categories made many economic processes tractable through relatively stable units. Chapter 1 has therefore reconstructed entity-centered economics as historically intelligible before identifying the contemporary conditions that expose some of its limits.

GRE emerges when several developments converge. Productive capacity becomes

increasingly distributed across relations among humans, organizations, software, data, platforms, and infrastructures. Scarcity becomes layered across material, informational, attentional, institutional, and generative domains. Economic subjecthood becomes less securely tied to inherited productive roles. Scientific and computational languages make dynamic relational structures more representable. Evidence infrastructures make some trajectories more observable. Legal and institutional forms provide partial precedents for collective, capability-based, fiduciary, cooperative, and adaptive arrangements. At the same time, uncertainty, normative plurality, surveillance, concentration, and classification power remain unresolved.

A historically situated theory must therefore remain open to historical revision. Concepts developed for the present configuration may lose explanatory value as productive forces, institutions, and social relations change. New evidence may show that a proposed generative mechanism is weak, domain-specific, or incorrectly specified. Other economic traditions may explain a problem more parsimoniously or reveal dimensions GRE has obscured. The framework's commitment to revisability follows from the same historical reasoning used to study economic categories themselves: a theory that treats economic forms as historically generated cannot coherently exempt its own categories from historical development.

This position also limits GRE's disciplinary ambition. The framework is proposed as a language and research programme for economic problems in which relational constitution, generative transformation, heterogeneous value, multiscale dynamics, and downstream possibility are analytically consequential. It does not require every economic problem to be reformulated relationally, and it does not claim to replace theories that perform well within their domains of adequacy. Its contribution must be demonstrated through comparative explanation, empirical research, formal development, historical interpretation, and practical analysis.

Chapter 1 has established the historical condition under which such a programme becomes thinkable. The next task is intellectual rather than chronological. The following chapter reconstructs the philosophical, social-theoretical, economic, and scientific lineages from which the conceptual resources of GRE can be assembled. The movement from historical conditions to intellectual genealogy preserves the chapter's central discipline: GRE is approached as a historically situated theoretical formation whose concepts require both ancestry and justification, and whose future development remains open to criticism, counterevidence, and revision.

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